



US012414807B2

(12) **United States Patent**
Babkin et al.

(10) **Patent No.:** **US 12,414,807 B2**

(45) **Date of Patent:** ***Sep. 16, 2025**

(54) **CRYOABLATION ELEMENT WITH
CONDUCTIVE LINER**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/448,910**

(22) Filed: **Aug. 12, 2023**

(65) **Prior Publication Data**

US 2023/0389977 A1 Dec. 7, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/958,589, filed as
application No. PCT/US2019/012754 on Jan. 8,
2019, now Pat. No. 11,751,930.

(60) Provisional application No. 62/615,573, filed on Jan.
10, 2018.

(51) **Int. Cl.**

A61B 18/02 (2006.01)

A61B 18/00 (2006.01)

(52) **U.S. Cl.**

CPC **A61B 18/02** (2013.01); **A61B 2018/00095**
(2013.01); **A61B 2018/00351** (2013.01); **A61B**
2018/00577 (2013.01); **A61B 2018/00714**
(2013.01); **A61B 2018/0212** (2013.01); **A61B**
2018/0262 (2013.01)

(58) **Field of Classification Search**

CPC **A61B 18/02**; **A61B 2018/00095**; **A61B**
2018/00351; **A61B 2018/00577**; **A61B**
2018/00714; **A61B 2018/00791**; **A61B**
2018/00839; **A61B 2018/0212**; **A61B**
2018/0262

See application file for complete search history.

(56)

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606/49
2016/0220294 A1 * 8/2016 Babkin **A61B 18/02**

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Primary Examiner — Tigist S Demie

(74) *Attorney, Agent, or Firm* — Batt IP A Law
Corporation; Richard Batt

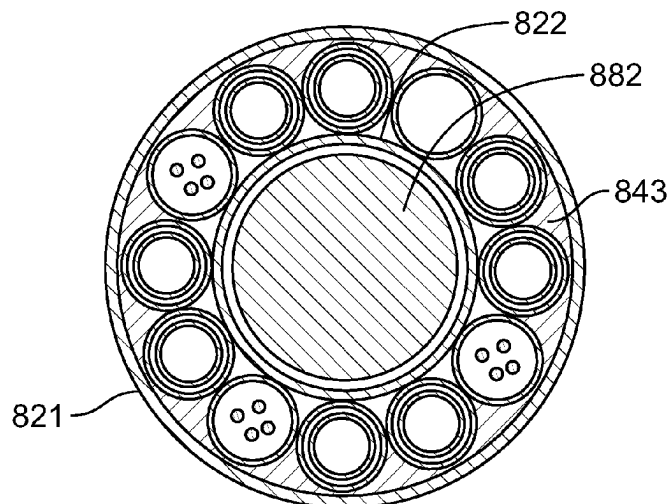
(57)

ABSTRACT

An ablation apparatus for creating a lesion in target tissue,
the ablation apparatus including a handle, an elongate shaft
extending from the handle to a distal tip, where the shaft
includes a first portion, an ablation portion distal to the first
portion and having an outer sheath, and at least one ablation
energy element disposed within the outer sheath, where a
space is formed between the at least one ablation energy
element and the outer sheath. The ablation apparatus further
includes a thermally conductive liner disposed within the
space.

25 Claims, 44 Drawing Sheets

880



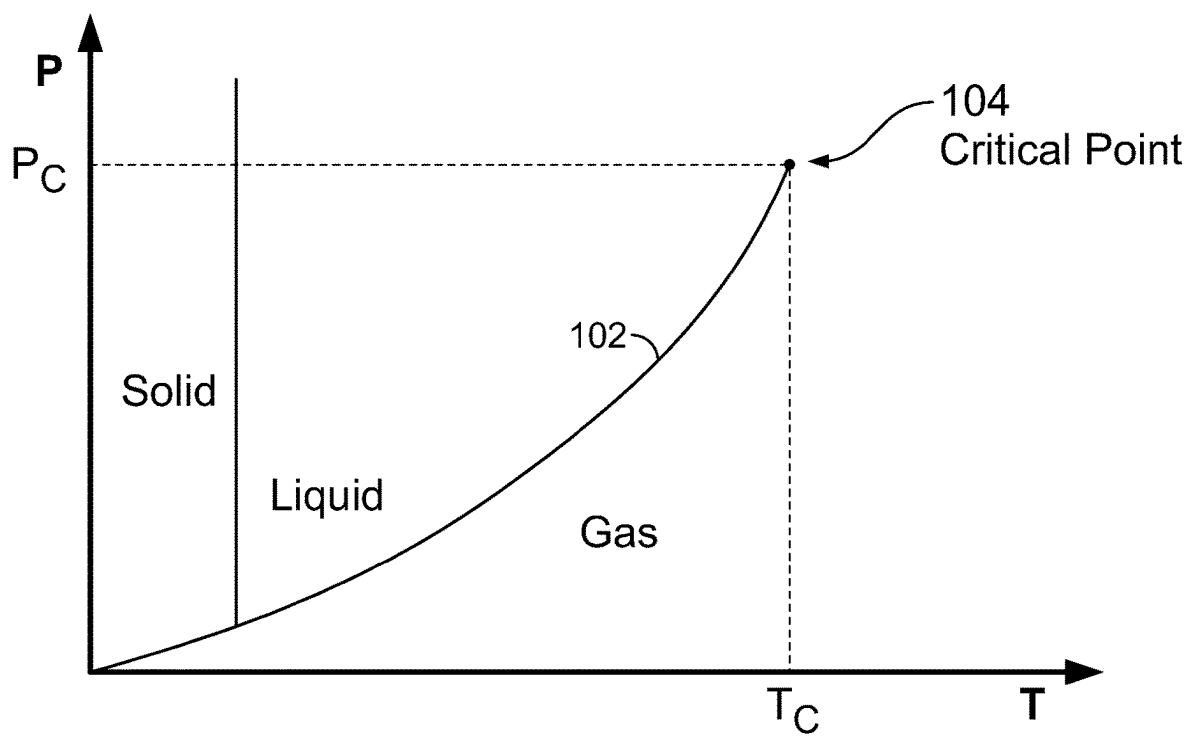
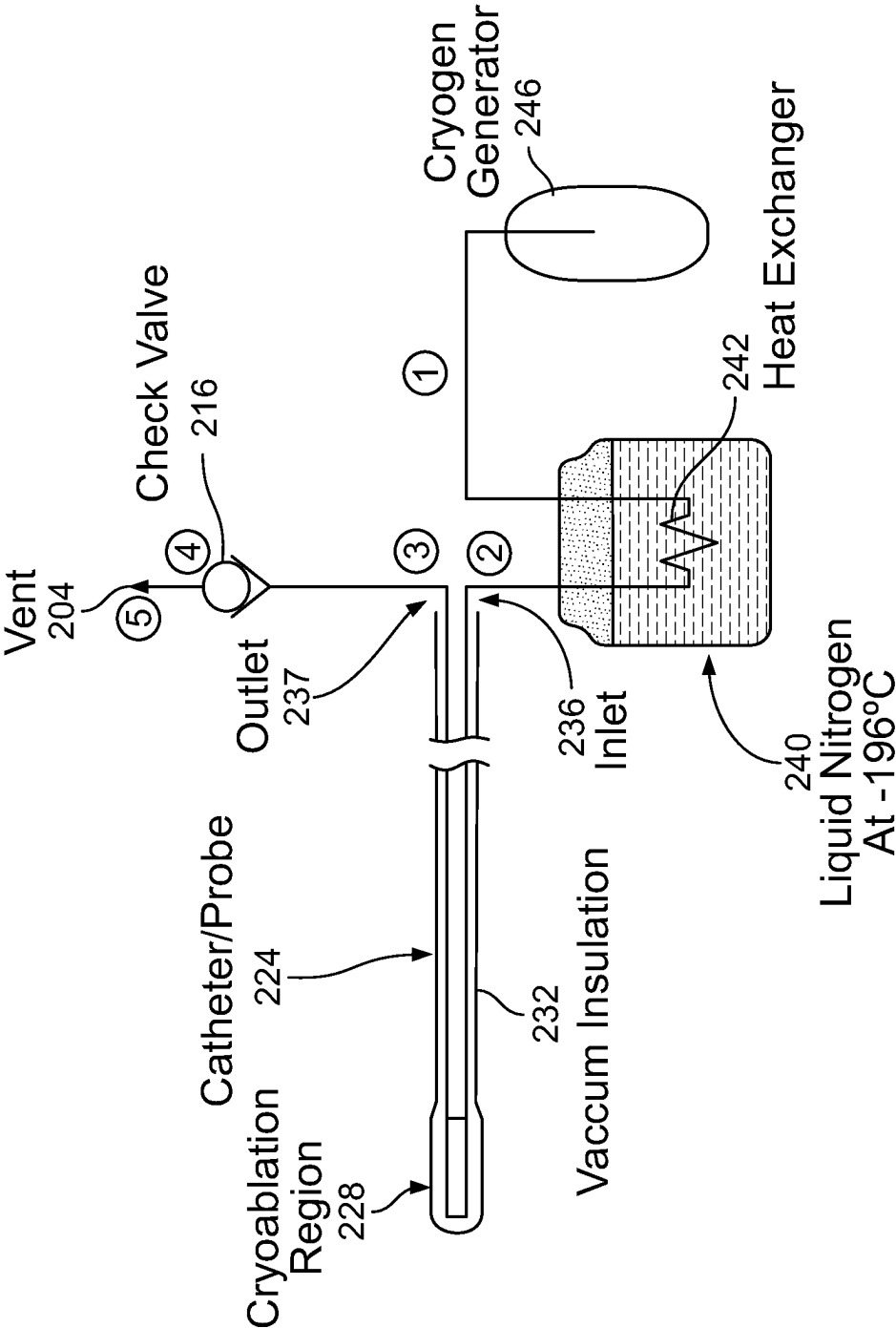


FIG. 1



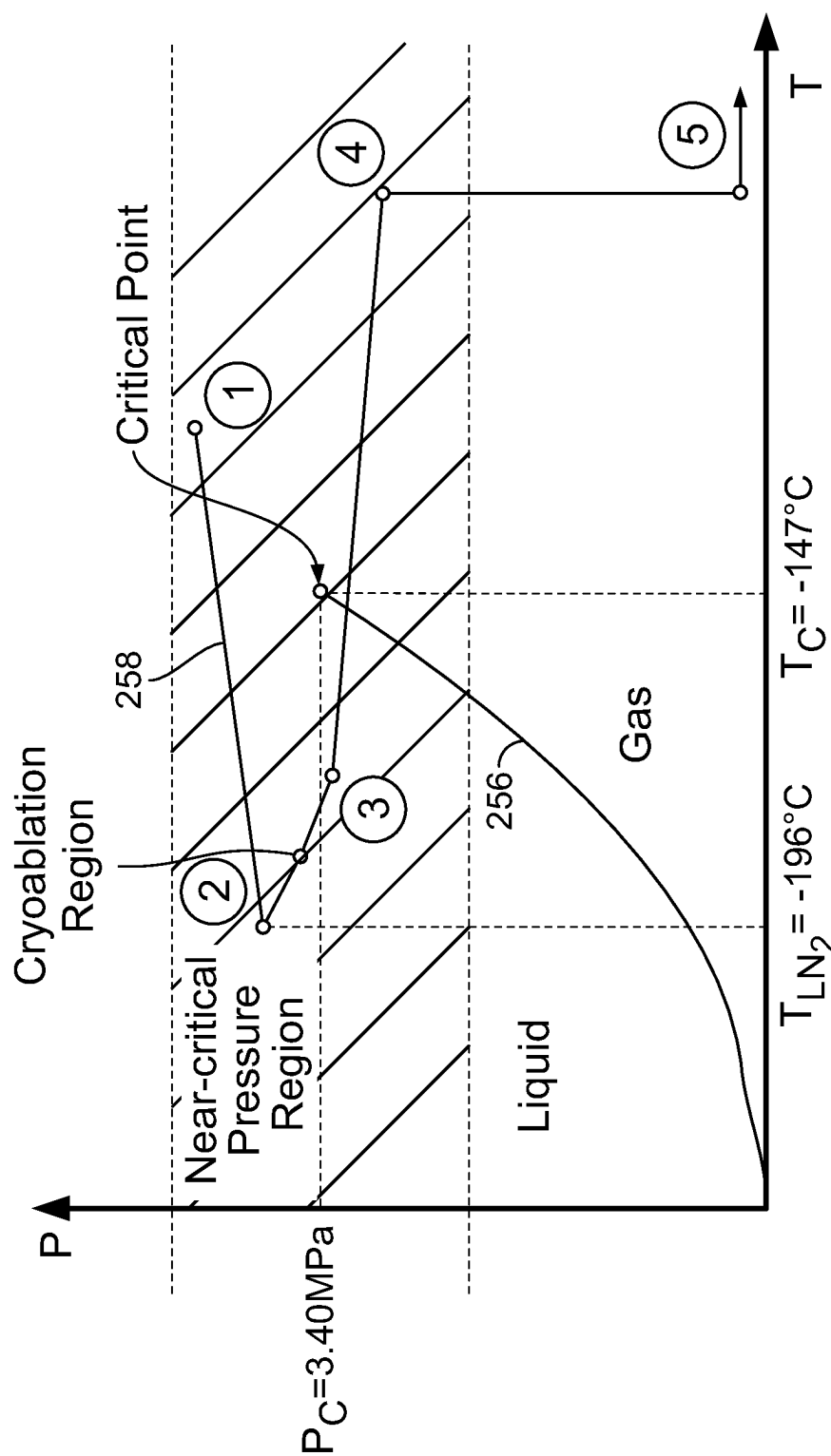


FIG. 3

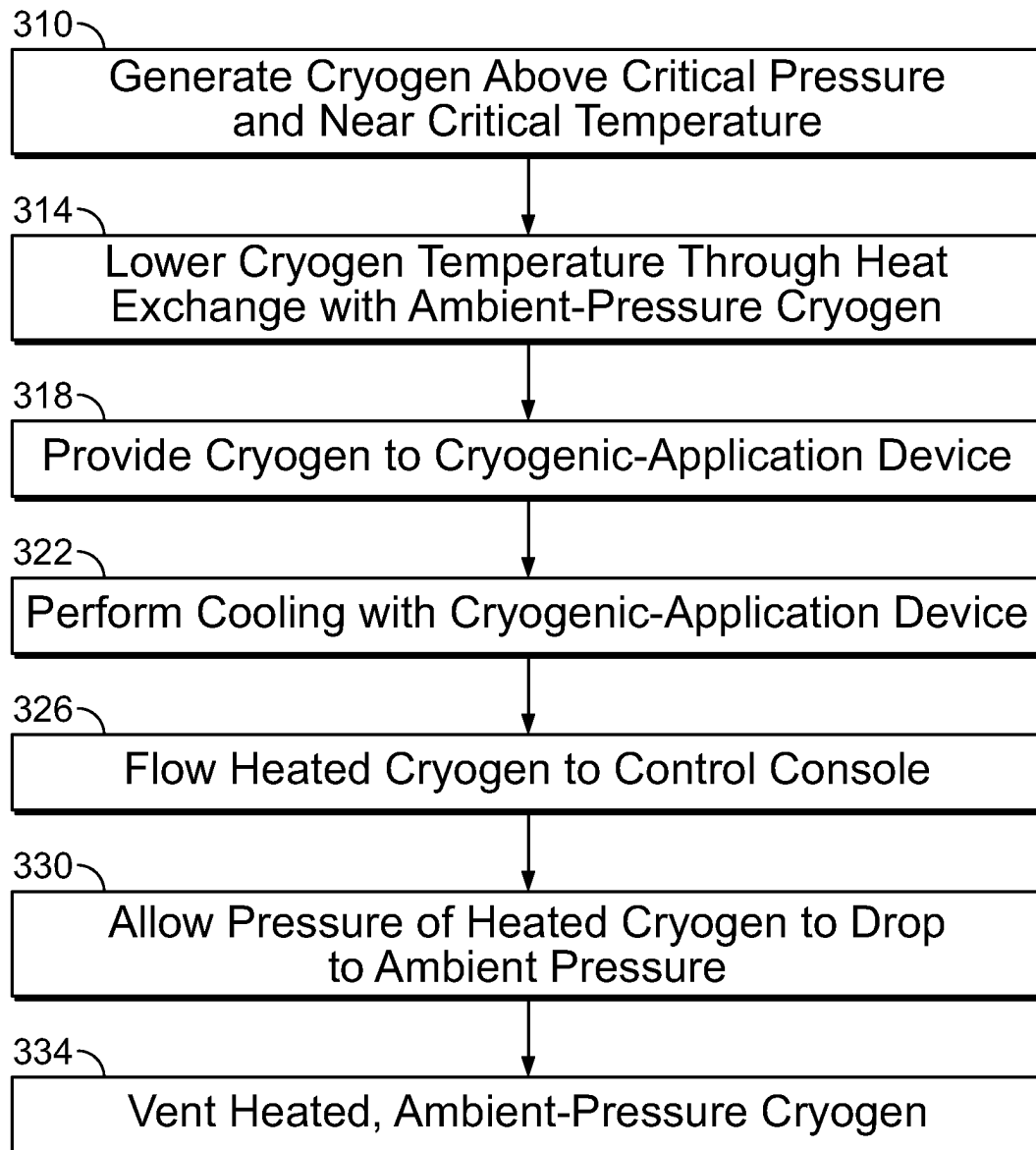


FIG. 4

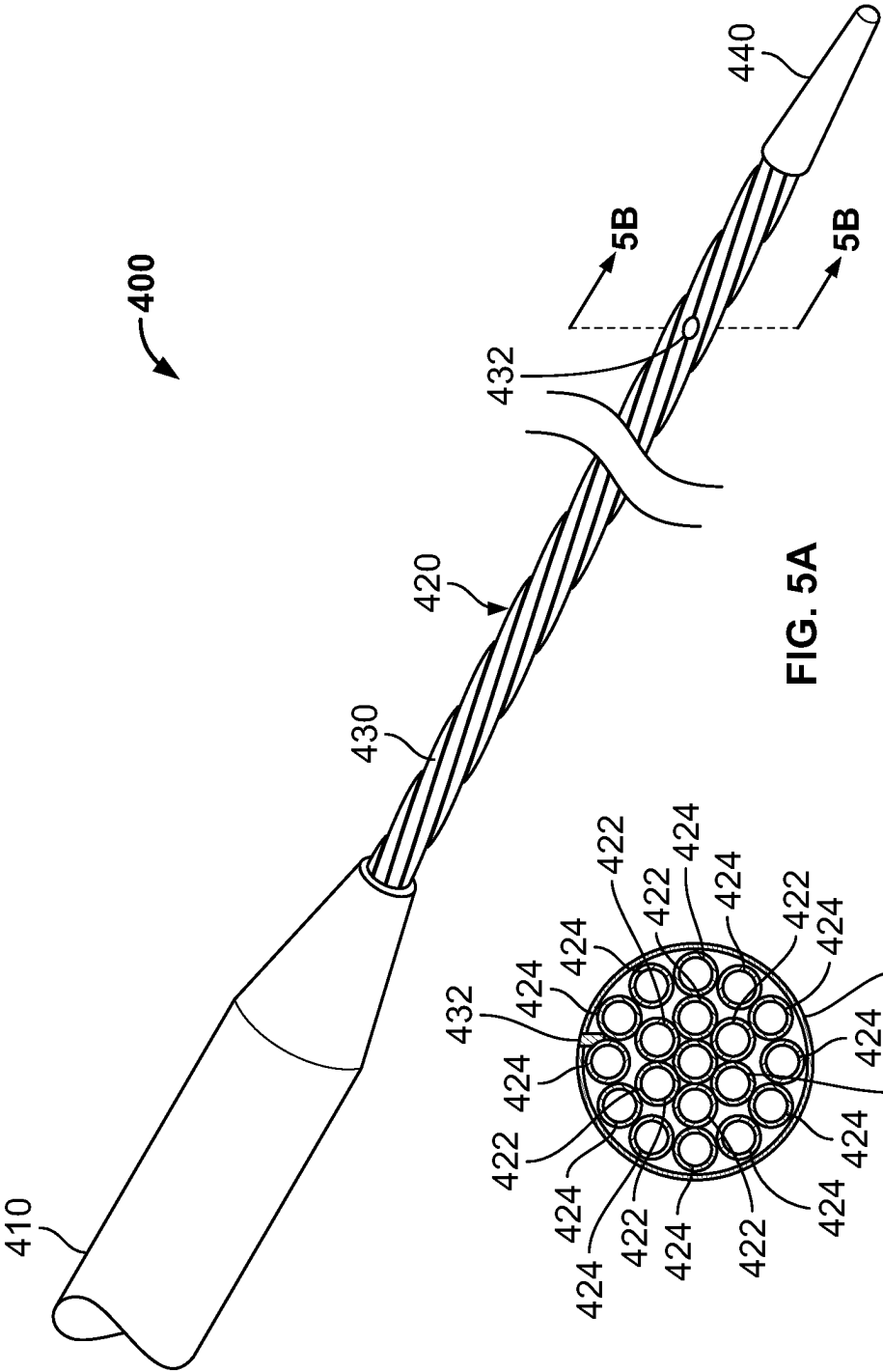


FIG. 5A

FIG. 5B

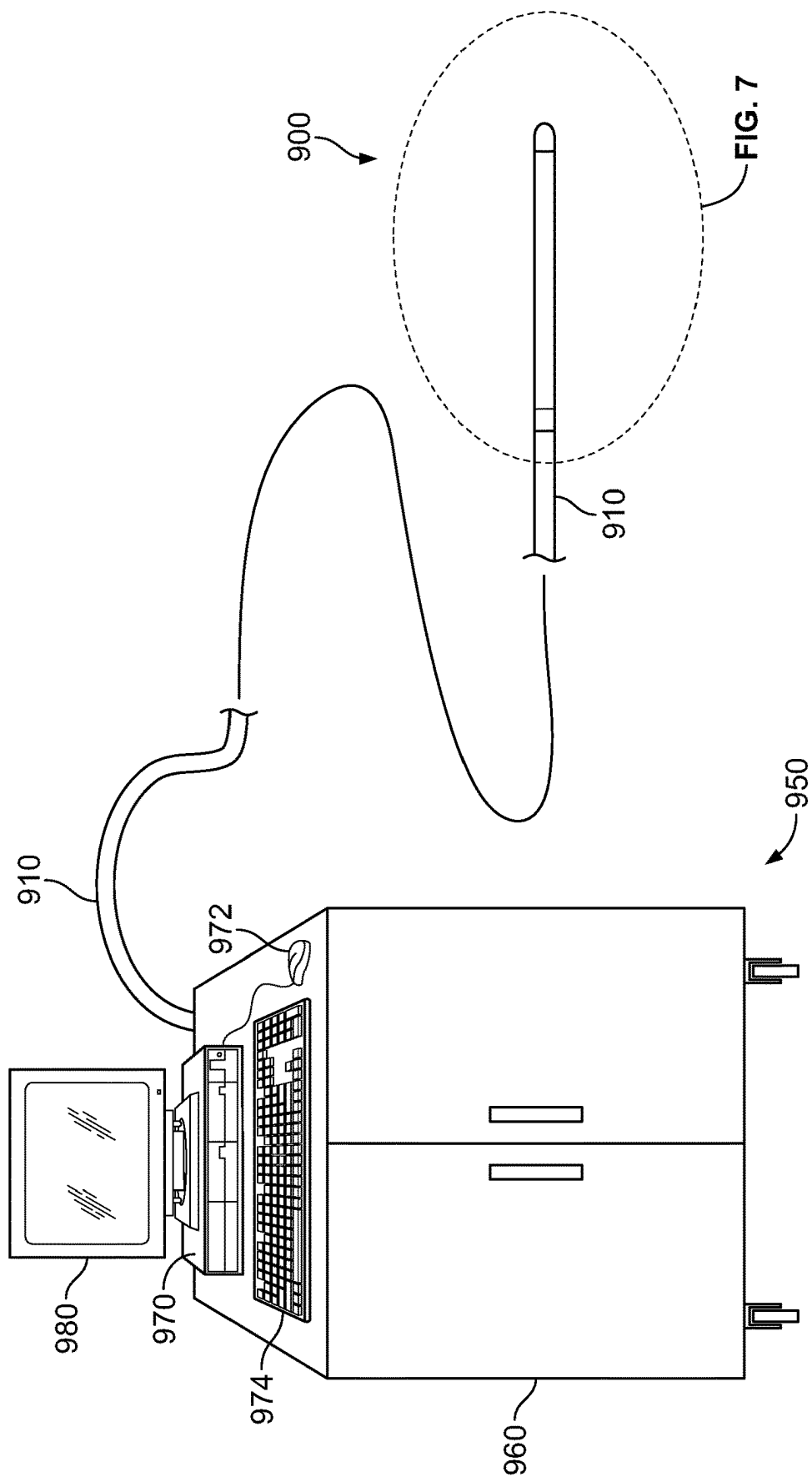


FIG. 6

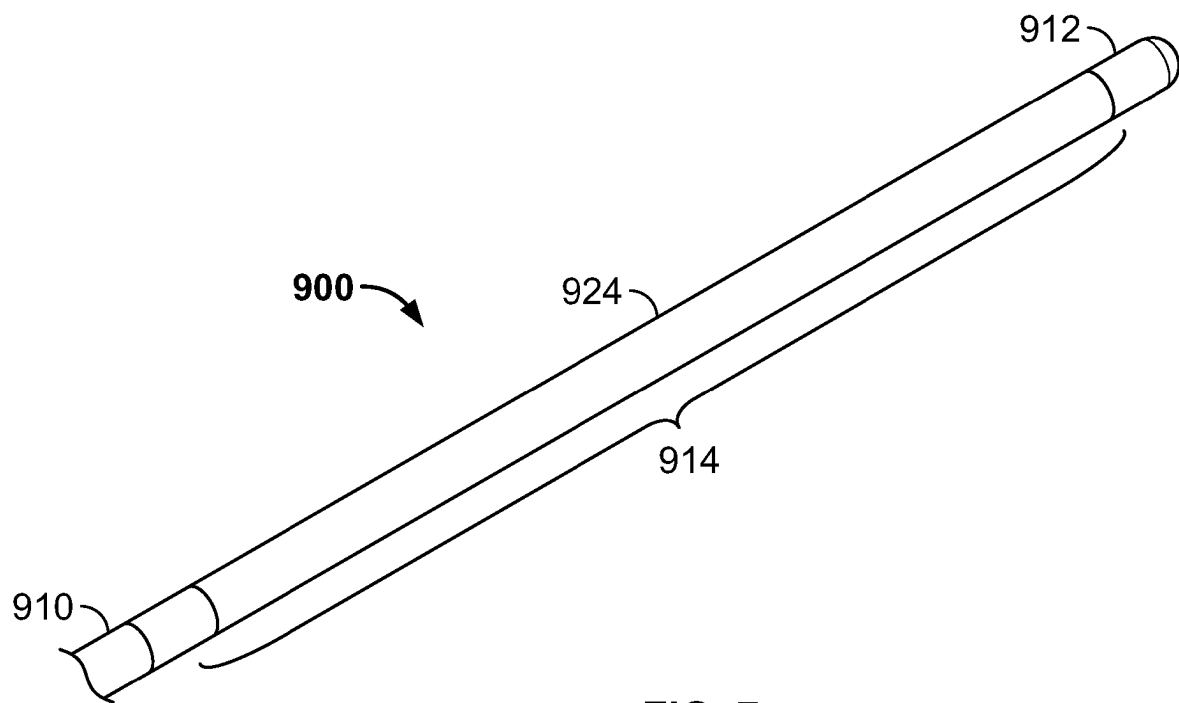


FIG. 7

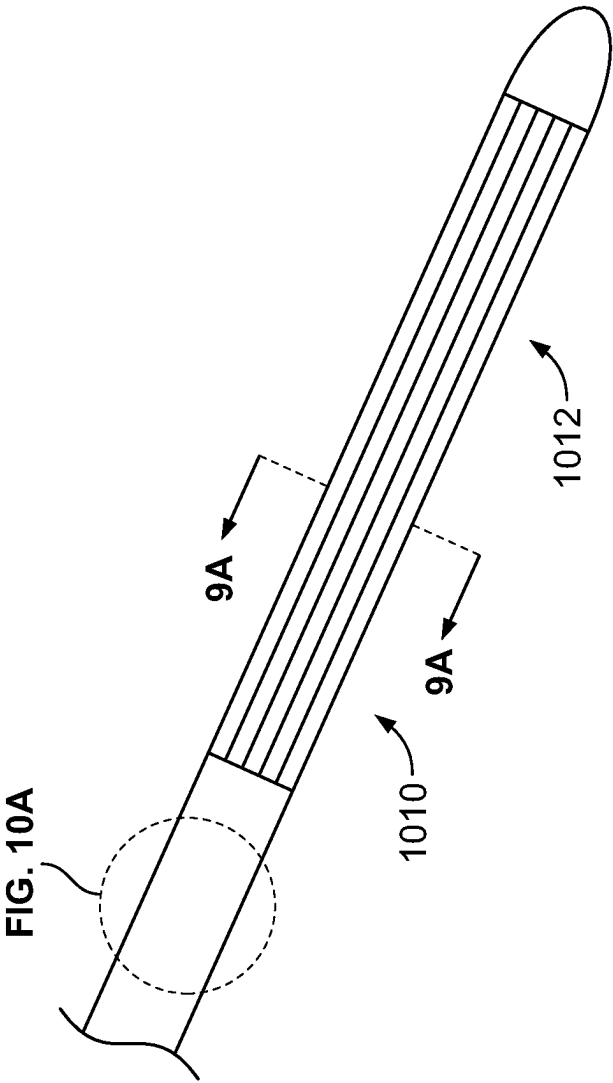


FIG. 8

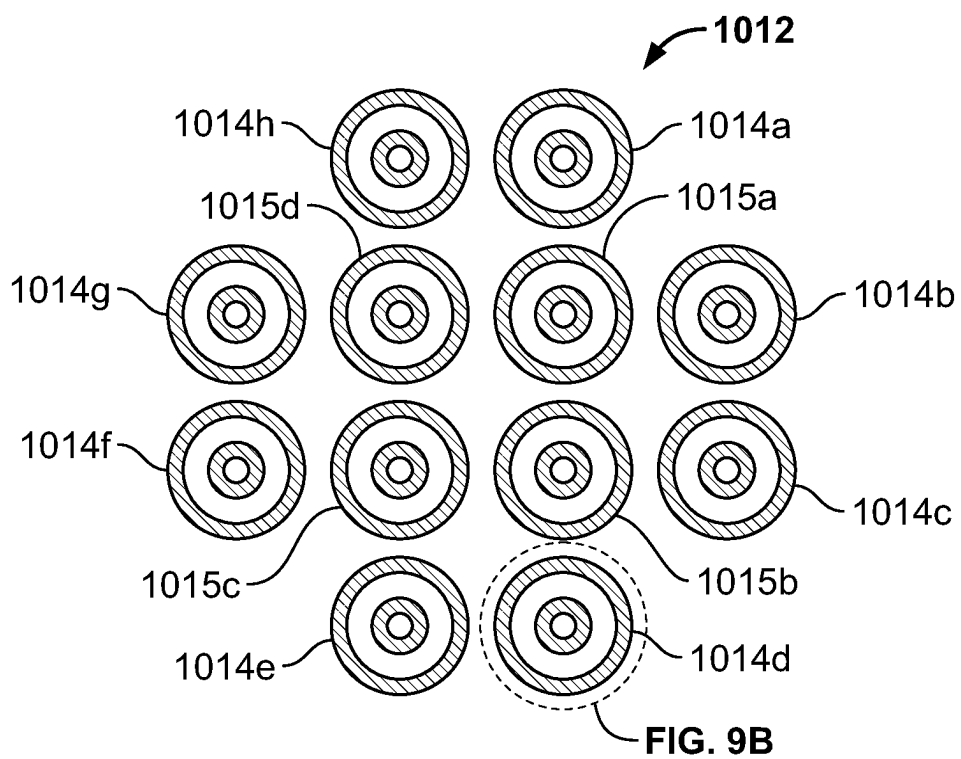


FIG. 9A

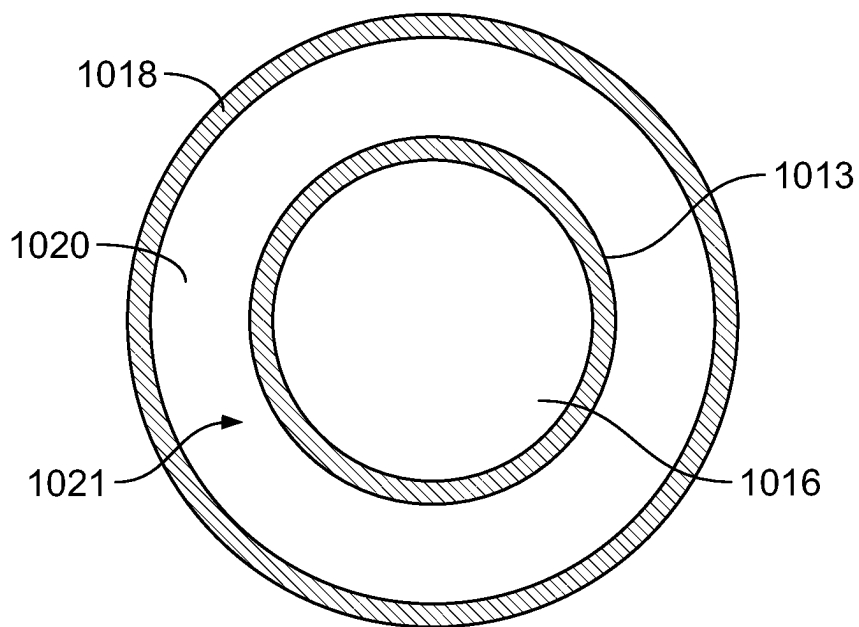


FIG. 9B

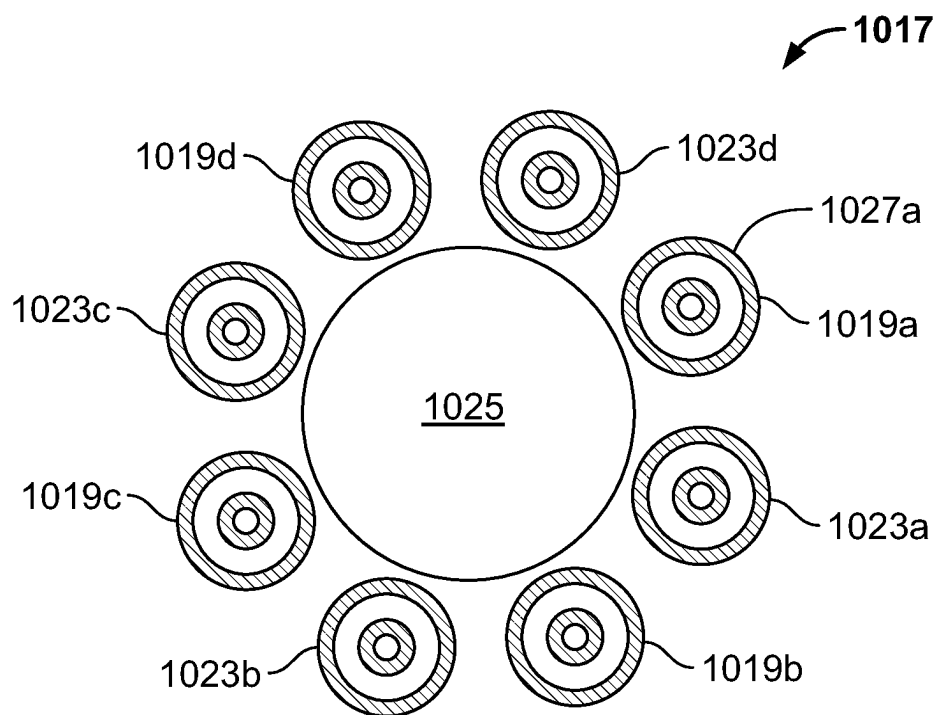


FIG. 9C

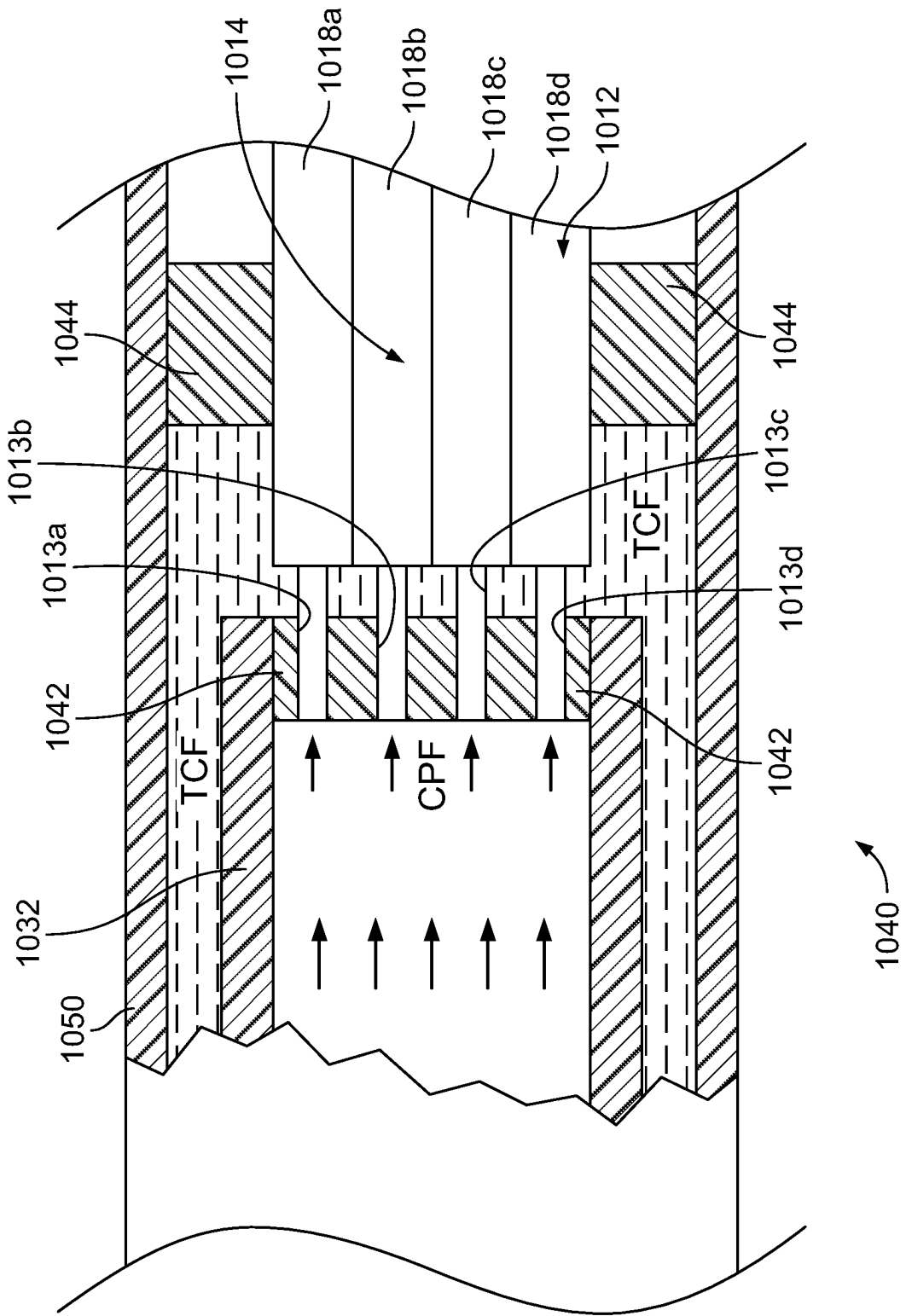


FIG. 10A

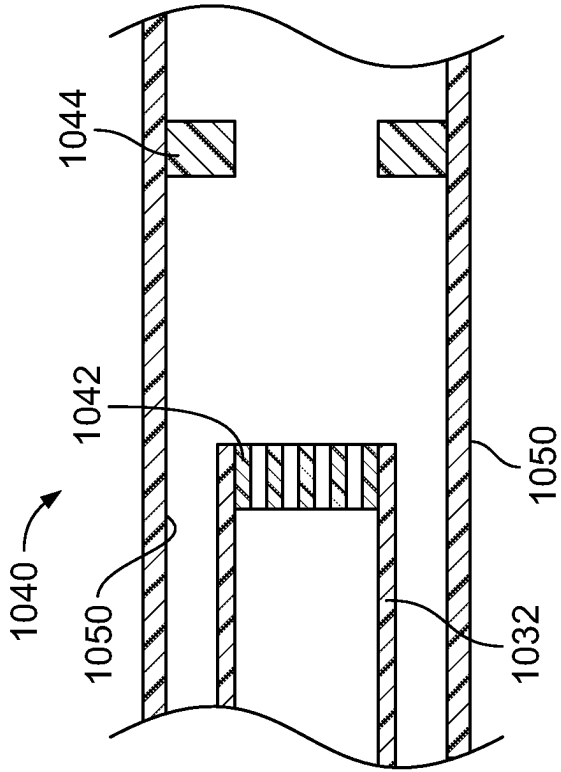
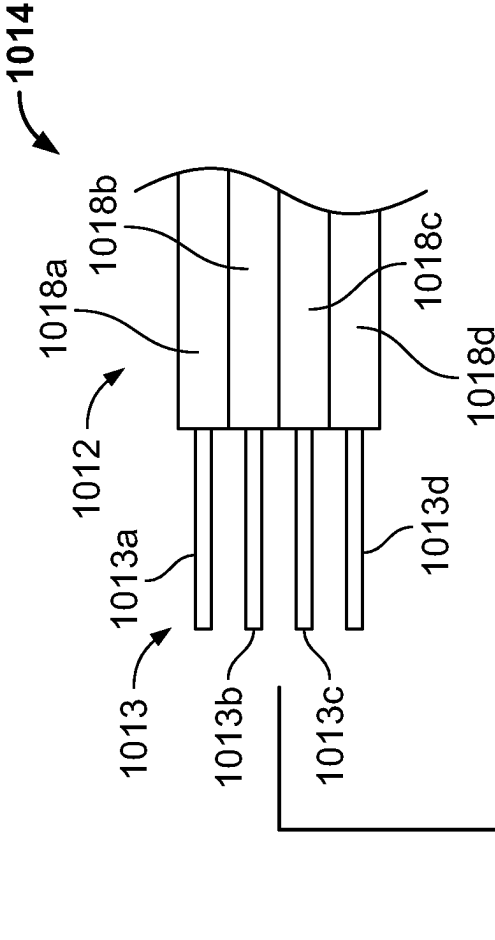
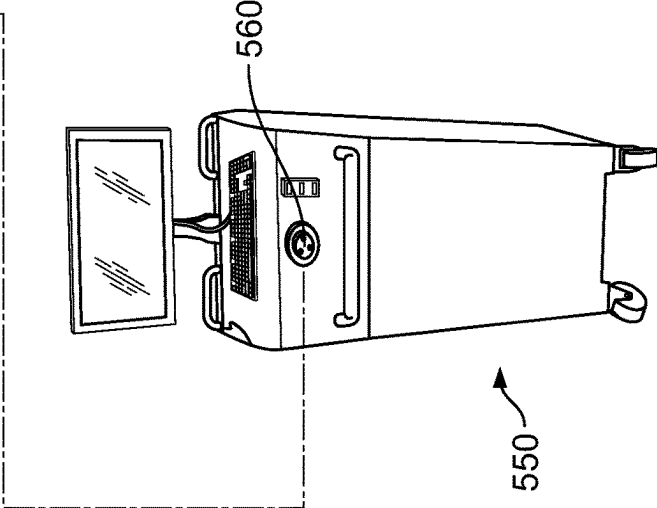
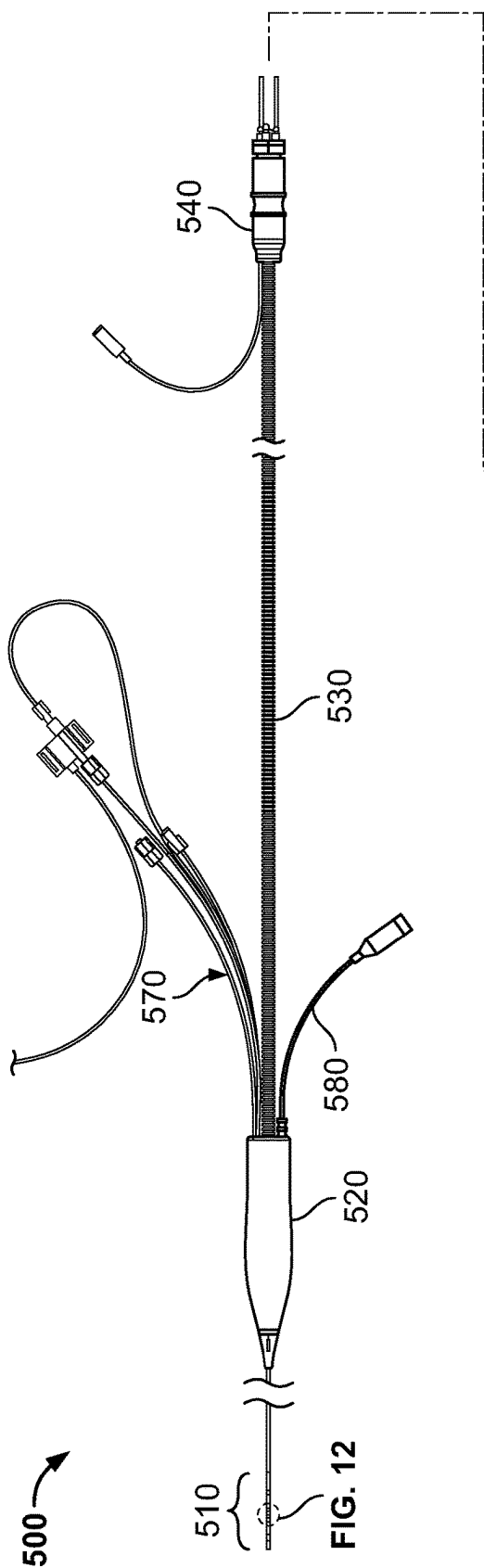


FIG. 10B



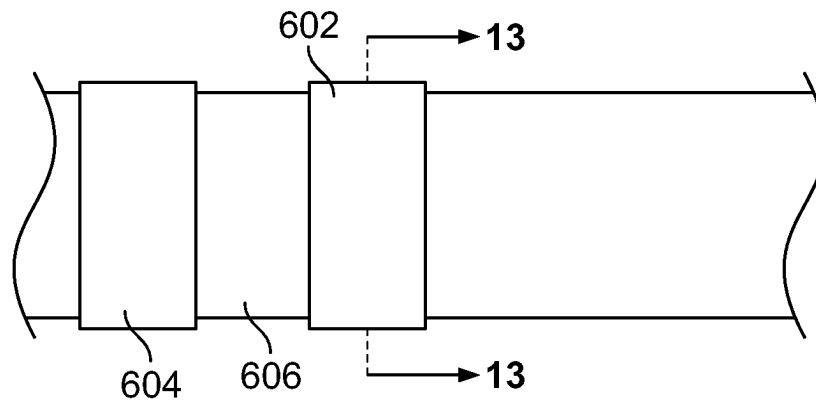


FIG. 12

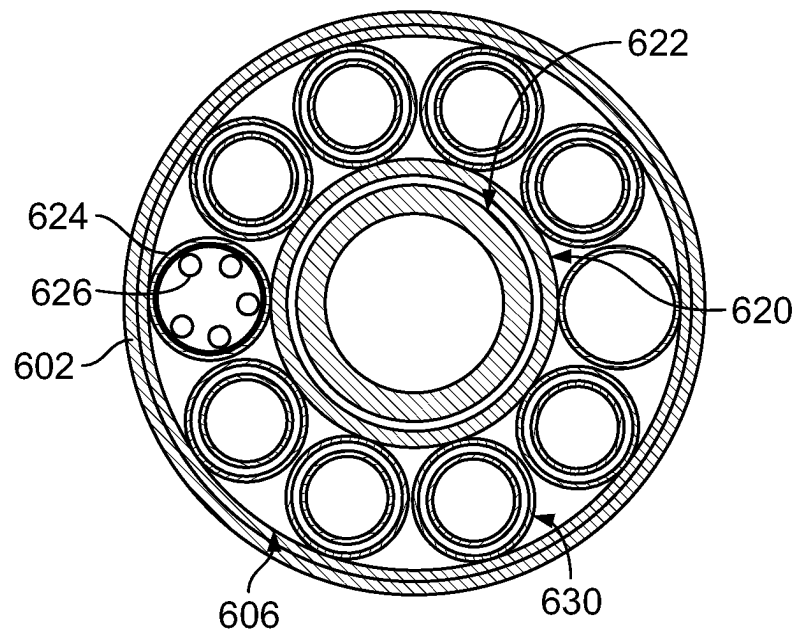


FIG. 13

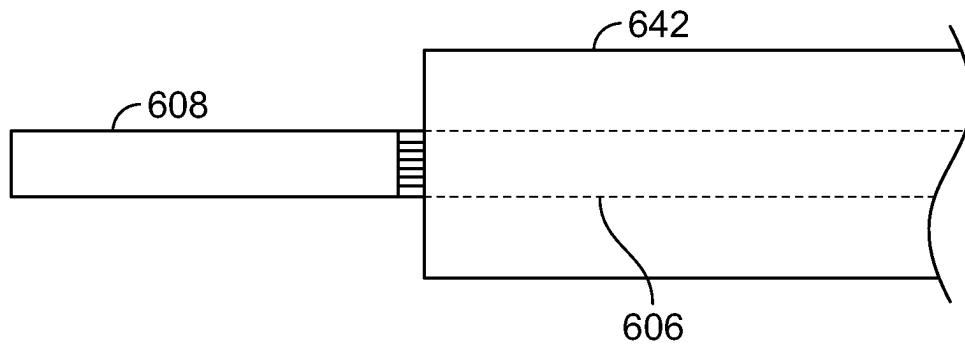


FIG. 14

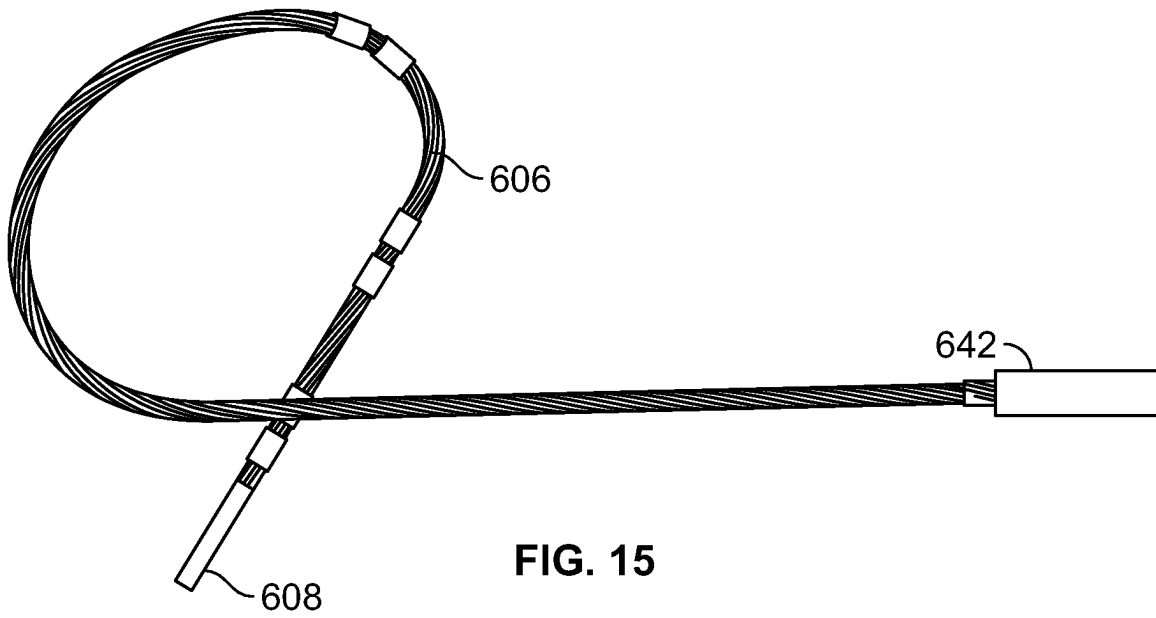


FIG. 15

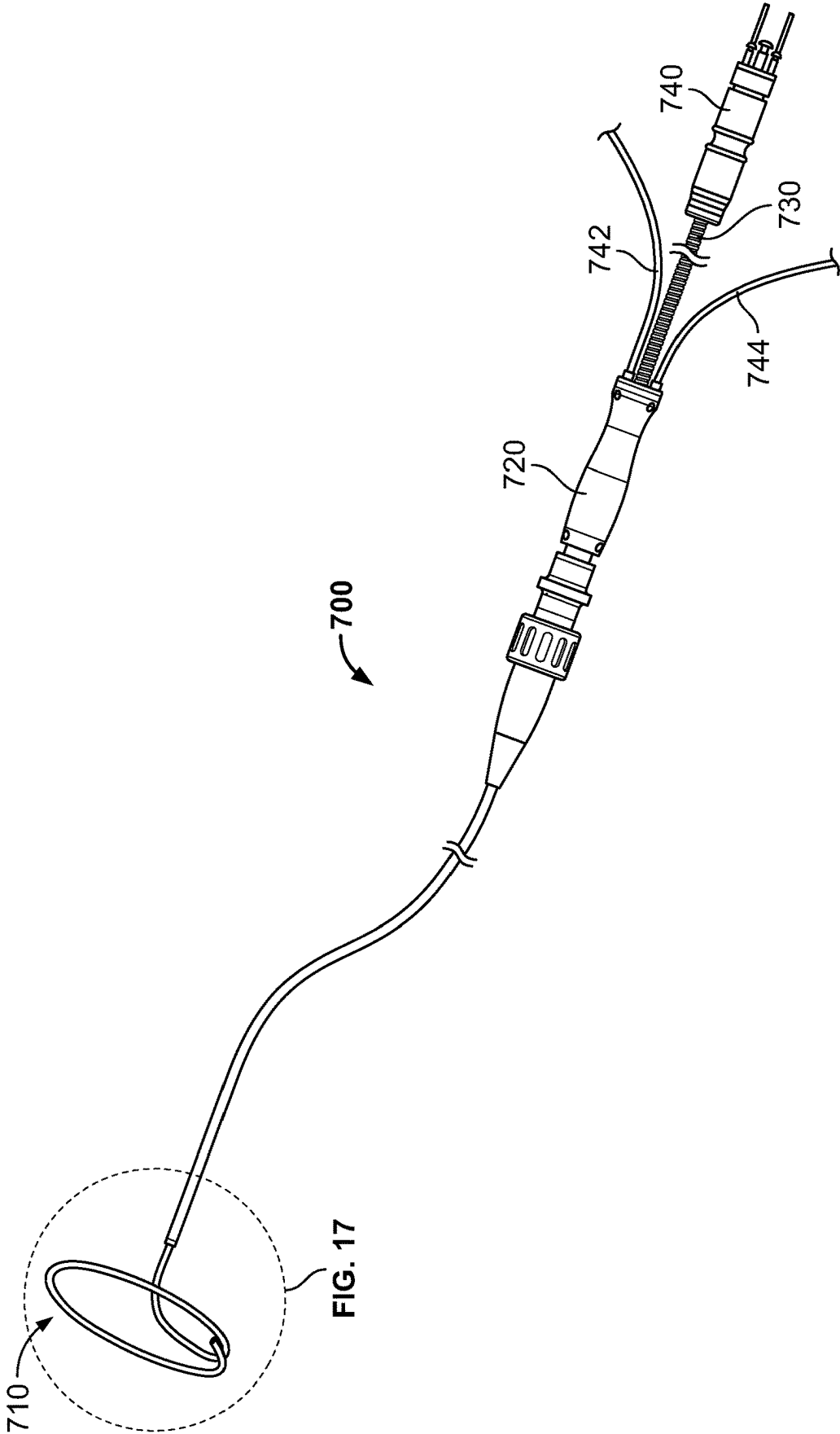


FIG. 16

FIG. 17

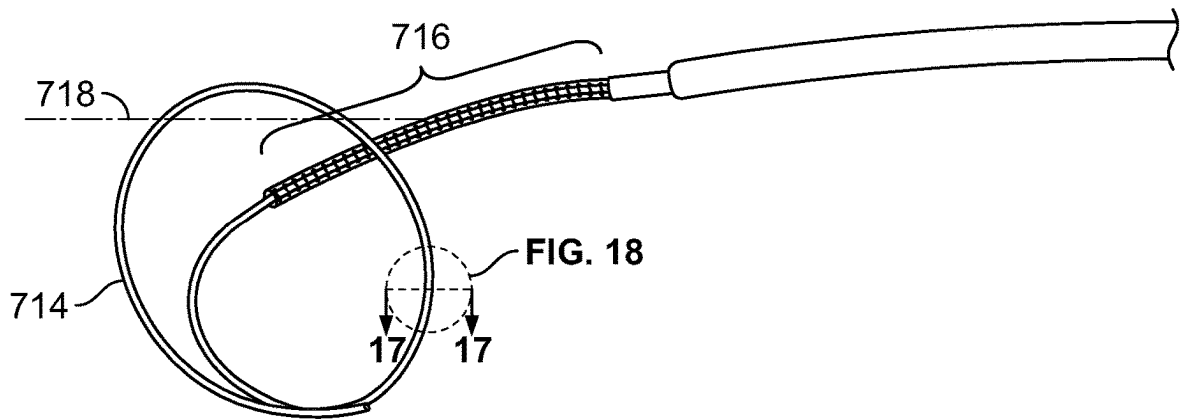


FIG. 17

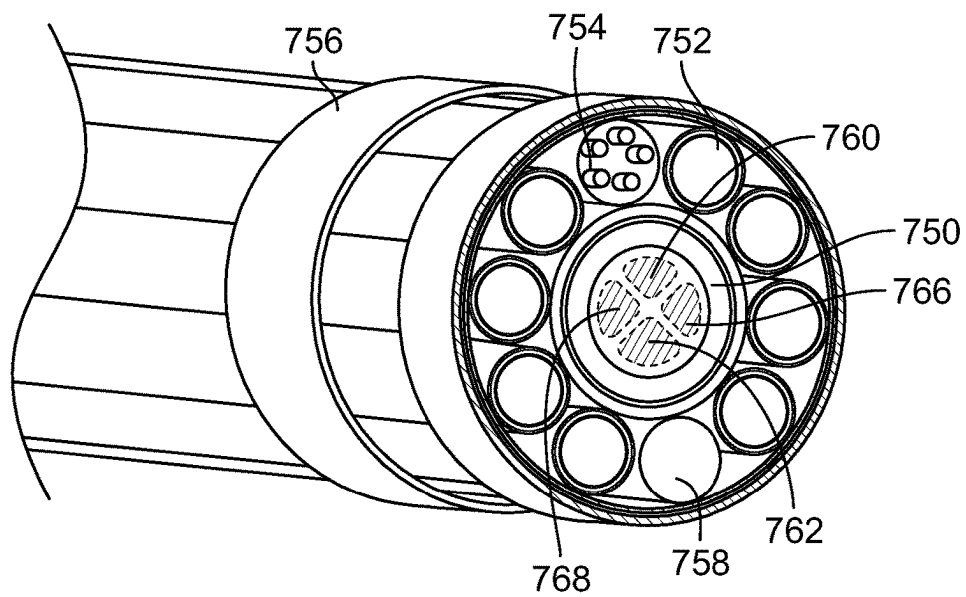


FIG. 18

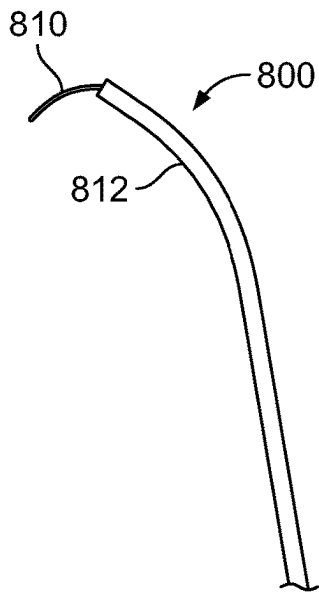


FIG. 19A

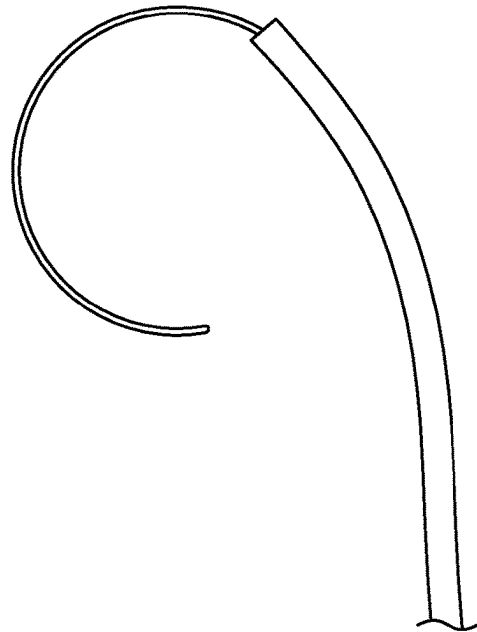


FIG. 19B

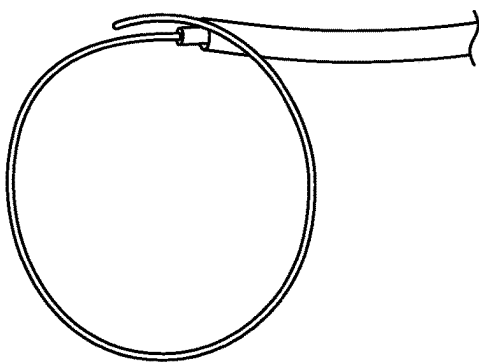


FIG. 19C

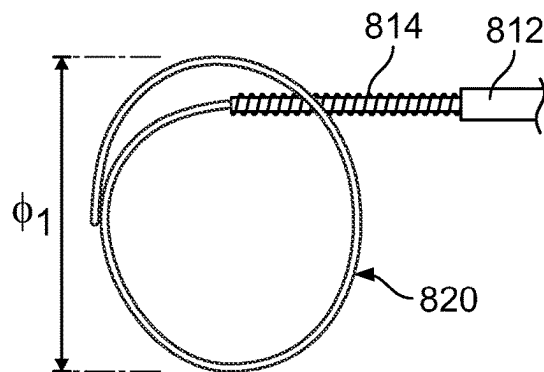


FIG. 19D

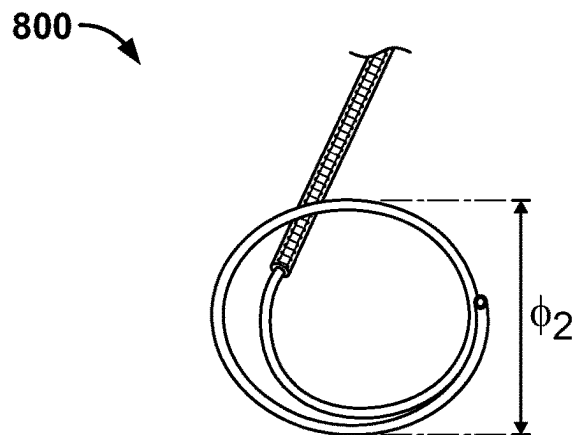


FIG. 20A

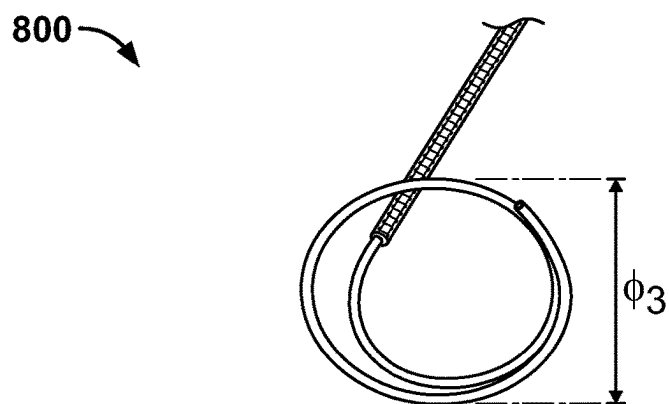


FIG. 20B

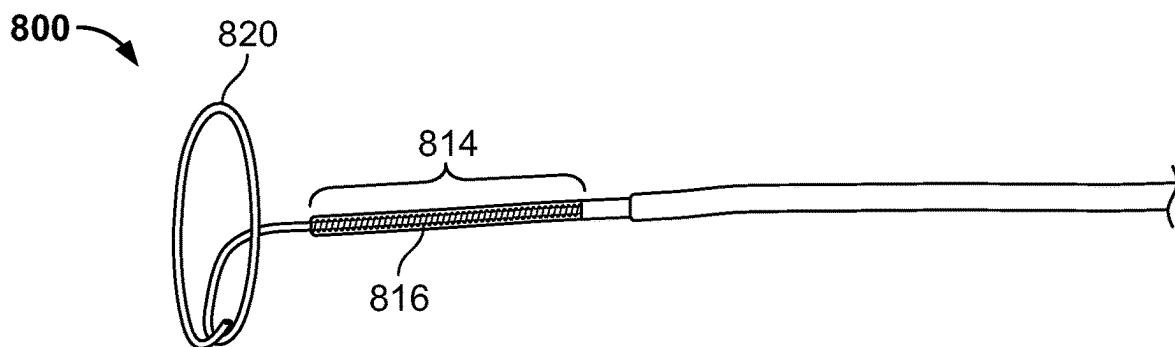


FIG. 21A

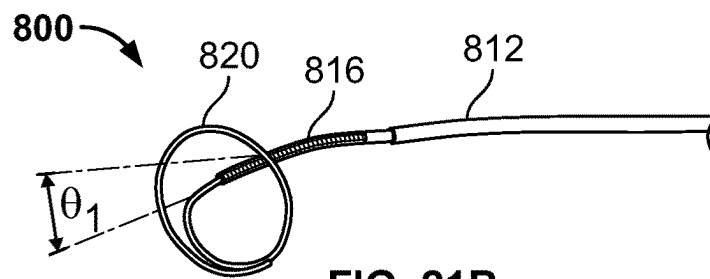


FIG. 21B

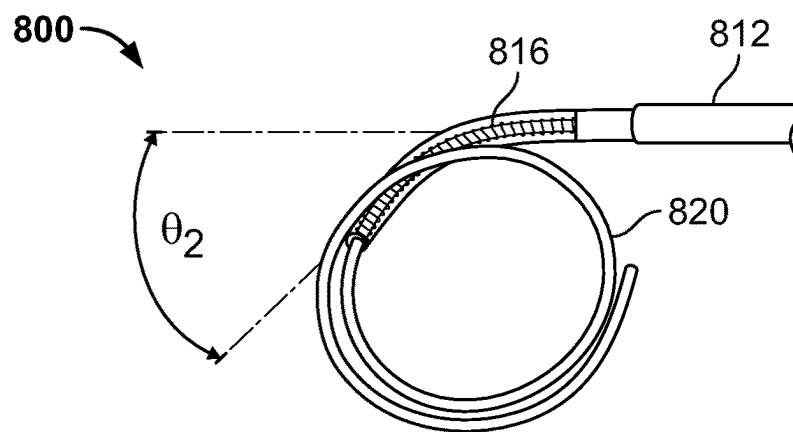


FIG. 21C

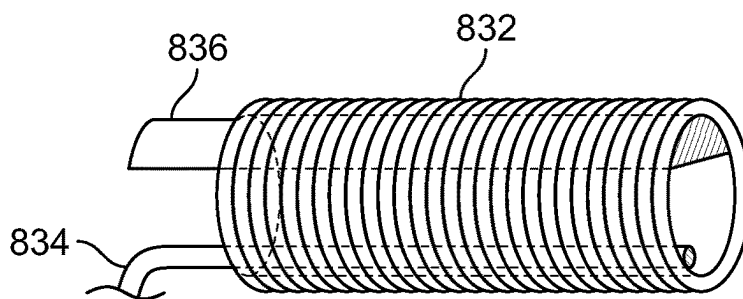


FIG. 22A

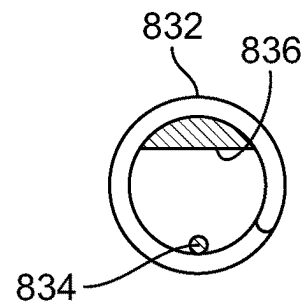


FIG. 22B

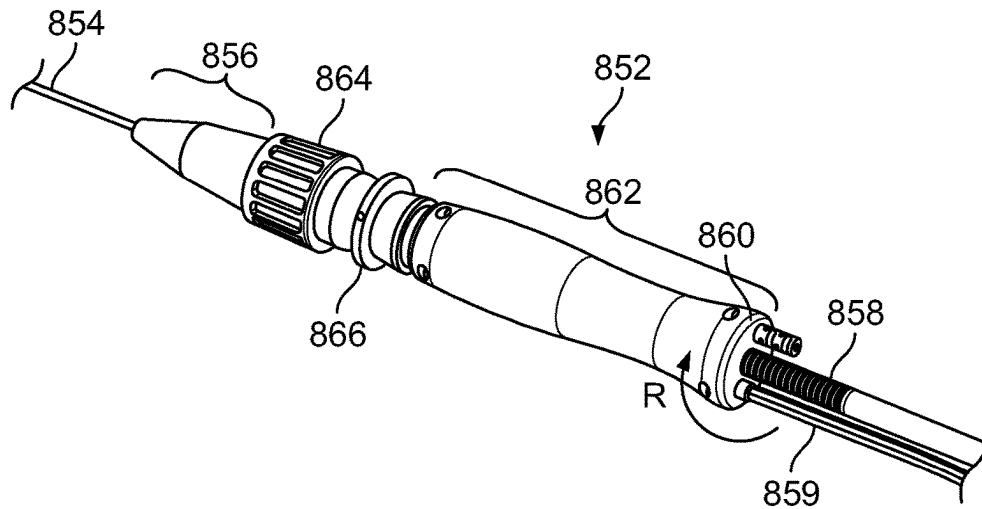


FIG. 23A

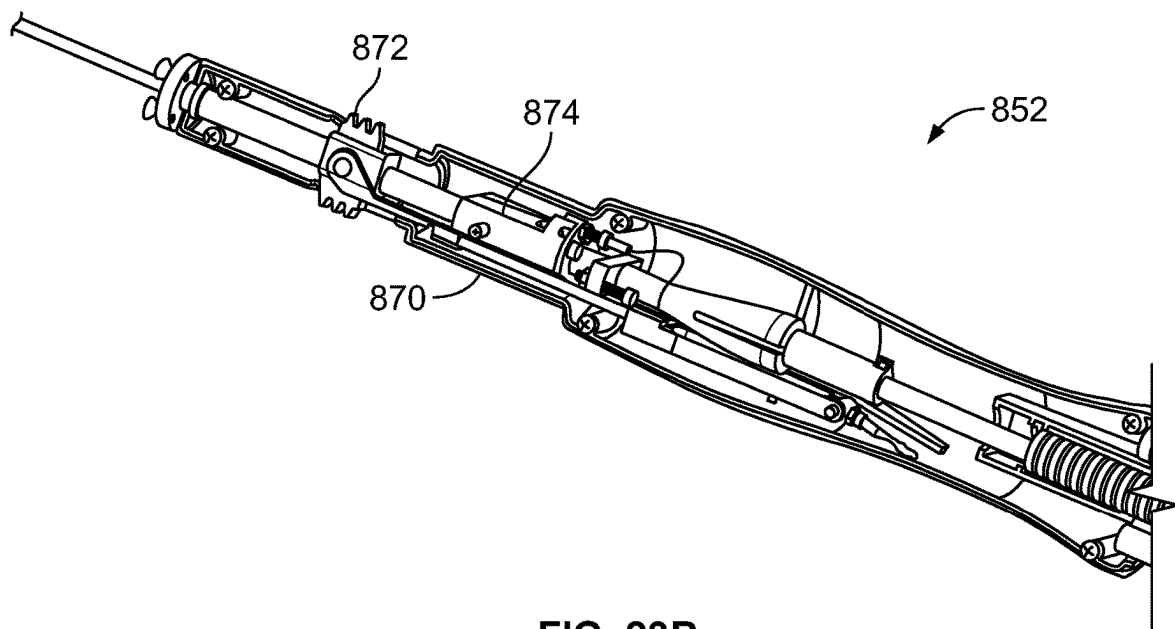


FIG. 23B

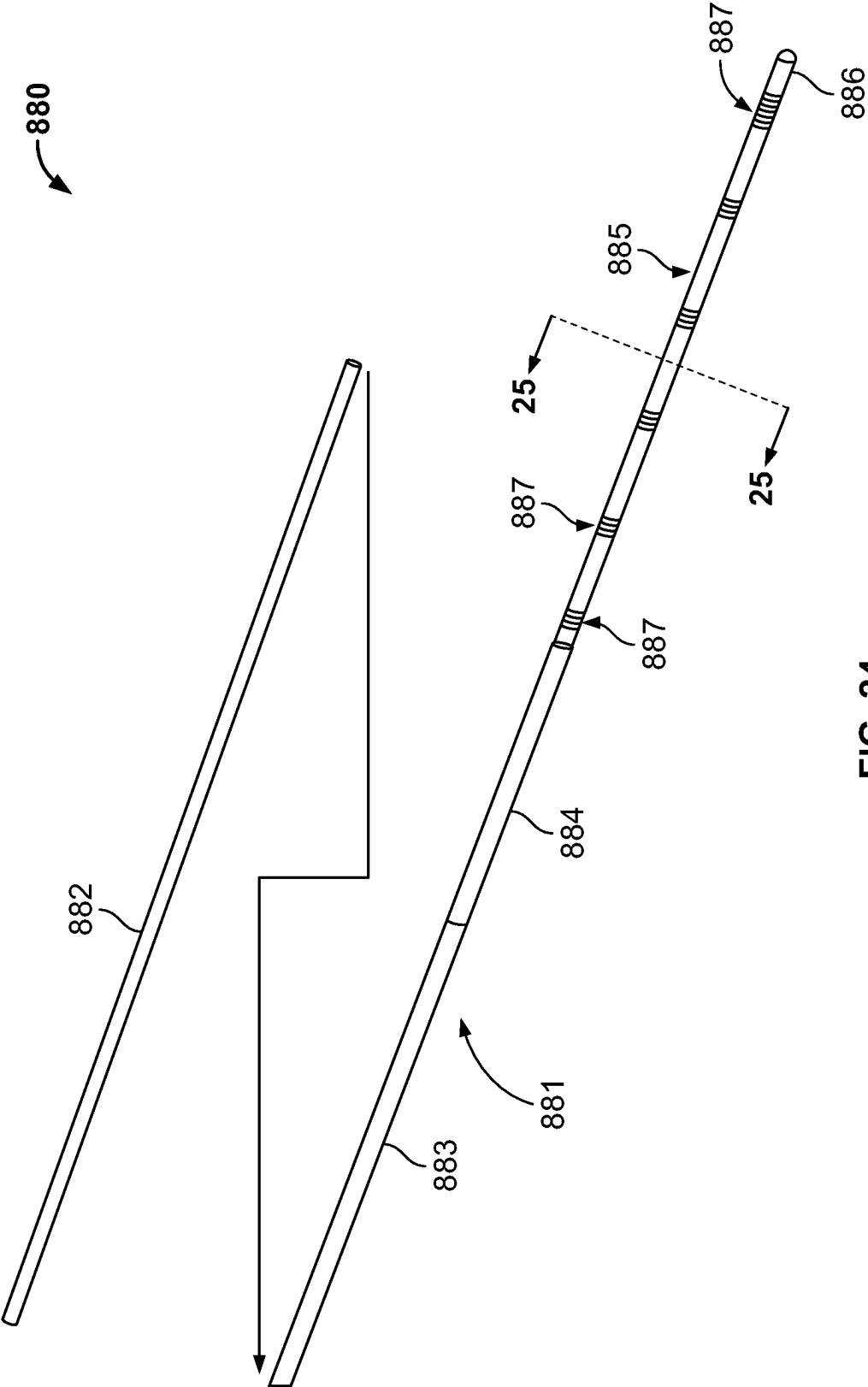


FIG. 24

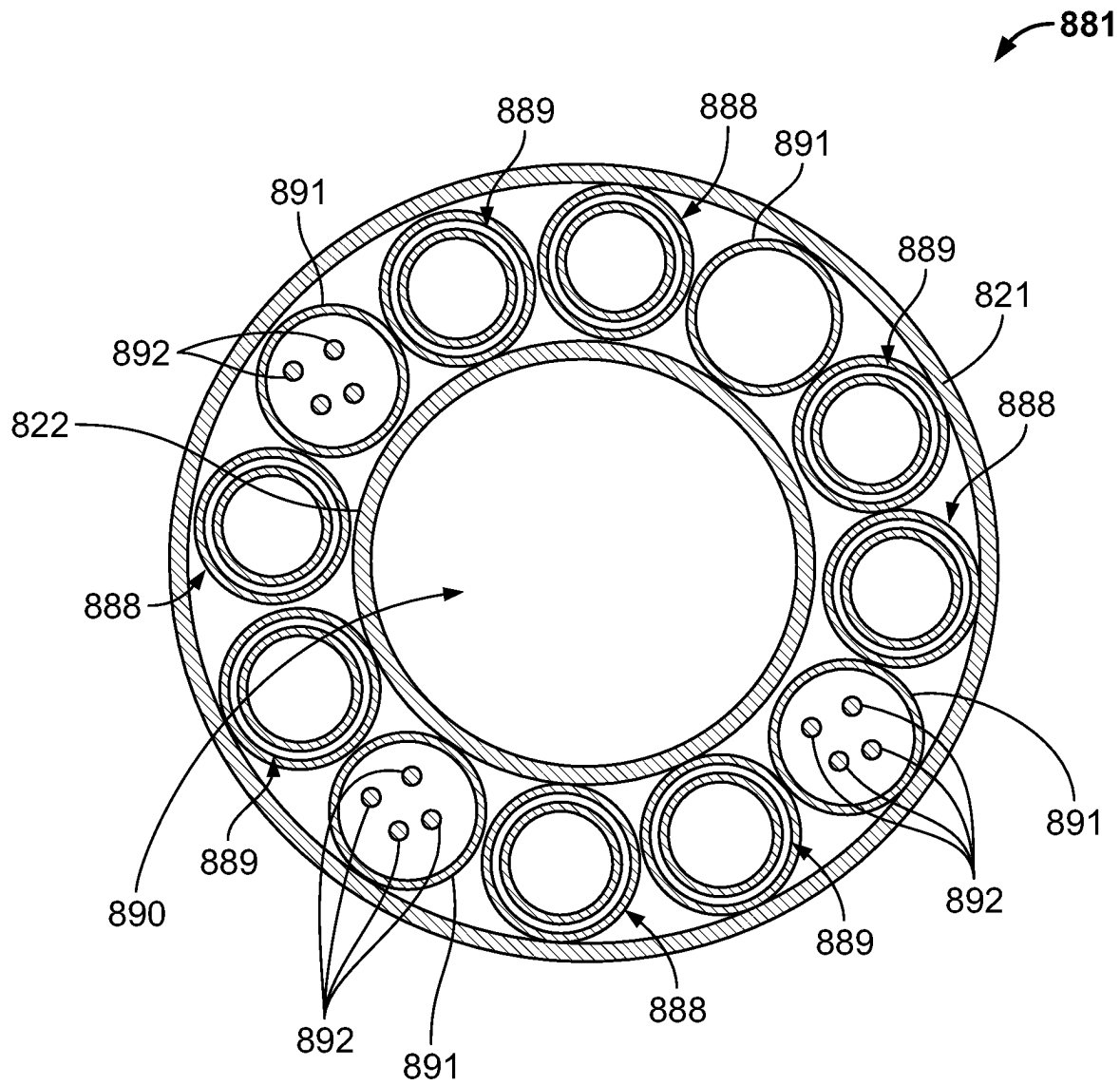


FIG. 25A

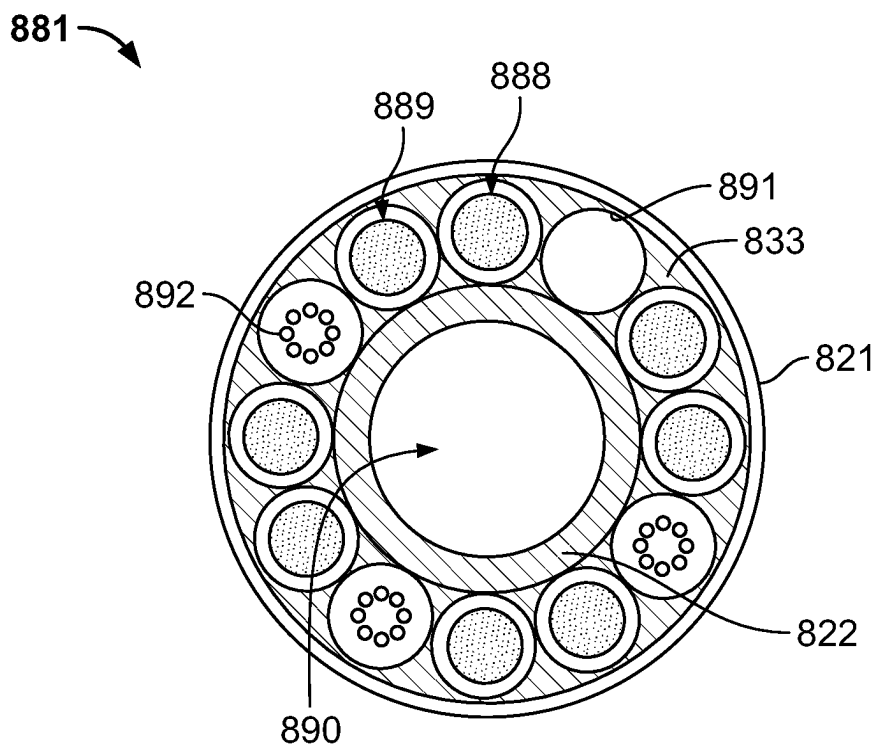


FIG. 25B

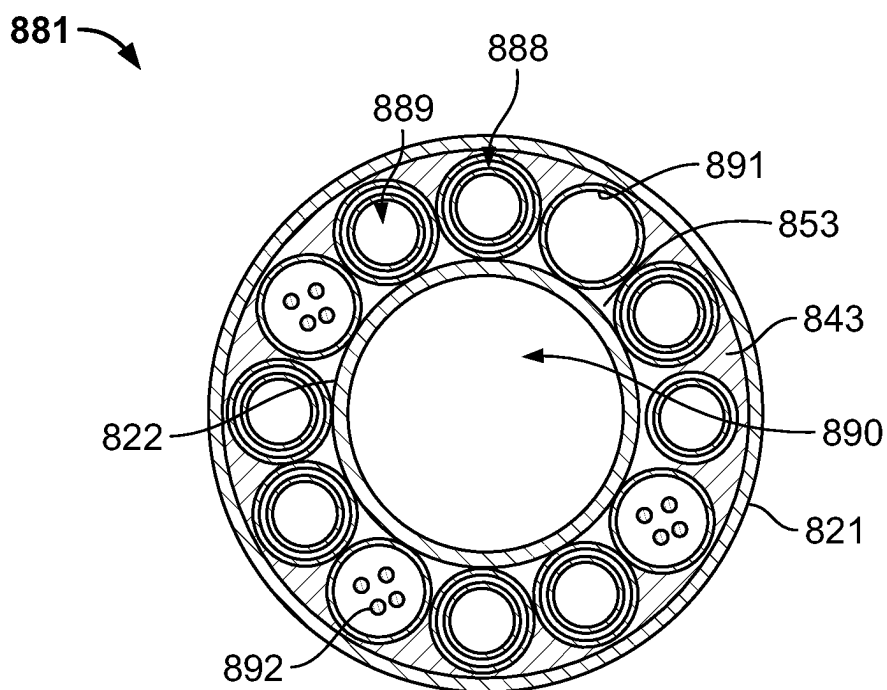


FIG. 25C

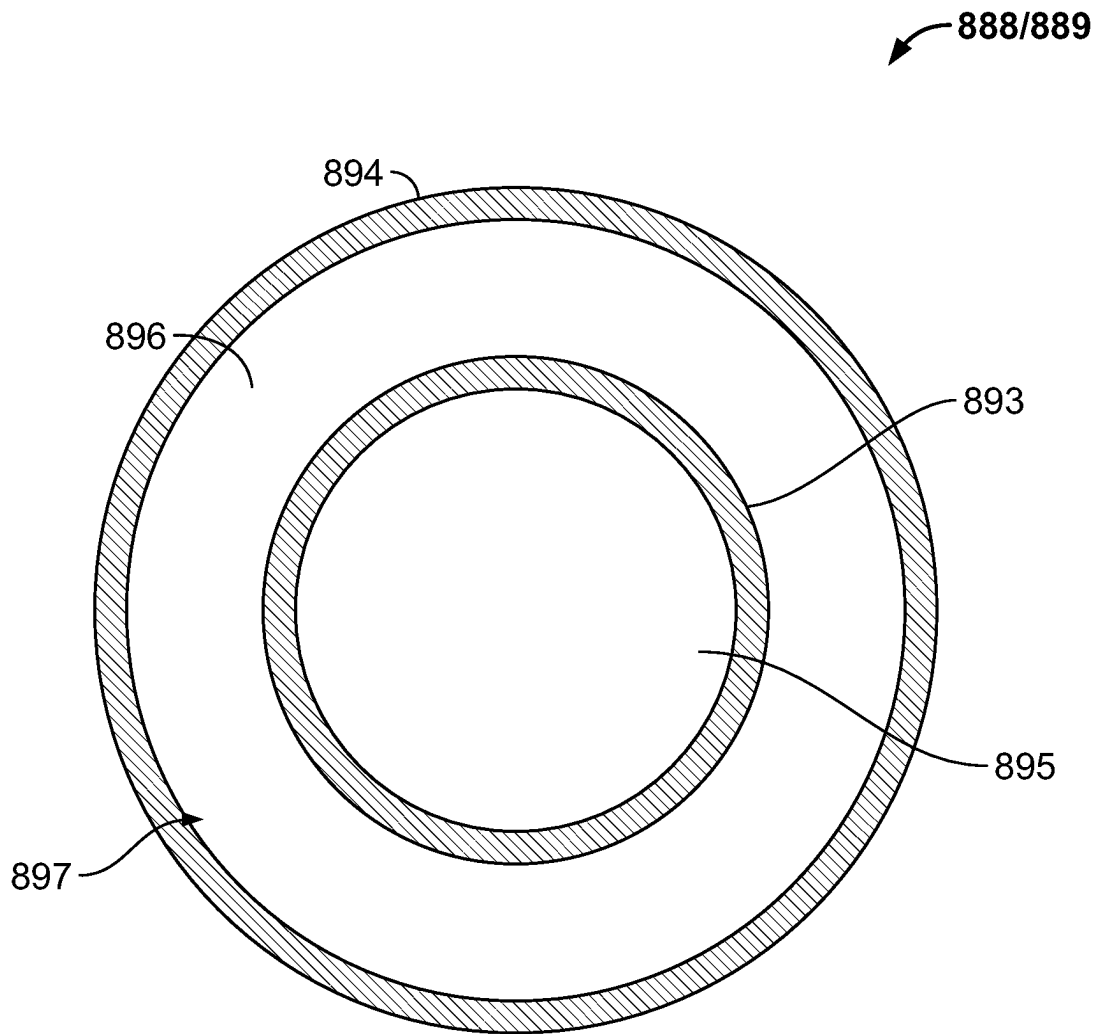


FIG. 26

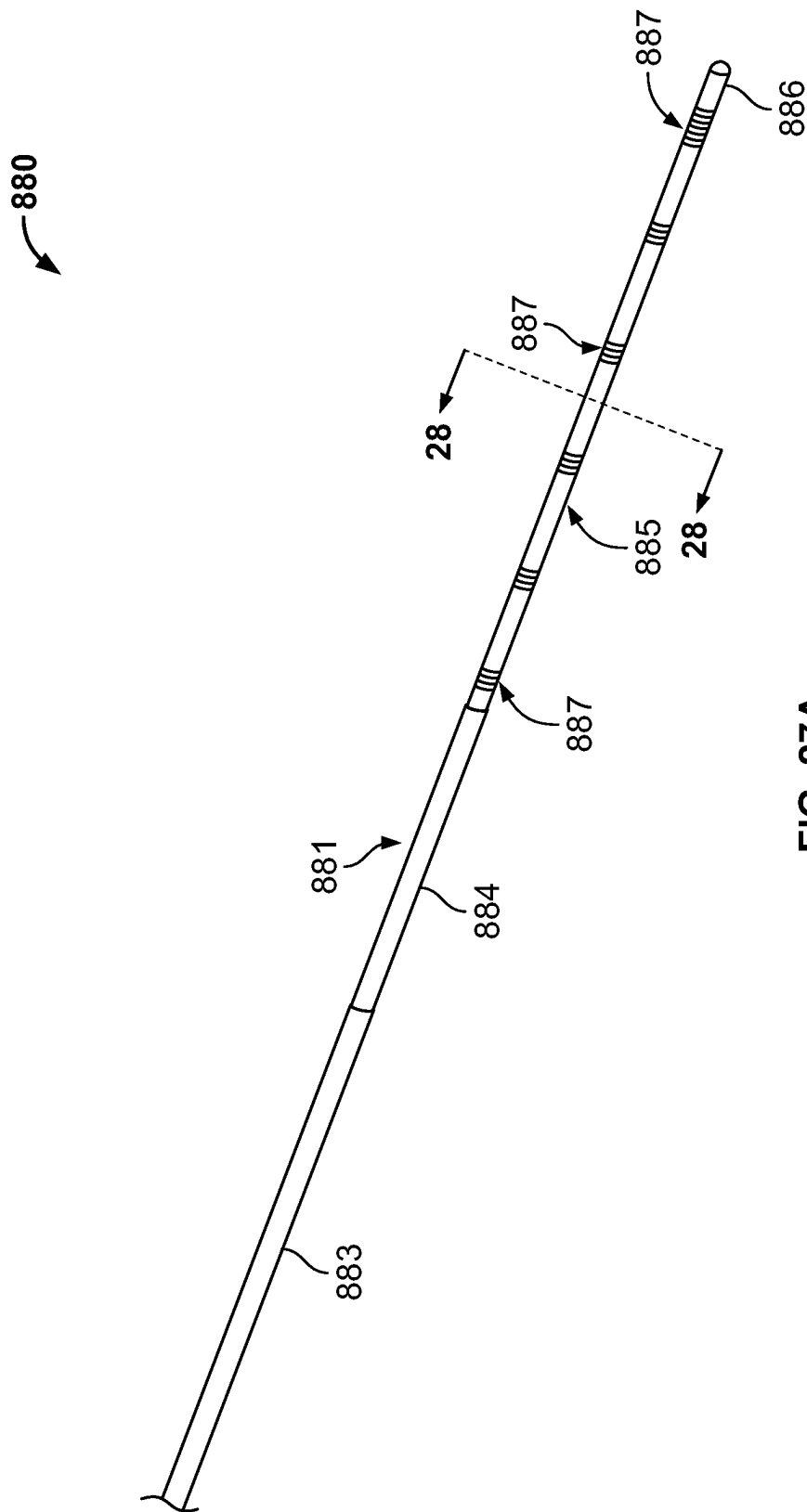


FIG. 27A

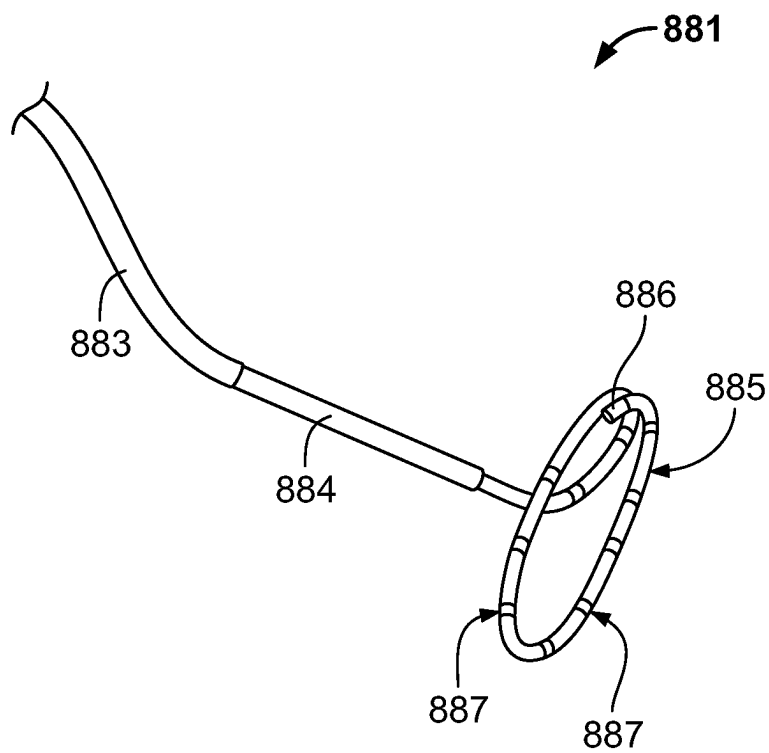


FIG. 27B

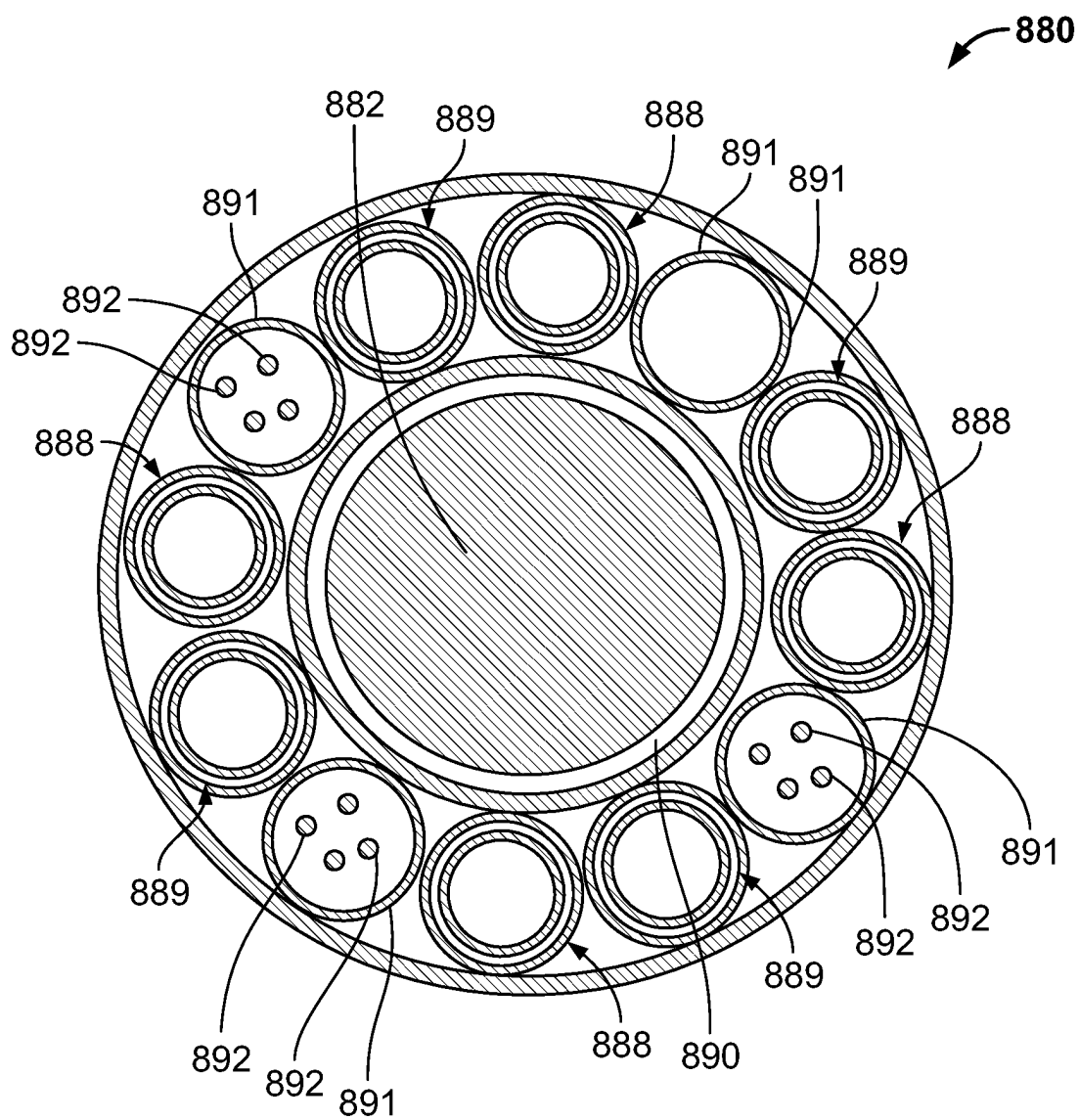


FIG. 28A

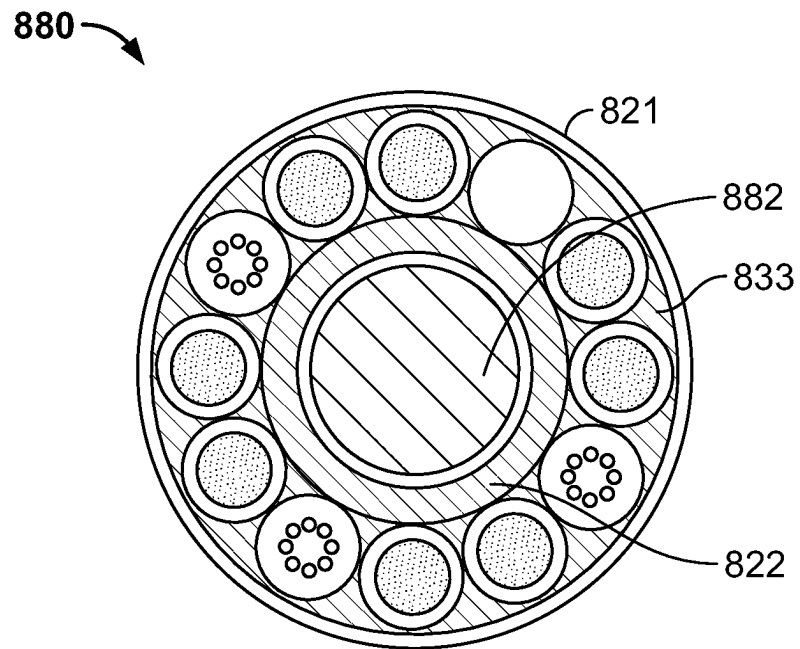


FIG. 28B

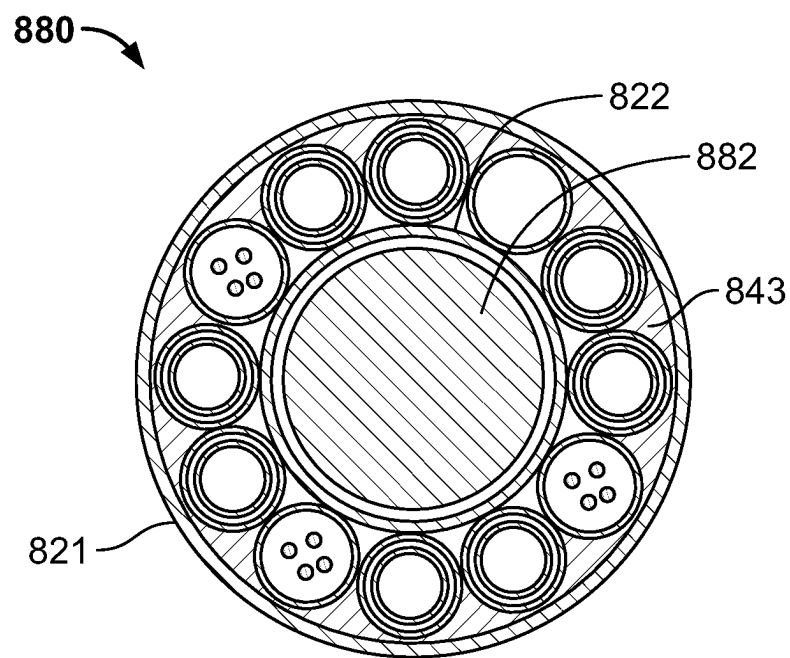


FIG. 28C

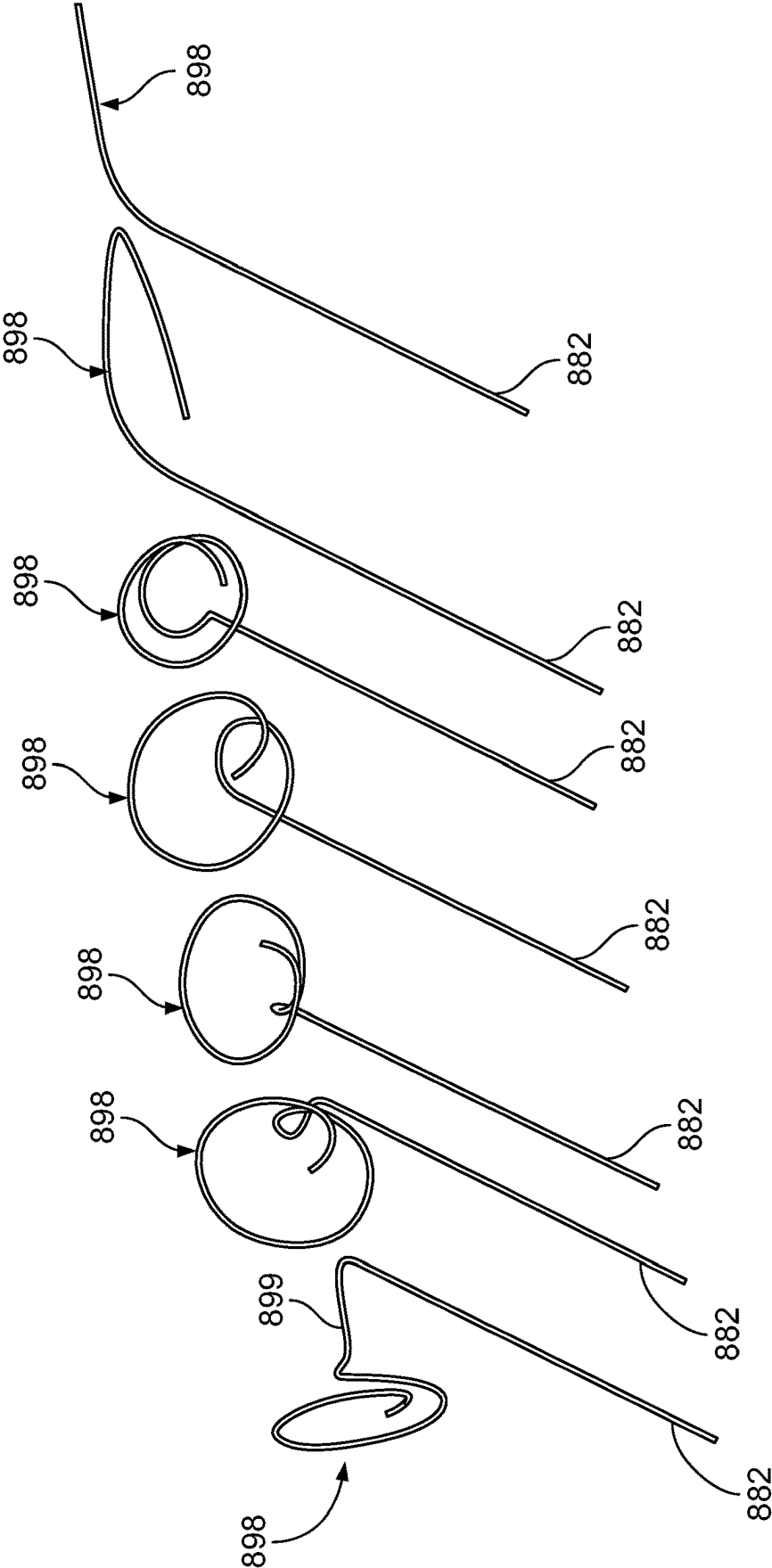


FIG. 29

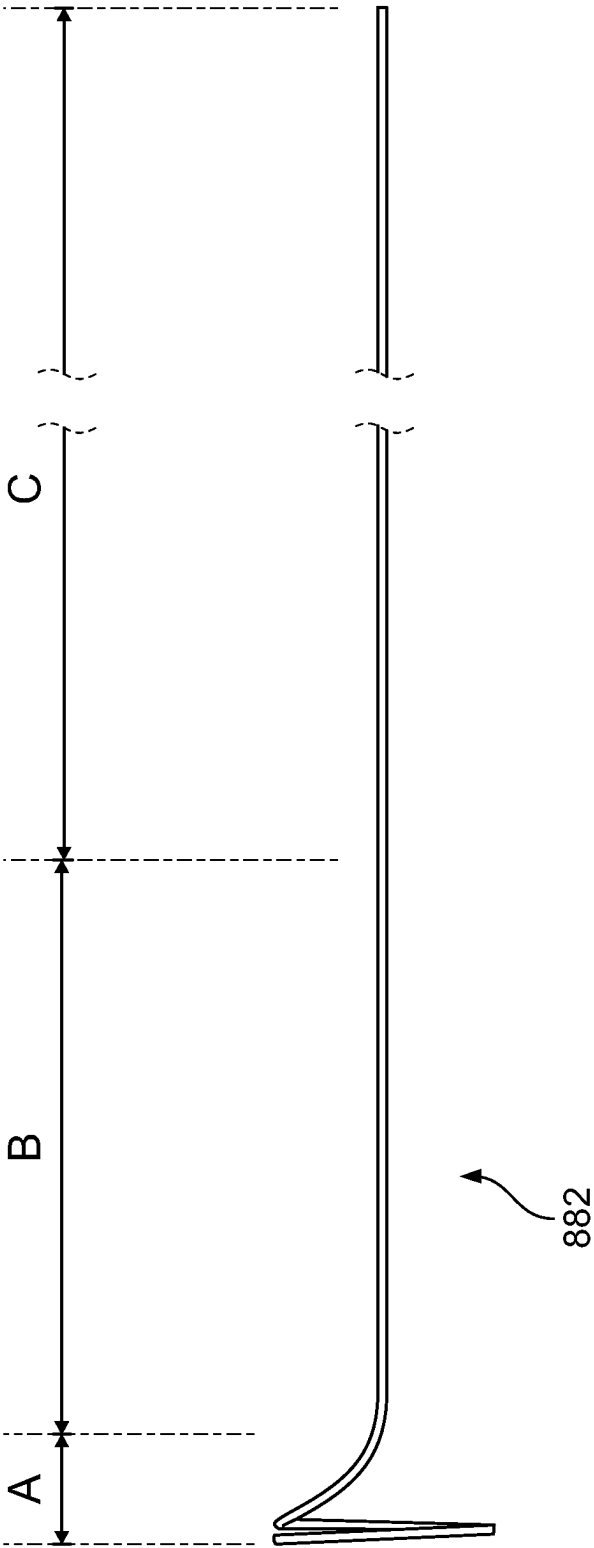


FIG. 30

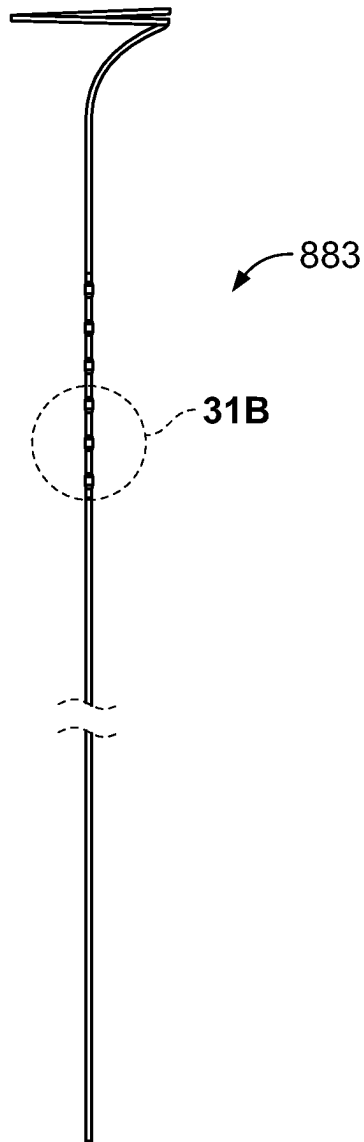


FIG. 31A

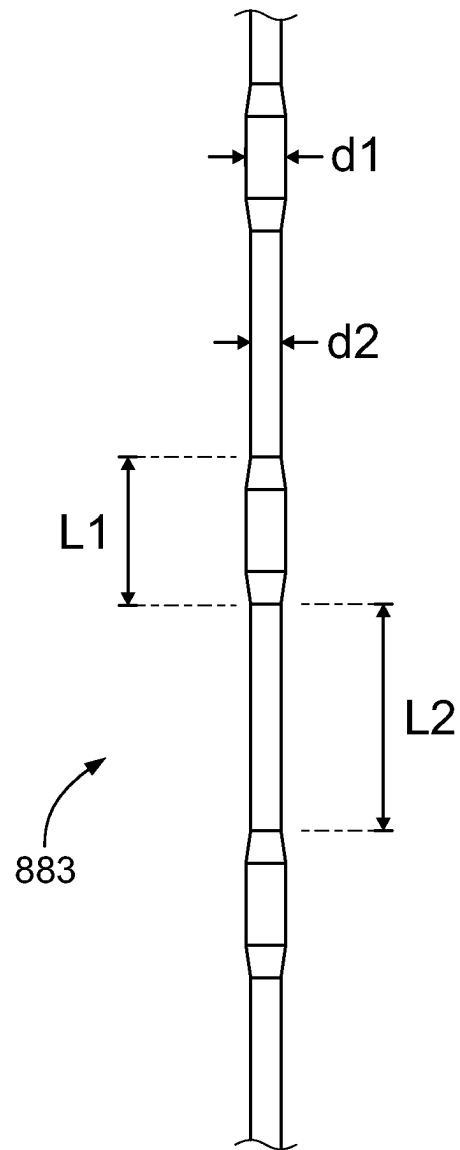


FIG. 31B

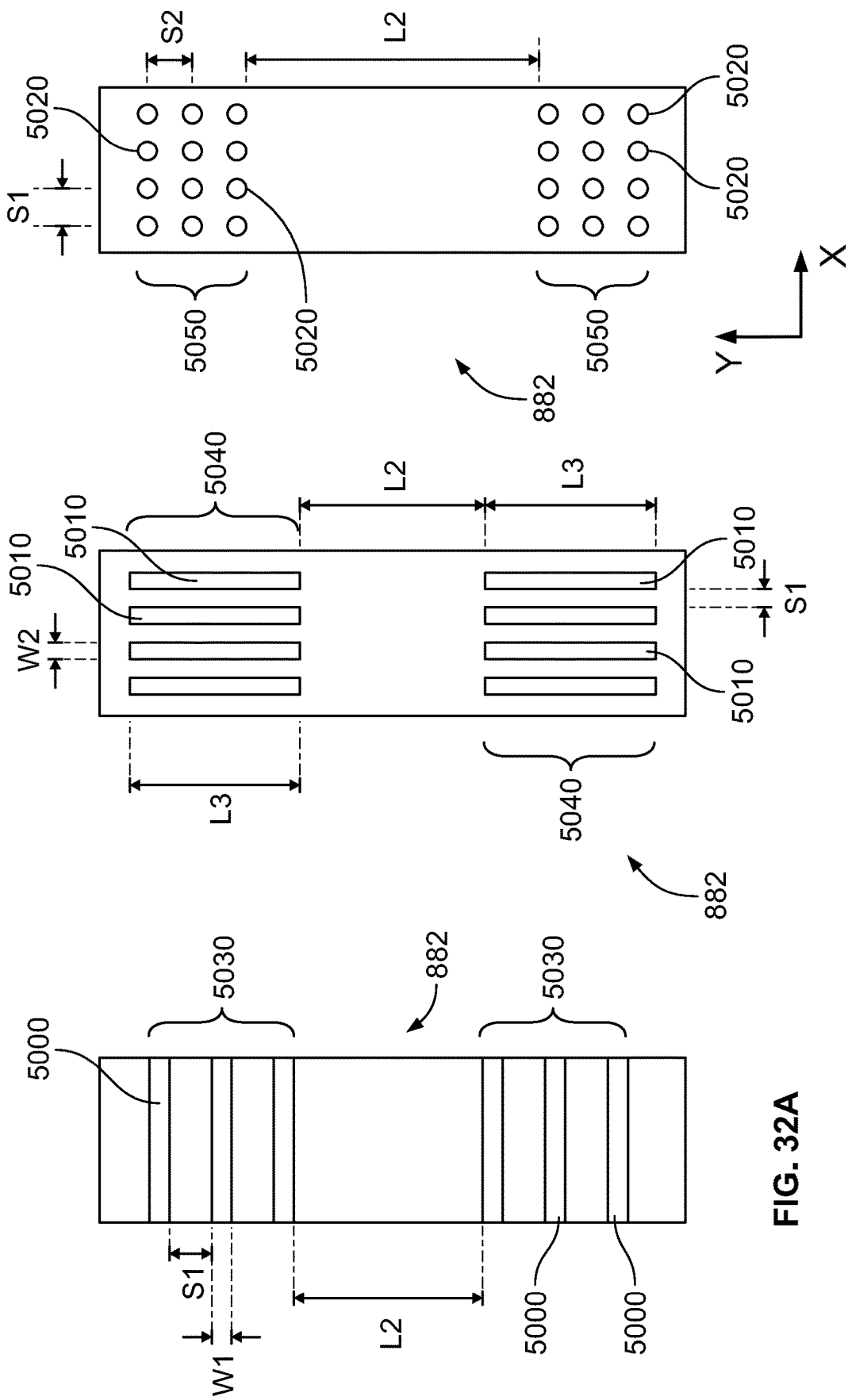


FIG. 32A

FIG. 32B

FIG. 32C

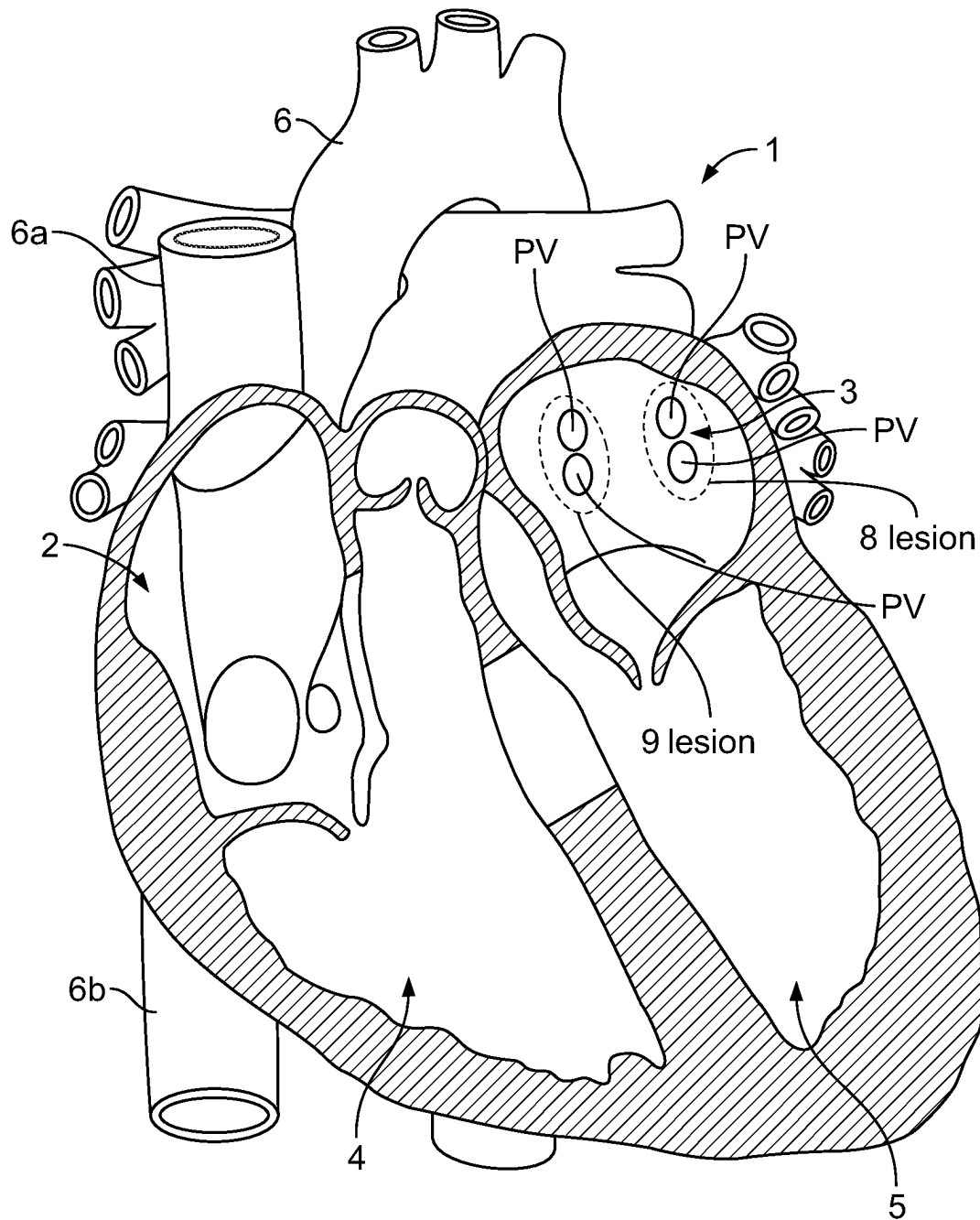


FIG. 33

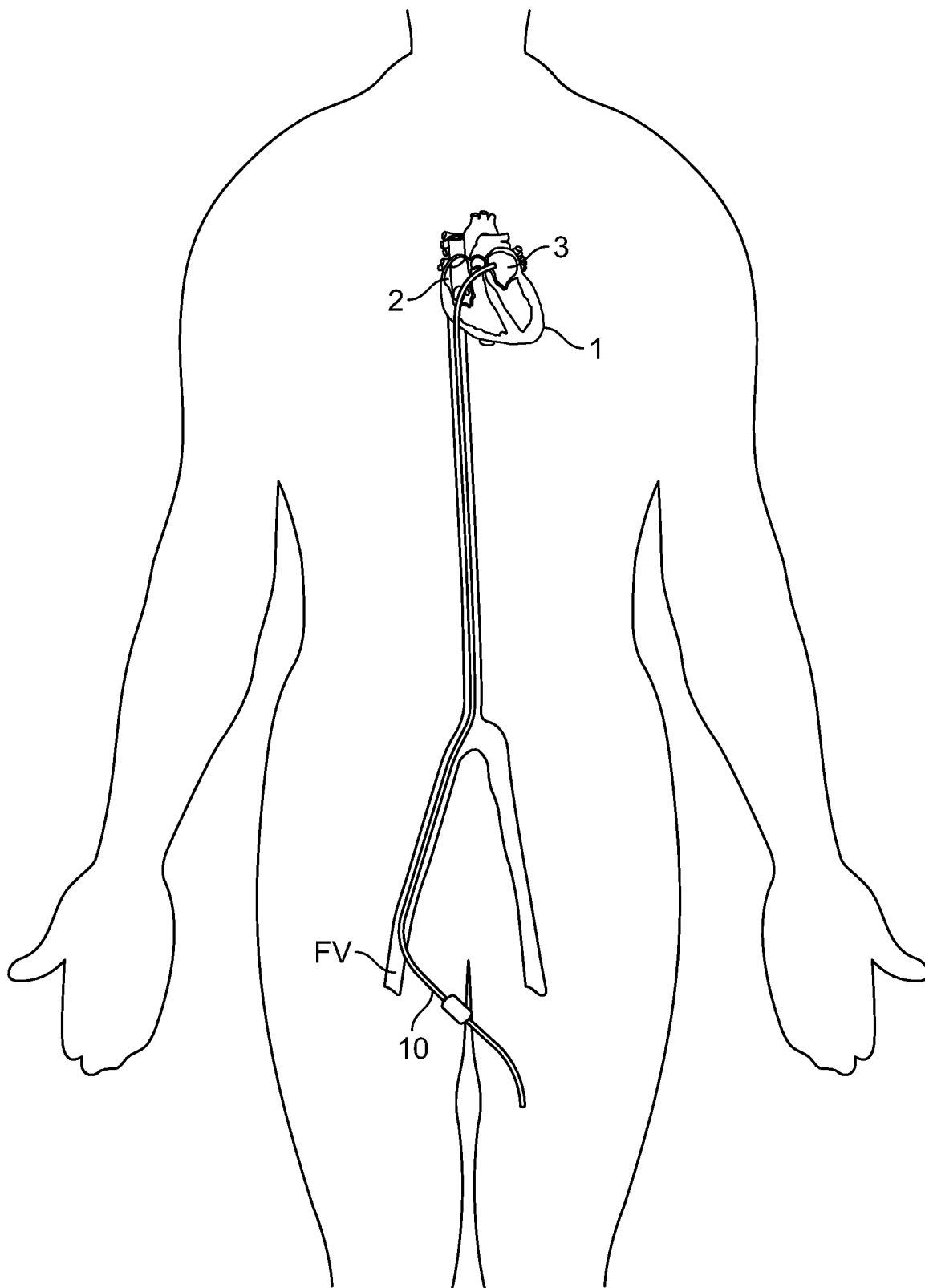


FIG. 34

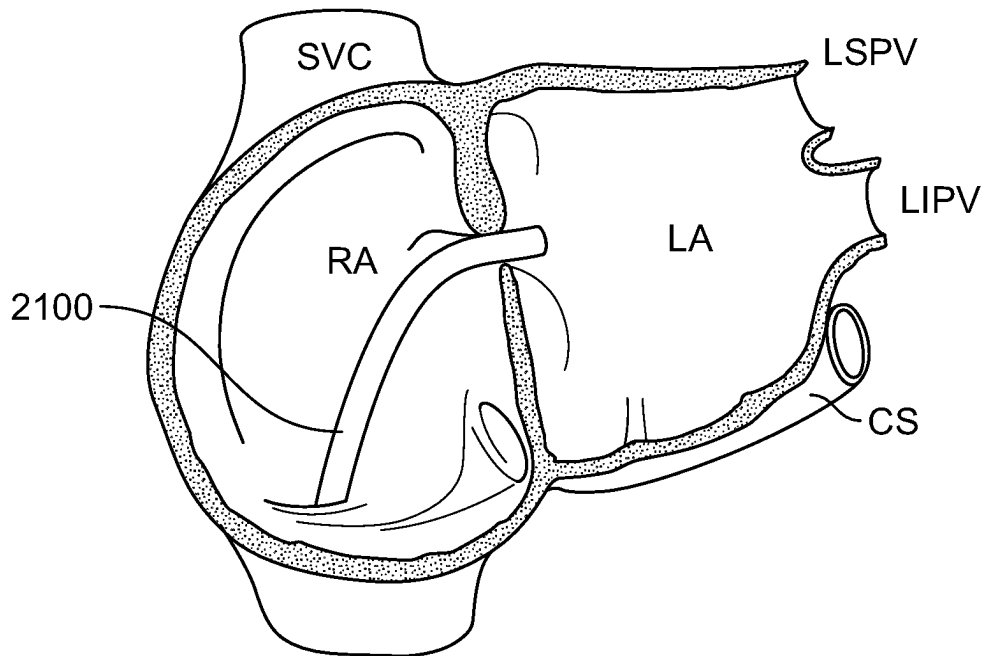


FIG. 35

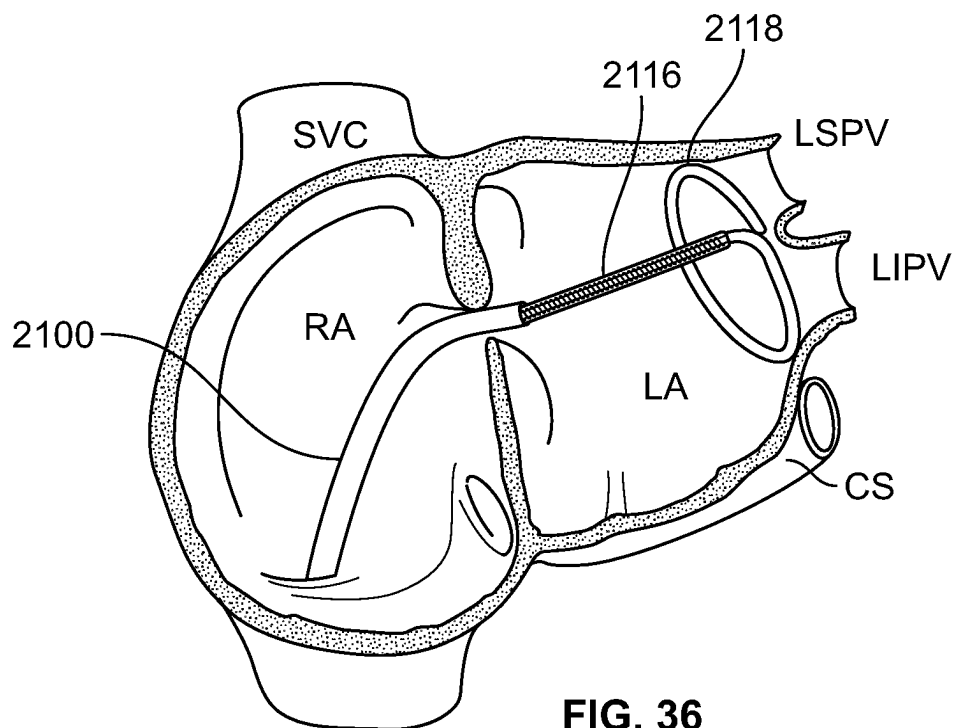


FIG. 36

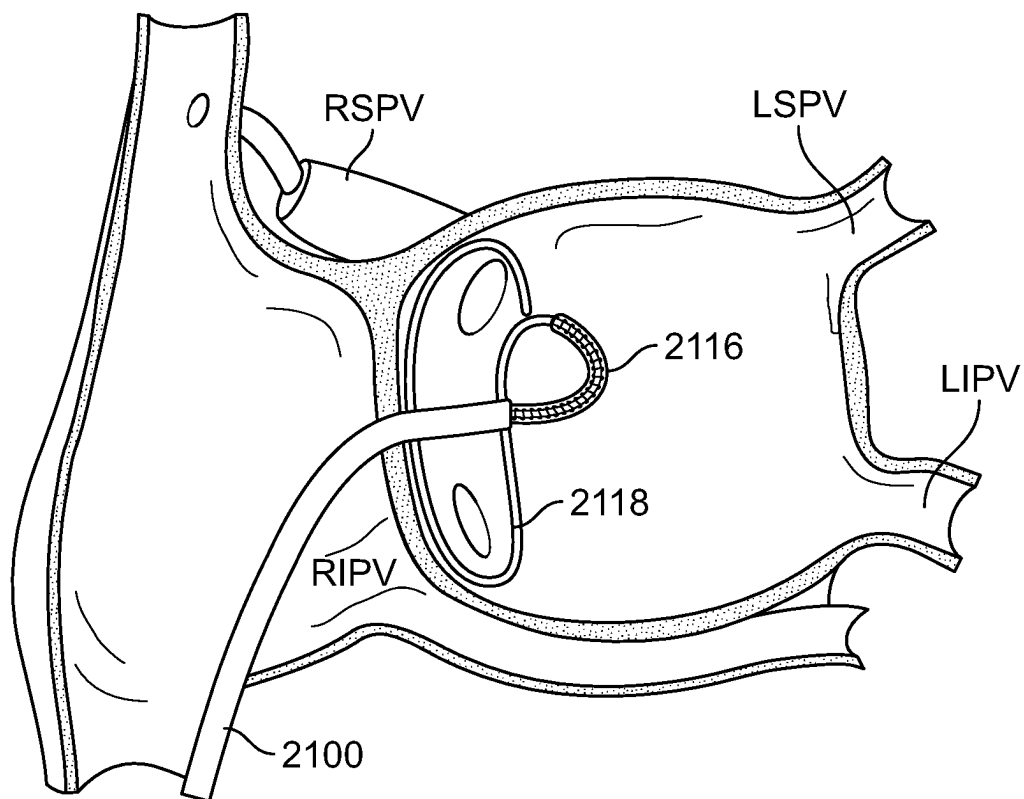


FIG. 37

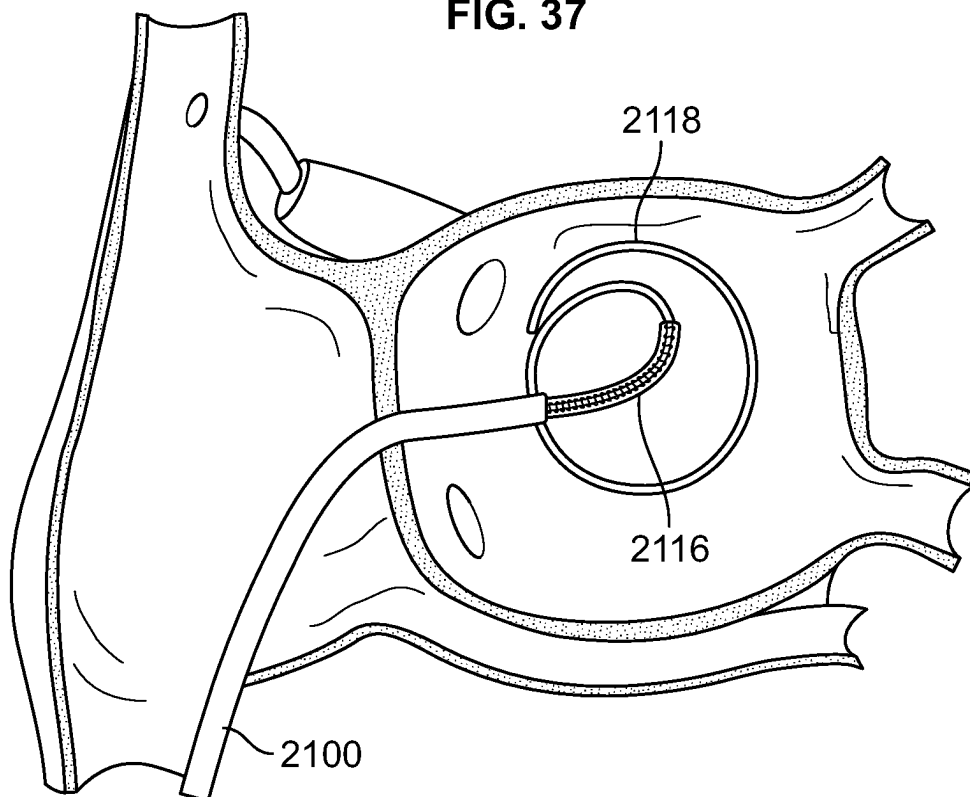


FIG. 38

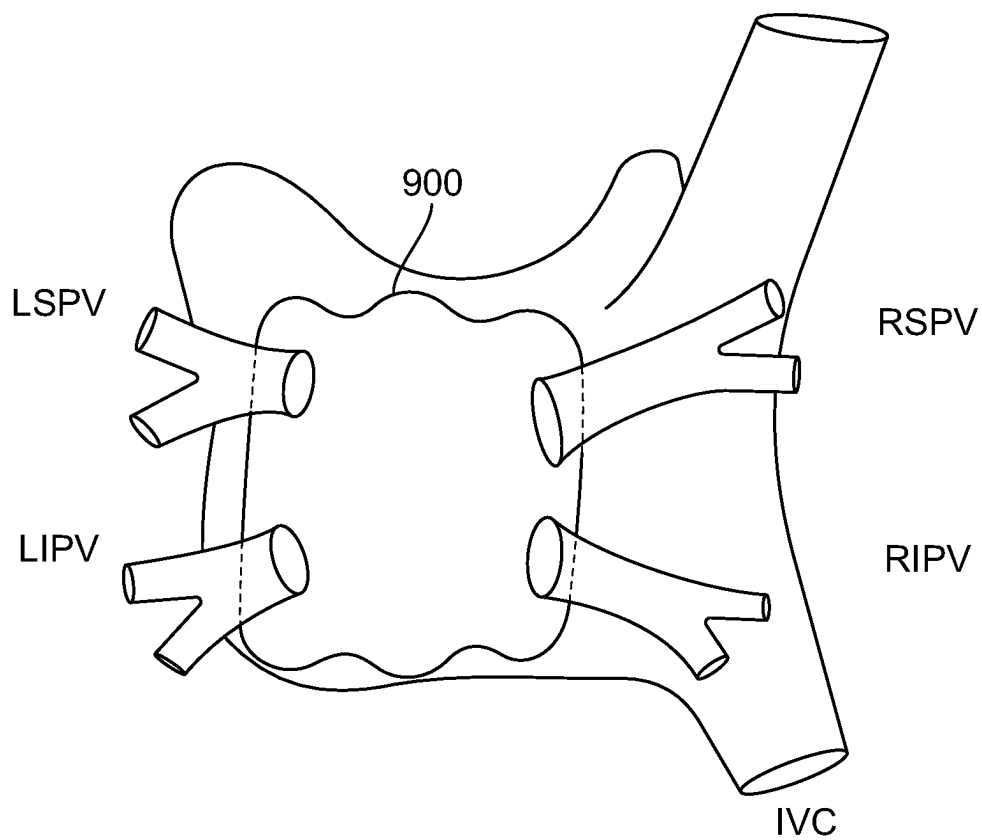


FIG. 39

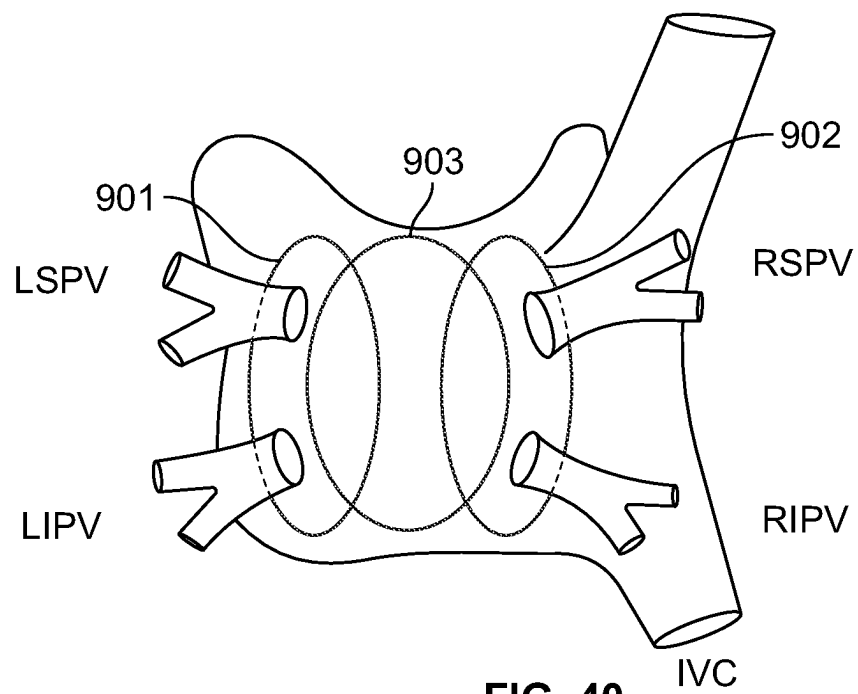


FIG. 40

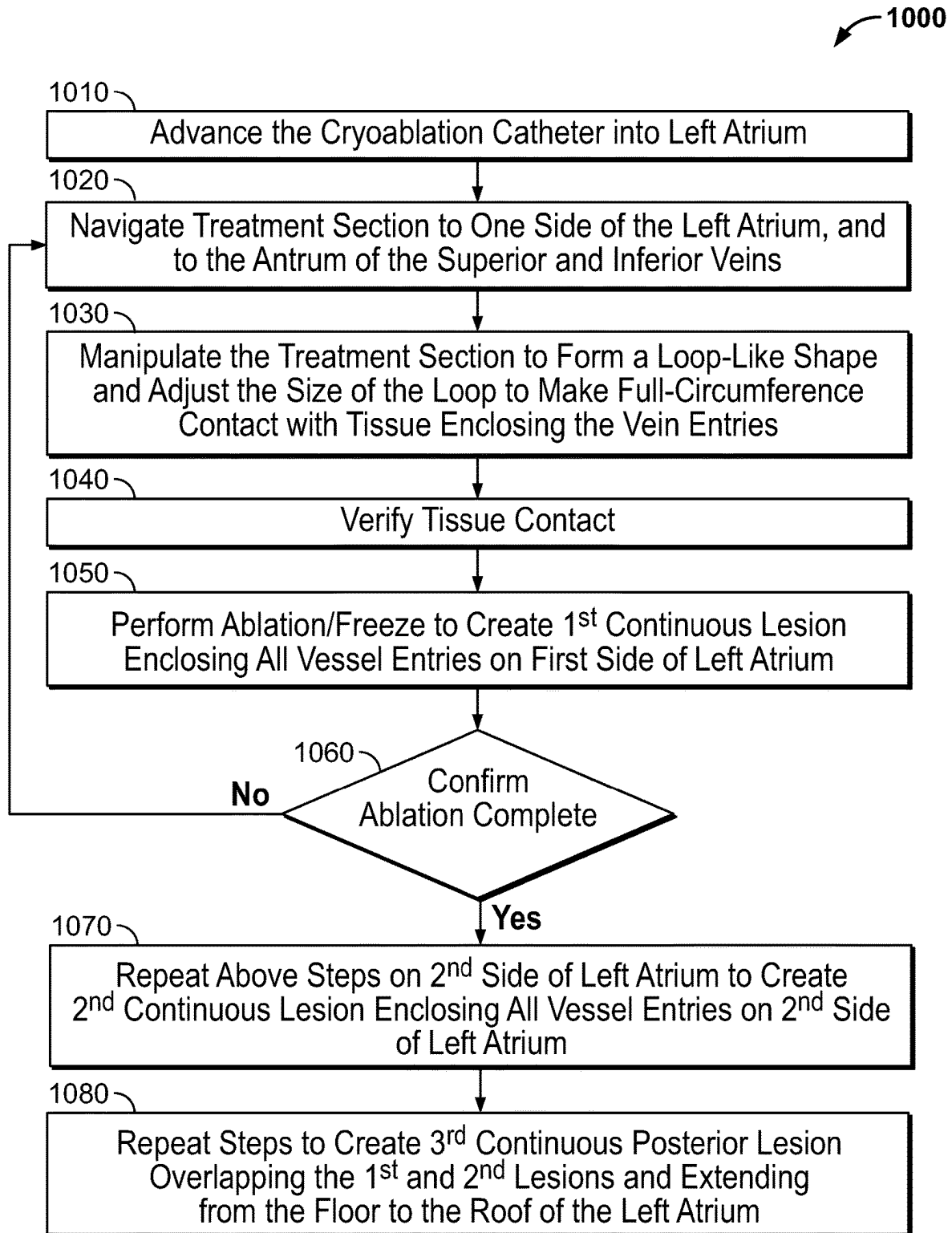


FIG. 41

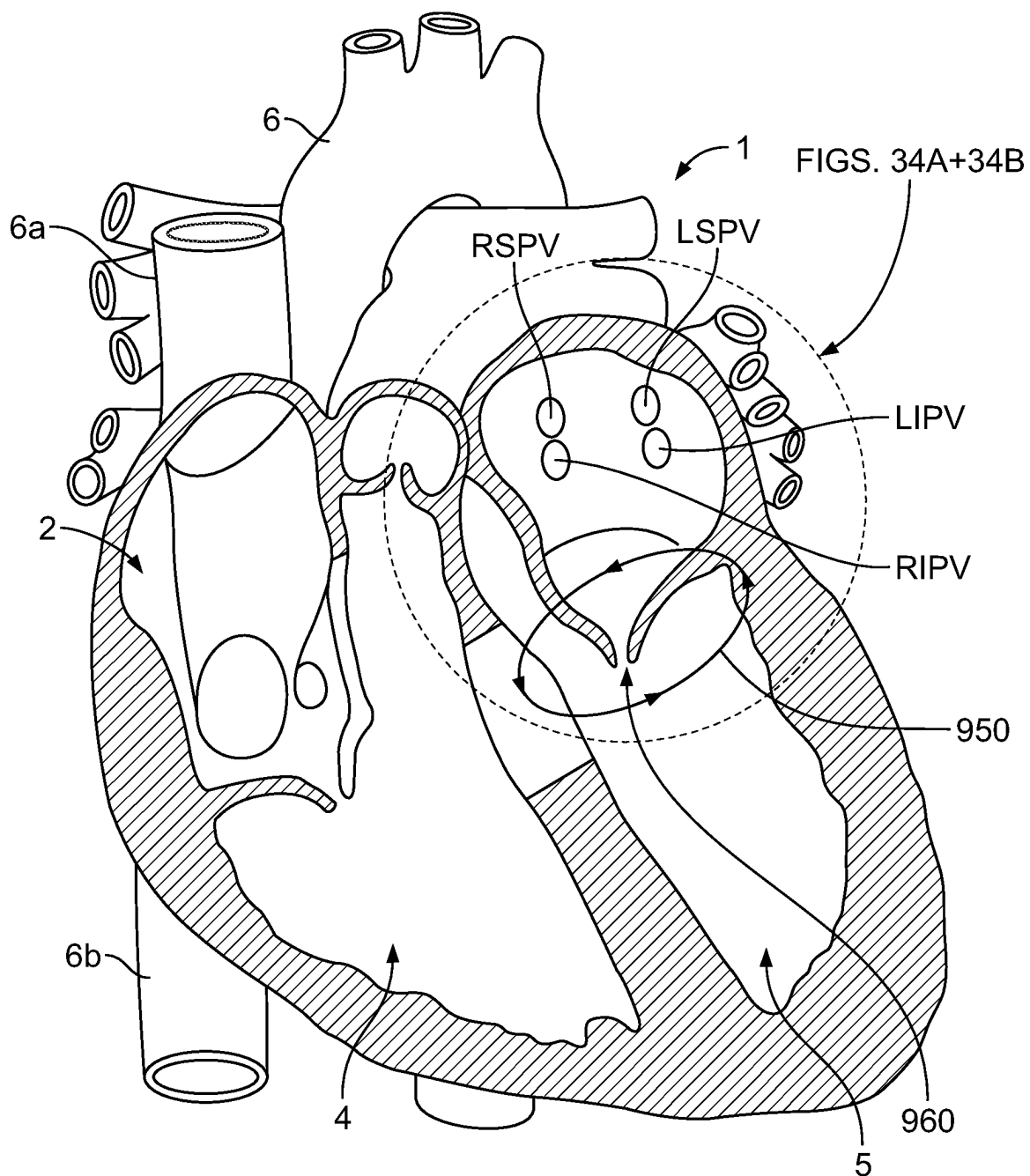


FIG. 42

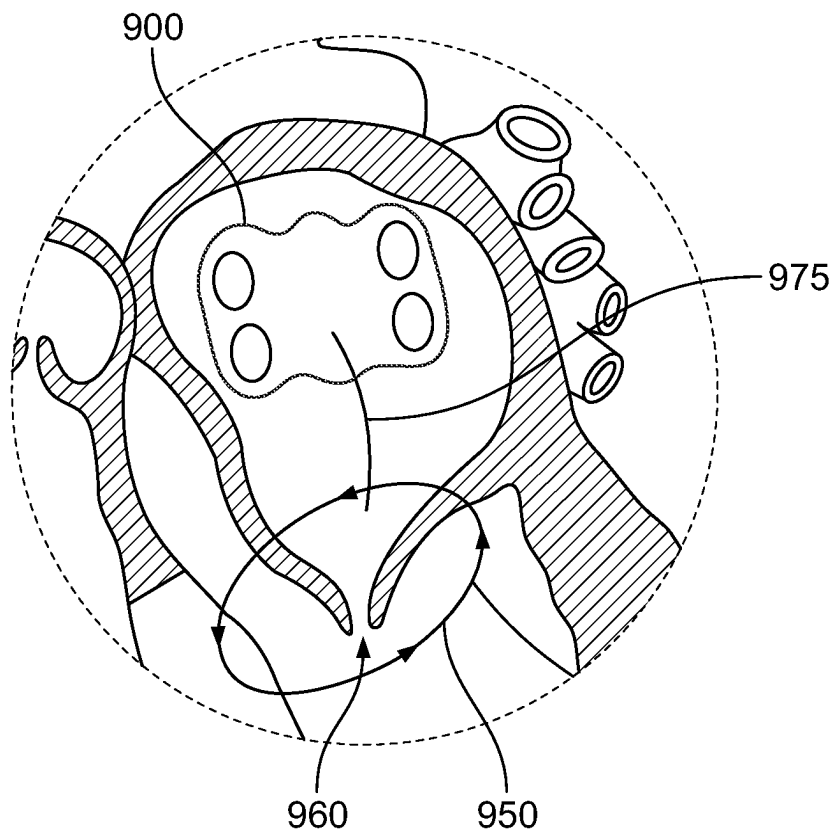


FIG. 43A

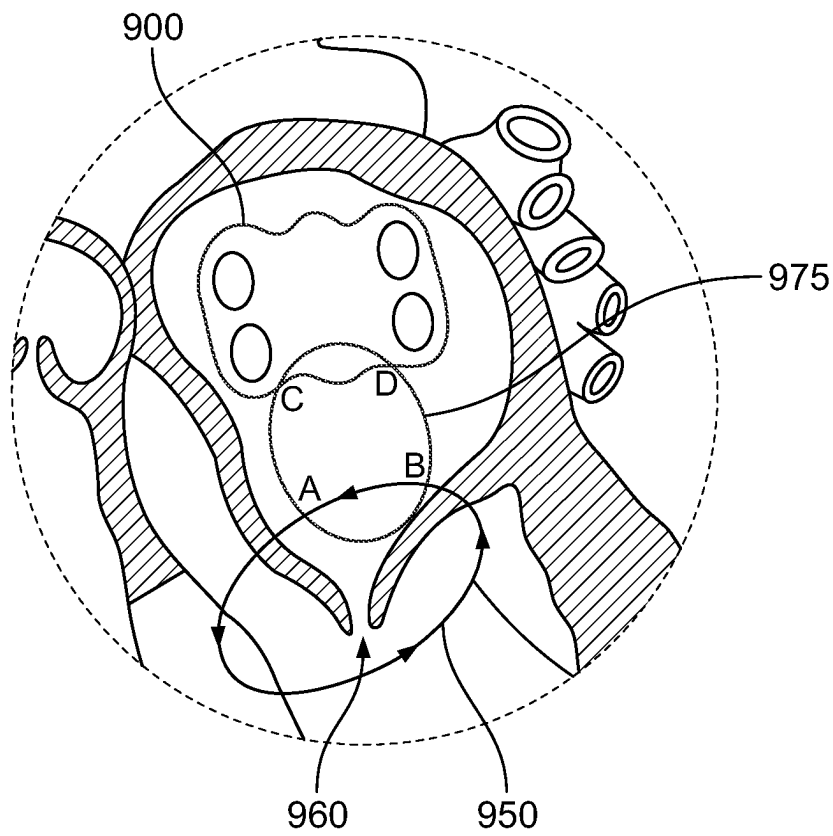


FIG. 43B

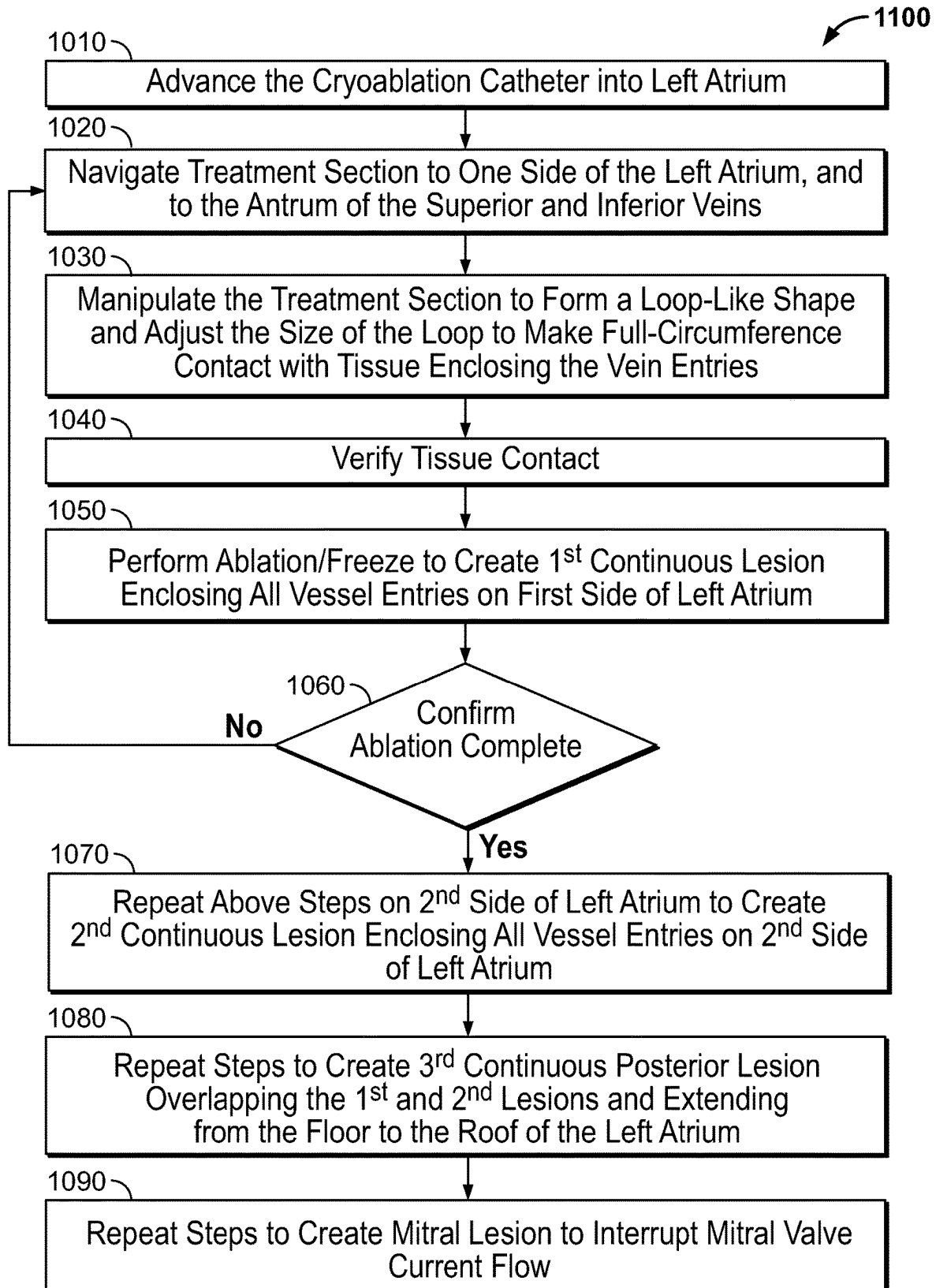


FIG. 44

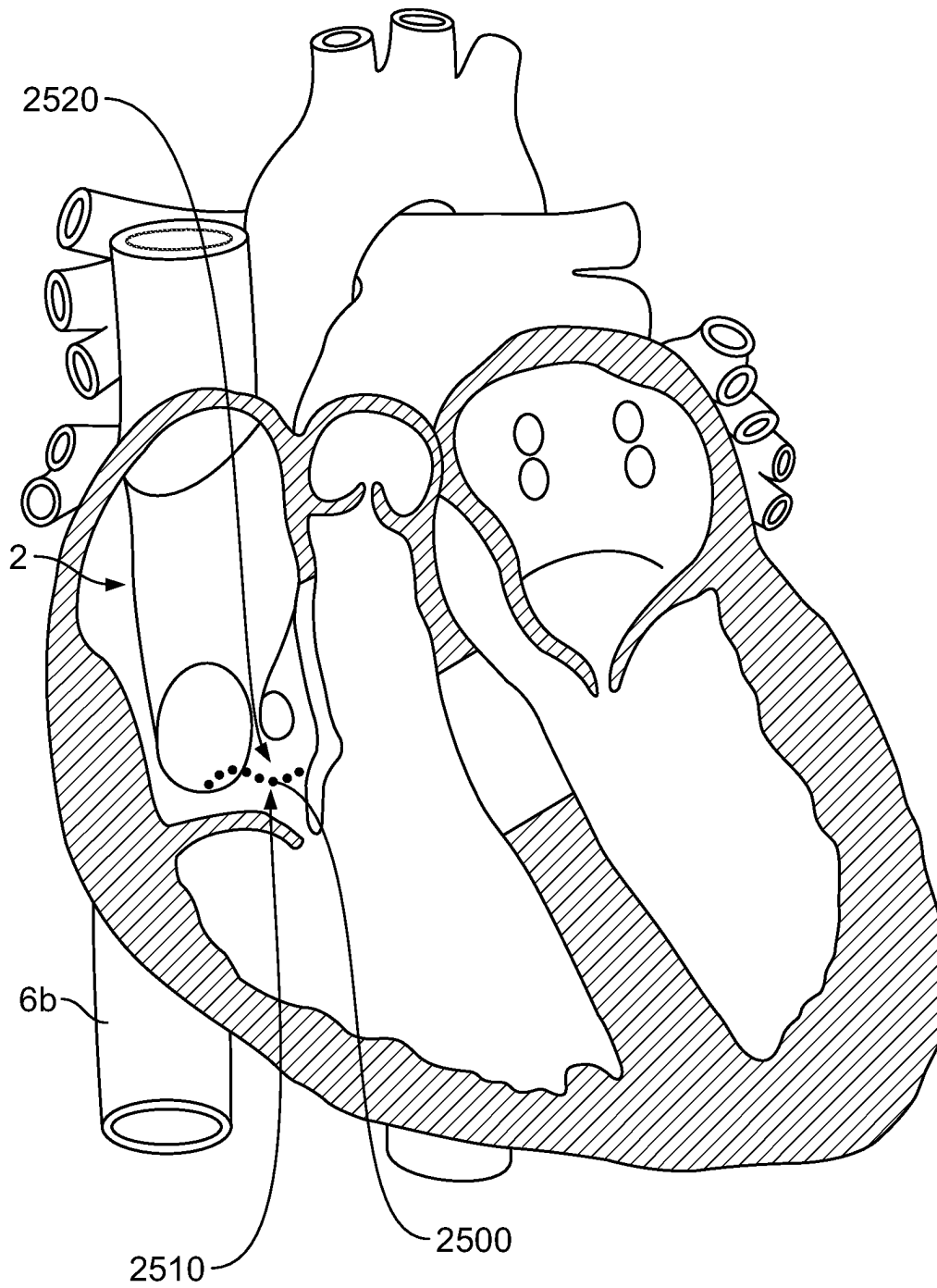


FIG. 45

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CRYOABLATION ELEMENT WITH CONDUCTIVE LINER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/958,589, filed Jun. 26, 2020, which is a national phase filing under 371 of international patent application number PCT/US19/12754, filed Jan. 8, 2019, and claims the benefit of U.S. Provisional Application No. 62/615,573, filed Jan. 10, 2018, the entire contents of which are incorporated herein by reference in their entirety for all purposes.

BACKGROUND

1. Field of the Invention

Embodiments of the invention relate to a surgical apparatus and more particularly to a surgical apparatus for applying thermal energy to ablate tissue.

2. Description of the Related Art

Atrial flutter and atrial fibrillation are heart conditions in which the left or right atrium of the heart beat improperly. Atrial flutter is a condition when the atria beat very quickly, but still evenly. Atrial fibrillation is a condition when the atria beat very quickly, but unevenly.

These conditions are often caused by aberrant electrical behavior of some portion of the atrial wall. Certain parts of the atria, or nearby structures such as the pulmonary veins, can misfire in their production or conduction of the electrical signals that control contraction of the heart, creating abnormal electrical signals that prompt the atria to contract between normal contractions caused by the normal cascade of electrical impulses. This can be caused by spots of ischemic tissue, referred to as ectopic foci, or by electrically active fibers in the pulmonary veins, for example.

Ventricular tachycardia (V-tach or VT) is a type of regular and fast heart rate that arises from improper electrical activity in the ventricles of the heart. In ventricular tachycardia, the abnormal electrical signals in the ventricles cause the heart to beat faster than normal, usually 100 or more beats a minute, out of sync with the upper chambers. When this happens, the heart may not be able to pump enough blood to the body and lungs because the chambers are beating so fast or out of sync with each other that the chambers do not have time to fill properly. Thus, V-tach may result in cardiac arrest and may turn into ventricular fibrillation.

Atrial fibrillation is one of the more prevalent types of heart conditions. Failing to treat atrial fibrillation can lead to a number of undesirable consequences including heart palpitations, shortness of breath, weakness and generally poor blood flow to the body.

Various techniques are practiced to treat atrial fibrillation. One technique to treat AF is pulmonary vein isolation (PVI). PVI is performed by creating lesions circumscribing the pulmonary veins. The PVI serves to block the errant or abnormal electrical signals.

A challenge in performing PVI, however, is to obtain a lasting or permanent isolation of the pulmonary veins. This shortcoming is highlighted in various studies. In one long-term follow-up study that investigated the rate of pulmonary vein reconnection after initial isolation, 53% of 161 patients were free of AF. In 66 patients, a repeat ablation was

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performed for repeat arrhythmia. The rate of pulmonary vein reconnection was high at 94% (62 of 66 patients). (Ouyang F, Tilz R, Chun J, et al. Long-term results of catheter ablation in paroxysmal atrial fibrillation: lessons from a 5-year follow-up. *Circulation* 2010; 122:2368-77.)

One reason that some PVI treatments are not durable is because of the phenomena of pulmonary vein (or electrical) reconnection. (Sawhney N, Anousheh R, Chen W C, et al. Five-year outcomes after segmental pulmonary vein isolation for paroxysmal atrial fibrillation. *Am J Cardiol* 2009; 104:366-72) (Callans D J, Gerstenfeld E P, Dixit S, et al. Efficacy of repeat pulmonary vein isolation procedures in patients with recurrent atrial fibrillation. *J Cardiovasc Electrophysiol* 2004; 15:1050-5) (Verma A, Kilicaslan F, Pisano E, et al. Response of atrial fibrillation to pulmonary vein antrum isolation is directly related to resumption and delay of pulmonary vein conduction. *Circulation* 2005; 112: 627-35)

Pulmonary vein reconnection may be attributed to gaps and incomplete or discontinuous isolation of the veins. (Bunch T J, Cutler M J. Is pulmonary vein isolation still the cornerstone in atrial fibrillation ablation? *J Thorac Dis*. 2015 February; 7(2):132-41). Incomplete isolation is a result of residual gap(s) within the encircling lesion or lack of transmural lesions. (McGann C J, Kholmovski E G, Oakes R S, et al. New magnetic resonance imaging-based method for defining the extent of left atrial wall injury after the ablation of atrial fibrillation. *J Am Coll Cardiol* 2008; 52:1263-71.) (Ranjan R, Kato R, Zviman M M, et al. Gaps in the ablation line as a potential cause of recovery from electrical isolation and their visualization using MRI. *Circ Arrhythm Electrophysiol* 2011; 4:279-86.)

Additionally, early recurrence of AF post ablation may be an early marker of incomplete pulmonary vein isolation. This is supported by a study of 12 patients that underwent a maze procedure after a failed radiofrequency ablation. Notably, myocardial biopsies showed anatomic gaps and/or non-transmural lesions in pulmonary veins that had reconnected. (Kowalski M, Grimes M M, Perez F J, et al. Histopathologic characterization of chronic radiofrequency ablation lesions for pulmonary vein isolation. *J Am Coll Cardiol* 2012; 59:930-8.)

This is further supported in a canine study in which endocardial conduction block was demonstrated and post procedural gaps were identified using MRI within the line of ablation. Long-term follow up data demonstrated that those pulmonary veins with the MRI-identified gaps were more likely to become electrically reconnected with symptomatic recurrences. (Ranjan R, Kato R, Zviman M M, et al. Gaps in the ablation line as potential cause of recovery from electrical isolation and their visualization using MRI. *Circ Arrhythm Electrophysiol* 2011; 4:279-86.)

Various attempts to solve the above referenced problem include making linear ablations in combination with circumferential pulmonary vein isolation (CPVI). One study, for example, compared clinical outcomes of CPVI with additional linear ablations and CPVI in a prospective randomized controlled study among patients with paroxysmal AF. The study enrolled 100 paroxysmal AF patients (male 75.0%, 56.4±11.6 years old) who underwent radio frequency circumferential ablation (RFCA) and were randomly assigned to the CPVI group (n=50) or the catheter Dallas lesion group (CPVI, posterior box lesion, and anterior linear ablation, n=50). The catheter Dallas lesion group required longer procedure (190.3±46.3 vs. 161.1±30.3 min, P<0.001) and ablation times (5345.4±1676.4 vs. 4027.2±878.0 s, P<0.001) than the CPVI group. Complete bidirectional

conduction block rate was 68.0% in the catheter Dallas lesion group and 100% in the CPVI group. Procedure-related complication rates were not significantly different between the catheter Dallas lesion (0%) and CPVI groups (4%, $P=0.157$). During the 16.3 ± 4.0 months of follow-up, the clinical recurrence rates were not significantly different between the two groups, regardless of complete bidirectional conduction block achievement after linear ablation. (Kim et al. Linear ablation in addition to circumferential pulmonary vein isolation (Dallas lesion set) does not improve clinical outcome in patients with paroxysmal atrial fibrillation: a prospective randomized study. *Europace*. 2015 March; 17(3):388-95.)

Thus, in view of the above referenced study, adding more ablation points around the vein entries, and/or attempting to add a linear lesion by using point by point ablation, does not appear to be an optimal solution to prevent gap(s) along the encircling lesion. Additionally, adding multiple points and lines undesirably increases the procedure time.

In view of the above shortcomings, various ablation catheters have been proposed for creation of the lesion, including flexible cryoprobes or cryocatheters, bipolar RF catheters, monopolar RF catheters (using ground patches on the patient's skin), microwave catheters, laser catheters, and ultrasound catheters. U.S. Pat. No. 6,190,382 to Ormsby and U.S. Pat. No. 6,941,953 to Feld, for example, describe RF ablation catheters for ablating heart tissue. These approaches are attractive because they are minimally invasive and can be performed on a beating heart. However, these approaches have a low success rate. The low success rate may be due to incomplete lesion formation. A fully transmural lesion is required to ensure that the electrical impulse causing atrial fibrillation are completely isolated from the remainder of the atrium, and this is difficult to achieve with beating heart procedures.

Thus, the challenge for the surgeon is to place the catheter/probe along the correct tissue contour such that the probe makes complete contact with the tissue. Due to the nature of the procedure and the anatomical locations where the lesions must be created, the catheter must be sufficiently flexible and adjustable such that they can match the shape and contour of the tissue to be ablated.

Malleable and flexible cryoprobes are described in U.S. Pat. Nos. 6,161,543 and 8,177,780, both to Cox, et al. The described probes have a malleable shaft. In embodiments, a malleable metal rod is coextruded with a polymer to form the shaft. The malleable rod permits the user to plastically deform the shaft into a desired shape so that a tip can reach the tissue to be ablated.

U.S. Pat. No. 5,108,390, issued to Potocky et al, discloses a highly flexible cryoprobe that can be passed through a blood vessel and into the heart without external guidance other than the blood vessel itself.

A challenge with some of the above apparatuses, however, is making continuous contact along the anatomical surface such that a continuous lesion may be created. This challenge is amplified not only because of the varying contours and shapes of the target tissue because of the location in the body but also because of variations in anatomy between patients. Thus, different treatment procedures and patient anatomy require different catheters to be designed and used. Another challenge is to be able to adjust the shape of the catheter in situ to address these variations in anatomy, etc.

Additional challenges with some of the above apparatuses is with efficient thermal conductivity, i.e., cooling/heat transfer, between the internal cooling/heating elements of the devices and the exterior jackets/sleeves of the devices. Thus,

freezing and heating temperatures may need be efficiently transferred to the tissue to be ablated.

Accordingly, there is a need for improved methods and systems for providing minimally invasive, adjustably shaped, safe and efficient cryogenic cooling of tissues. These improved systems include improved apparatuses and methods to form continuous lesions in target tissue regardless of the condition being treated and variations in patient anatomy.

There is also a need for an improved apparatus and method to treat AF, atrial flutter and V-tach and to achieve more complete, durable, and safe electrical signal isolation within the various chambers of the heart, including pulmonary vein isolation.

SUMMARY

An ablation apparatus for creating a lesion in target tissue, the ablation apparatus including a handle, an elongate shaft extending from the handle to a distal tip, where the shaft comprises a first portion, an ablation portion distal to the first portion and an outer sheath. The ablation apparatus further includes at least one ablation energy element disposed within the outer sheath, where a space is formed between the at least one ablation energy element and the outer sheath, and a thermally conductive liner disposed within the space.

In embodiments, the thermally conductive liner is a thermoplastic elastomer or thermoplastic urethane loaded with a thermally conductive material.

In embodiments, the thermally conductive liner is a polyether block amide (PEBA) loaded with aluminum oxide. In embodiments, the PEBA is loaded with aluminum oxide in the range from about 50-70% by weight. In some embodiments, the PEBA is loaded with boron nitride in the range from about 50-70% by weight.

In embodiments, the thermally conductive liner is disposed in the space by flow melting to substantially fill the space and surround each of the ablation energy elements.

In embodiments, the ablation energy element is linear or elongate and disposed longitudinally within the outer sheath.

In embodiments, the at least one ablation energy element comprises at least one ablation energy delivery lumen and at least one ablation energy return lumen. In embodiments, each of the ablation energy delivery lumen and the at least one ablation energy return lumen comprise an inner tube having an outer tube surrounding the inner tube thereby defining a gap between the inner tube and the outer tube. The gap is capable of being filled with a thermally conducting liquid.

In embodiments, the ablation apparatus comprises a plurality of ablation energy delivery lumens and a plurality of ablation energy return lumens.

In embodiments, the ablation apparatus further comprises at least one electrode on an exterior surface of the outer sheath and at least one service lumen to provide electrical conductors or other functional elements to the ablation portion.

In embodiments, the ablation energy is provided by a cryogen being transported through the ablation energy delivery lumen and at least one ablation energy return lumen. In embodiments, the cryogen is nitrogen or nitrogen in a near critical state.

In embodiments, the ablation apparatus further comprises a stylet lumen that extends substantially along a length of the shaft from the handle to at least the ablation portion; and a stylet capable of being inserted into the stylet lumen. In embodiments, the stylet comprises a shape-memory mate-

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rial, and wherein the stylet has a plurality of flexibilities along its length. In embodiments, at least a distal portion of the stylet is pre-set with a shape that corresponds to a desired shape of the lesion to be formed.

In embodiments, the ablation apparatus is used to treat a condition selected from the group consisting of atrial fibrillation, atrial flutter and ventricular tachycardia.

In embodiments, the thermally conductive liner entirely surrounds each ablation energy element.

In embodiments, the at least one ablation energy element decreases the temperature of the target tissue to cause ablation. In some embodiments, the at least one ablation energy element increases the temperature of the target tissue.

In some embodiments, an ablation system for creating a lesion in target tissue is disclosed. The ablation system comprises a catheter having a handle, a distal tip, and an elongate shaft extending from the handle to the distal tip. The shaft comprises a first portion an ablation portion distal to the first portion that includes an outer sheath, and at least one ablation energy element disposed within the outer sheath, wherein a space is formed between the at least one ablation energy element and the outer sheath. The shaft also includes a first thermally conductive media disposed within said space. The system also includes an energy generator coupled to the catheter to deliver and control the ablation energy from the at least one ablation energy element to the target tissue.

In embodiments, a method for manufacturing an ablation catheter having an elongate ablation portion comprising an outer sheath and an inner ablation element extending longitudinally therethrough, the method comprising flowing a conductive liner between a space defined between the inner ablation element and the outer sheath. In embodiments, the step of flowing is performed by flowing a thermoplastic loaded with a thermally conductive agent.

In embodiments, the step of flowing is performed by flowing a thermoplastic loaded with a low linear coefficient of thermal expansion agent.

In embodiments, a method for ablating tissue comprises any one or combination of the steps described herein except where such steps are exclusive of one another.

In embodiments, an apparatus for ablating tissue comprises any one or combination of the structures described herein except where such structures are exclusive of one another.

In embodiments, a system for ablating tissue comprises any one or combination of the components described herein except where such components are exclusive of one another.

In another embodiment, an ablation catheter for creating a lesion in target tissue is disclosed. The ablation catheter comprises an elongate shaft extending from the handle to a distal tip, the shaft including an ablation portion distal to the first portion where the ablation portion also includes an outer sheath, at least one ablation energy element disposed within the outer sheath and defining a space between the at least one ablation energy element and the outer sheath, and a thermally conductive liner disposed within the space.

Some embodiments are directed to a thermally-conductive material that comprises a base material selected from the group comprising thermoplastic elastomers (TPE) and thermoplastic urethanes (TPU) and a thermally conductive filler selected from the group comprising aluminum, aluminum oxide, boron nitride, copper, silver and gold. In some embodiments, the base material is polyether block amide (PEBA). In some embodiments, the thermally conductive filler is aluminum oxide (Al₂O₃) or boron nitride (BN). In

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some embodiments, the PEBA is loaded with about 10-70% of the aluminum oxide (Al₂O₃) or boron nitride (BN) by weight.

The description, objects and advantages of embodiments of the present invention will become apparent from the detailed description to follow, together with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The above-mentioned aspects, as well as other features, aspects and advantages of the present technology will now be described in connection with various embodiments, with reference to the accompanying drawings. The illustrated embodiments, however, are merely examples and are not intended to be limiting. Throughout the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Note that the relative dimensions of the following figures may not be drawn to scale.

FIG. 1 illustrates a typical cryogen phase diagram;

FIG. 2 is a schematic illustration of a cryogenic cooling system;

FIG. 3 is a cryogen phase diagram corresponding to the system shown in FIG. 2 where the cryogen is N₂;

FIG. 4 provides a flow diagram that summarizes aspects of the cooling system of FIG. 2;

FIG. 5A is a perspective view of a cryoablation catheter, according to an embodiment of the invention;

FIG. 5B is a cross-sectional view taken along line 5B-5B of FIG. 5A;

FIG. 6 is an illustration of a cryoablation system including a cryoablation catheter, according to an embodiment of the invention;

FIG. 7 is an enlarged perspective view of a distal section of the cryoablation catheter shown in FIG. 6.

FIG. 8 is a perspective view of another embodiment of a cryoablation catheter having a flexible distal treatment section;

FIG. 9A is a cross-sectional view of an embodiment of a catheter shown in FIG. 8 taken along line 9A-9A in FIG. 9;

FIG. 9B is an enlarged view of one of the multi-layered tubes shown in FIG. 9A;

FIG. 9C is a cross sectional view of another embodiment of a cryoablation catheter;

FIG. 10A is a partial sectional view of an embodiment of a catheter shown in FIG. 8;

FIG. 10B is a partial exploded view of the proximal ends of the tube elements and the distal end of the intermediate section of an embodiment of a catheter shown in FIG. 8;

FIG. 11 is a perspective view of another embodiment of a cryoablation catheter having a flexible distal treatment section;

FIG. 12 is an enlarged view of a portion of the distal section shown in FIG. 11;

FIG. 13 is a cross sectional view of the catheter shown in FIG. 12 taken along line 13-13 in FIG. 12;

FIGS. 14-15 illustrate sequential deployment of the distal section of catheter shown in FIG. 11 from an outer sheath member;

FIG. 16 is a perspective view of another embodiment of a cryoablation catheter having a flexible distal treatment section;

FIG. 17 is an enlarged view of the distal section of the catheter shown in FIG. 16;

FIG. 18 is a cross sectional view of the catheter shown in FIG. 17 taken along line 17-17 in FIG. 17;

FIGS. 19A-19D show deployment of a distal section of the catheter, according to an embodiment of the invention;

FIGS. 20A-20B show reducing the diameter of the preset loop shape of the catheter shown in FIG. 19D;

FIGS. 21A-21C show articulation of a catheter shaft, according to an embodiment of the invention;

FIGS. 22A-22B show components of an intermediate section of the catheter;

FIG. 23A shows a perspective view of a handle for an ablation catheter, according to an embodiment of the invention;

FIG. 23B shows a partial perspective view of the handle shown in FIG. 23A with the exterior removed;

FIG. 24 is a perspective view of another embodiment of a cryoablation catheter having an internal stylet;

FIGS. 25A-25C are cross sectional views of different embodiments of the invention corresponding to line 24A-24A in FIG. 24;

FIG. 26 is an enlarged view of the multi-layered cryogen delivery/return tubes shown in FIG. 25A;

FIG. 27A is a perspective view of the cryoablation catheter depicted in FIG. 24 with the internal stylet inserted;

FIG. 27B is a perspective view of the cryoablation catheter depicted in FIG. 24 with the internal stylet inserted with the flexible distal ablation portion of the ablation shaft/sleeve transformed into the curved configuration of the stylet;

FIGS. 28A-28C are cross sectional views of various embodiments of the invention corresponding to line 27A-27A in FIG. 27A;

FIG. 29 depicts sample shapes for the stylet;

FIG. 30 depicts a stylet having multiple flexibilities long its length, according to an embodiment of the invention;

FIG. 31 A depicts a method of altering the flexibility of a portion of a stylet, according to an embodiment of the invention;

FIG. 31 B depicts View A in FIG. 31A, according to an embodiment of the invention;

FIG. 32A depicts a method of altering the flexibility of a portion of a stylet, according to an embodiment of the invention;

FIG. 32B depicts a method of altering the flexibility of a portion of a stylet, according to an embodiment of the invention;

FIG. 32C depicts a method of altering the flexibility of a portion of a stylet, according to an embodiment of the invention;

FIG. 33 is an illustration of a heart, and locations of various lesions according to an embodiment of the invention;

FIG. 34 is an illustration of an embodiment of endovascular catheterization to access the heart;

FIGS. 35-36 are illustrations of a procedure to place a distal section of a cryoablation catheter against the endocardial wall in the left atrium, circumscribing the left superior and inferior pulmonary vein entries, according to an embodiment of the invention;

FIGS. 37-38 are illustrations of a procedure to place a distal section of a cryoablation catheter against the endocardial wall in the left atrium, circumscribing the right superior and inferior pulmonary vein entries, according to an embodiment of the invention.

FIGS. 39-40 illustrate a method for creating a box-shaped lesion, according to an embodiment of the invention, where the figures depict the left atrium as viewed from the back of a patient;

FIG. 41 is flow diagram showing a method of creating a box-shaped lesion to enclose multiple PVs in the left atrium, according to an embodiment of the invention;

FIG. 42 is an illustration of a heart showing mitral valve electrical activity;

FIG. 43A depicts formation of a lesion to interrupt mitral valve electrical activity, according to an embodiment of the invention;

FIG. 43B depicts formation of a lesion to interrupt mitral valve electrical activity, according to an embodiment of the invention;

FIG. 44 is flow diagram showing a method of creating a box-shaped lesion to enclose multiple PVs in the left atrium and a lesion to interrupt mitral valve electrical activity, according to an embodiment of the invention; and

FIG. 45 depicts formation of a lesion to interrupt electrical activity in the right atrium, according to an embodiment of the invention.

DETAILED DESCRIPTION

It is to be understood that the embodiments of the invention described herein are not limited to particular variations set forth herein as various changes or modifications may be made to the embodiments of the invention described and equivalents may be substituted without departing from the spirit and scope of the embodiments of the invention. As will be apparent to those of skill in the art upon reading this disclosure, each of the individual embodiments described and illustrated herein has discrete components and features that may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the embodiments of the present invention. In addition, many modifications may be made to adapt a particular situation, material, composition of matter, process, process act(s) or step(s) to the objective(s), spirit or scope of the embodiments of the present invention. All such modifications are intended to be within the scope of the claims made herein.

Moreover, while methods may be depicted in the drawings or described in the specification in a particular order, such methods need not be performed in the particular order shown or in sequential order, and that all methods need not be performed, to achieve desirable results. Other methods that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional methods can be performed before, after, simultaneously, or between any of the described methods. Further, the methods may be rearranged or reordered in other implementations. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products. Additionally, other implementations are within the scope of this disclosure.

Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include or do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is other-

wise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

Reference to a singular item, includes the possibility that there are plural of the same items present. More specifically, as used herein and in the appended claims, the singular forms “a,” “an,” “said” and “the” include plural referents unless the context clearly dictates otherwise. It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely,” “only” and the like in connection with the recitation of claim elements, or use of a “negative” limitation.

It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, if an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

It will also be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, a first element could be termed a second element without departing from the teachings of the present invention.

Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially,” represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than or equal to 10% of, within less than or equal to 5% of, within less than or equal to 1% of, within less than or equal to 0.1% of, and within less than or equal to 0.01% of the stated amount. If the stated amount is 0 (e.g., none, having no), the above recited ranges can be specific ranges, and not within a particular % of the value. Additionally, numeric ranges are inclusive of the numbers defining the range, and any individual value provided herein can serve as an endpoint for a range that includes other individual values provided herein. For example, a set of values such as 1, 2, 3, 8, 9, and 10 is also a disclosure of a range of numbers from 1-10, from 1-8, from 3-9, and so forth.

Some embodiments have been described in connection with the accompanying drawings. The figures are drawn to scale, but such scale should not be limiting, since dimensions and proportions other than what are shown are contemplated and are within the scope of the disclosed inventions. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with various embodiments can be used in all other embodiments set forth herein. Additionally, it will be recognized that any methods described herein may be practiced using any device suitable for performing the recited steps.

While a number of embodiments and variations thereof have been described in detail, other modifications and methods of using the same will be apparent to those of skill in the

art. Accordingly, it should be understood that various applications, modifications, materials, and substitutions can be made of equivalents without departing from the unique and inventive disclosure herein or the scope of the claims.

All existing subject matter mentioned herein (e.g., publications, patents, patent applications and hardware) is incorporated by reference herein in its entirety except insofar as the subject matter may conflict with that of the present invention (in which case what is present herein shall prevail).

Embodiments of the invention make use of thermodynamic processes using cryogens that provide cooling without encountering the phenomenon of vapor lock.

Cryogen Phase Diagram and Near Critical Point

This application uses phase diagrams to illustrate various thermodynamic processes. An example phase diagram is shown in FIG. 1. The phase diagram includes axes that correspond to pressure P and temperature T, and a phase line 102 that delineates the locus of all (P, T) points where liquid and gas coexist. For (P, T) values to the left of the phase line 102, the cryogen is in a liquid state, generally achieved with higher pressures and lower temperatures, while (P, T) values to the right of the phase line 102 define regions where the cryogen is in a gaseous state, generally achieved with lower pressures and higher temperatures. The phase line 102 ends abruptly in a single point known as the critical point 104. In the case of nitrogen N₂, the critical point is at P_c=3.396 MPa and T_c=-147.15° C.

When a fluid has both liquid and gas phases present during a gradual increase in pressure, the system moves up along the liquid-gas phase line 102. In the case of N₂, the liquid at low pressures is up to two hundred times more dense than the gas phase. A continual increase in pressure causes the density of the liquid to decrease and the density of the gas phase to increase, until they are equal only at the critical point 104. The distinction between liquid and gas disappears at the critical point 104. The blockage of forward flow by gas expanding ahead of the liquid cryogen (“vapor lock”) is thus avoided when a cryogen flows at conditions surrounding the critical point, defined herein as “near-critical conditions.” Factors that allow greater departure from the critical point while maintaining a functional flow include greater speed of cryogen flow, larger diameter of the flow lumen and lower heat load upon the thermal exchanger, or cryo-treatment region.

As the critical point is approached from below, the vapor phase density increases and the liquid phase density decreases until right at the critical point, where the densities of these two phases are exactly equal. Above the critical point, the distinction of liquid and vapor phases vanishes, leaving only a single, supercritical phase, where the fluid has the properties of both a liquid and a gas (i.e., a dense fluid without surface tension capable of frictionless flow).

Van der Waals thermodynamic equation of state is a well-established equation for describing gases and liquids:

$$(p + 3/v^2)(3v - 1) = 8t \quad [\text{Eq. 1}]$$

where $p = P/P_c$, $v = V/V_c$, and $t = T/T_c$, and P_c , V_c , and T_c are the critical pressure, critical molar volume, and the critical temperature respectively.

The variables v , p , and t are often referred to as the “reduced molar volume,” the “reduced pressure,” and the “reduced temperature,” respectively. Hence, any two substances with the same values of p , v , and t are in the same thermodynamic state of fluid near its critical point. Eq. 1 is thus referred to as embodying the “Law of Corresponding States.” This is described more fully in H. E. Stanley,

Introduction to Phase Transitions and Critical Phenomena (Oxford Science Publications, 1971), the entire disclosure of which is incorporated herein by reference in its entirety for all purposes.

In embodiments of the present invention, the reduced pressure p is fixed at a constant value of approximately one, and hence at a fixed physical pressure near the critical pressure, while the reduced temperature t varies with the heat load applied to the device. If the reduced pressure p is a constant set by the engineering of the system, then the reduced molar volume v is an exact function of the reduced temperature t .

In other embodiments of the present invention, the operating pressure p may be adjusted so that over the course of variations in the temperature t of the device, v is maintained below some maximum value at which the vapor lock condition will result. It is generally desirable to maintain p at the lowest value at which this is true because boosting the pressure to achieve higher values of p may involve use of a more complex and more expensive compressor, resulting in more expensive procurement and maintenance of the entire apparatus support system and lower overall cooling efficiency.

The conditions for v depend in a complex way on the volume flow rate dV/dt , the heat capacity of the liquid and vapor phases, and the transport properties such as the thermal conductivity, viscosity, etc., in both the liquid and the vapor. The exact relationship is not derived here in closed form algebraically, but may be determined numerically by integrating the model equations that describe mass and heat transport within the cooling device. Conceptually, vapor lock occurs when the rate of heating of the tip (or other device structure for transporting the cryogen and cooling the tissue) produces the vapor phase. The cooling power of this vapor phase, which is proportional to the flow rate of the vapor multiplied by its heat capacity divided by its molar volume, is not able to keep up with the rate of heating to the tip. When this occurs, more and more of the vapor phase is formed in order to absorb the excess heat through the conversion of the liquid phase to vapor in the cryogen flow. This creates a runaway condition where the liquid converts into vapor phase to fill the tip, and effectively all cryogen flow stops due to the large pressure that results in this vapor phase as the heat flow into the tip increases its temperature and pressure rapidly. This condition is called "vapor lock."

In accordance with one embodiment of the present invention, the liquid and vapor phases are substantially identical in their molar volume. The cooling power is at the critical point, and the cooling system avoids vapor lock. Additionally, at conditions slightly below the critical point, the apparatus may avoid vapor lock as well.

Cryoablation System

FIG. 2 provides a schematic illustration of a structural arrangement for a cryogenic system in one embodiment, and FIG. 3 provides a phase diagram that illustrates a thermodynamic path taken by the cryogen when the system of FIG. 2 is operated. The circled numerical identifiers in the two figures correspond so that a physical position is indicated in FIG. 2 where operating points identified along the thermodynamic path are achieved. The following description thus sometimes makes simultaneous reference to both the structural drawing of FIG. 2 and to the phase diagram of FIG. 3 in describing physical and thermodynamic aspects of the cooling flow.

For purposes of illustration, both FIGS. 2 and 3 make specific reference to a nitrogen cryogen, but this is not intended to be limiting. Embodiments of the invention may

more generally be used with any suitable cryogen such as, for example, argon, neon, helium, hydrogen, and oxygen.

In FIG. 3, the liquid-gas phase line is identified with reference label 256 and the thermodynamic path followed by the cryogen is identified with reference label 258.

A cryogenic generator 246 is used to supply the cryogen at a pressure that exceeds the critical-point pressure P_c for the cryogen at its outlet, referenced in FIGS. 2 and 3 by label ①. The cooling cycle may generally begin at any point in the phase diagram having a pressure above or slightly below P_c , although it is advantageous for the pressure to be near the critical-point pressure P_c . The cooling efficiency of the process described herein is generally greater when the initial pressure is near the critical-point pressure P_c so that at higher pressures there may be increased energy requirements to achieve the desired flow. Thus, embodiments may sometimes incorporate various higher upper boundary pressure but generally begin near the critical point, such as between 0.8 and 1.2 times P_c , and in one embodiment at about 0.85 times P_c .

As used herein, the term "near critical" is meant to refer to near the liquid-vapor critical point. Use of this term is equivalent to "near a critical point" and it is the region where the liquid-vapor system is adequately close to the critical point, where the dynamic viscosity of the fluid is close to that of a normal gas and much less than that of the liquid; yet, at the same time its density is close to that of a normal liquid state. The thermal capacity of the near critical fluid is even greater than that of its liquid phase. The combination of gas-like viscosity, liquid-like density and very large thermal capacity makes it a very efficient cooling agent. Reference to a near critical point refers to the region where the liquid-vapor system is adequately close to the critical point so that the fluctuations of the liquid and vapor phases are large enough to create a large enhancement of the heat capacity over its background value. The near critical temperature is a temperature within $\pm 10\%$ of the critical point temperature. The near critical pressure is between 0.8 and 1.2 times the critical point pressure.

Referring again to FIG. 2, the cryogen is flowed through a tube, at least part of which is surrounded by a reservoir 240 of the cryogen in a liquid state, reducing its temperature without substantially changing its pressure. In FIG. 2, reservoir is shown as liquid N_2 , with a heat exchanger 242 provided within the reservoir 240 to extract heat from the flowing cryogen. Outside the reservoir 240, thermal insulation may be provided around the tube to prevent unwanted warming of the cryogen as it is flowed from the cryogen generator 246. At point ②, after being cooled by being brought into thermal contact with the liquid cryogen, the cryogen has a lower temperature but is at substantially the initial pressure. In some instances, there may be a pressure change, as is indicated in FIG. 3 in the form of a slight pressure decrease, provided that the pressure does not drop substantially below the critical-point pressure P_c , i.e. does not drop below the determined minimum pressure. In the example shown in FIG. 3, the temperature drop as a result of flowing through the liquid cryogen is about 50°C .

The cryogen is then provided to a device for use in cryogenic applications. In the exemplary embodiment shown in FIG. 2, the cryogen is provided to an inlet 236 of a catheter 224, such as may be used in medical cryogenic endovascular applications, but this is not a requirement.

Indeed, the form of the medical device may vary widely and include without limitation: instruments, appliances, catheters, devices, tools, apparatus', and probes regardless of whether such probe is short and rigid, or long and flexible,

and regardless of whether it is intended for open, minimal, non-invasive, manual or robotic surgeries.

In embodiments, the cryogen may be introduced through a proximal portion of a catheter, continue along a flexible intermediate section of the catheter, and into the distal treatment section of the catheter. As the cryogen is transported through the catheter, and across the cryoablation treatment region **228**, between labels ② and ③ in FIGS. **2** and **3**, there may be a slight change in pressure and/or temperature of the cryogen as it moves through the interface with the device, e.g. cryoablation region **228** in FIG. **2**. Such changes may typically show a slight increase in temperature and a slight decrease in pressure. Provided the cryogen pressure remains above the determined minimum pressure (and associated conditions), slight increases in temperature do not significantly affect performance because the cryogen simply moves back towards the critical point without encountering the liquid-gas phase line **256**, thereby avoiding vapor lock.

Flow of the cryogen from the cryogen generator **246** through the catheter **224** or other device may be controlled in the illustrated embodiment with an assembly that includes a check valve **216**, a flow impedance, and/or a flow controller. The catheter **224** itself may comprise a vacuum insulation **232** (e.g., a cover or jacket) along its length and may have a cold cryoablation region **228** that is used for the cryogenic applications. Unlike a Joule-Thomson probe, where the pressure of the working cryogen changes significantly at the probe tip, these embodiments of the invention provide relatively little change in pressure throughout the apparatus. Thus, at point ④, the temperature of the cryogen has increased approximately to ambient temperature, but the pressure remains elevated. By maintaining the pressure above or near the critical-point pressure P_c as the cryogen is transported through the catheter, vapor lock are avoided.

The cryogen pressure returns to ambient pressure at point ⑤. The cryogen may then be vented through vent **204** at substantially ambient conditions.

Examples of cryoablation systems, their components, and various arrangements are described in the following commonly-assigned U.S. patents and U.S. patent applications: U.S. patent application Ser. No. 10/757,768, which issued as U.S. Pat. No. 7,410,484, on Aug. 12, 2008 entitled "CRYOTHERAPY PROBE," filed Jan. 14, 2004 by Peter J. Littrup et al.; U.S. patent application Ser. No. 10/757,769, which issued as U.S. Pat. No. 7,083,612 on Aug. 1, 2006, entitled "CRYOTHERAPY SYSTEM," filed Jan. 14, 2004 by Peter J. Littrup et al.; U.S. patent application Ser. No. 10/952,531, which issued as U.S. Pat. No. 7,273,479 on Sep. 25, 2007 entitled "METHODS AND SYSTEMS FOR CRYOGENIC COOLING," filed Sep. 27, 2004 by Peter J. Littrup et al.; U.S. patent application Ser. No. 11/447,356, which issued as U.S. Pat. No. 7,507,233 on Mar. 24, 2009 entitled "CRYOTHERAPY SYSTEM," filed Jun. 6, 2006 by Peter Littrup et al.; U.S. patent application Ser. No. 11/846,226, which issued as U.S. Pat. No. 7,921,657 on Apr. 12, 2011 entitled "METHODS AND SYSTEMS FOR CRYOGENIC COOLING," filed Aug. 28, 2007 by Peter Littrup et al.; U.S. patent application Ser. No. 12/018,403, which issued as U.S. Pat. No. 8,591,503 on Nov. 26, 2013 entitled "CRYOTHERAPY PROBE," filed Jan. 23, 2008 by Peter Littrup et al.; U.S. patent application Ser. No. 13/046,274, which issued as U.S. Pat. No. 8,387,402 on Mar. 5, 2013 entitled "METHODS AND SYSTEMS FOR CRYOGENIC COOLING," filed Mar. 11, 2011 by Peter Littrup et al.; U.S. patent application Ser. No. 14/087,947, which is pending entitled "CRYOTHERAPY PROBE," filed Nov. 22, 2013 by Peter Littrup et

al.; U.S. patent application Ser. No. 12/744,001, which issued as U.S. Pat. No. 8,740,891, on Jun. 3, 2014 entitled "FLEXIBLE MULTI-TUBULAR CRYOPROBE," filed Jul. 29, 2010 by Alexei Babkin et al.; U.S. patent application Ser. No. 12/744,033, which issued as U.S. Pat. No. 8,740,892, on Jun. 3, 2014 entitled "EXPANDABLE MULTI-TUBULAR CRYOPROBE," filed Jul. 29, 2010 by Alexei Babkin et al. and U.S. patent application Ser. No. 14/915,632 entitled "ENDOVASCULAR NEAR CRITICAL FLUID BASED CRYOABLATION CATHETER AND RELATED METHODS," filed Sep. 22, 2014 by Alexei Babkin, et al., the contents of each of the above-identified U.S. patents/applications are incorporated herein by reference in their entireties for all purposes.

A method for cooling a target tissue in which the cryogen follows a thermodynamic path similar to that shown in FIG. **3** is illustrated with the flow diagram of FIG. **4**. At block **310**, the cryogen is generated with a pressure that exceeds the critical-point pressure and is near the critical-point temperature. The temperature of the generated cryogen is lowered at block **314** through heat exchange with a substance having a lower temperature. In some instances, this may conveniently be performed by using heat exchange with an ambient-pressure liquid state of the cryogen, although the heat exchange may be performed under other conditions in different embodiments. For example, a different cryogen might be used in some embodiments, such as by providing heat exchange with liquid nitrogen when the working fluid is argon. Also, in other alternative embodiments, heat exchange may be performed with a cryogen that is at a pressure that differs from ambient pressure, such as by providing the cryogen at lower pressure to create a colder ambient.

The further cooled cryogen is provided at block **318** to a cryogenic-application device, which may be used for a cooling application at block **322**. The cooling application may comprise chilling and/or freezing, depending on whether an object is frozen with the cooling application. The temperature of the cryogen is increased as a result of the cryogen application, and the heated cryogen is flowed to a control console at block **326**. While there may be some variation, the cryogen pressure is generally maintained greater than the critical-point pressure throughout blocks **310-326**; the principal change in thermodynamic properties of the cryogen at these stages is its temperature. At block **330**, the pressure of the heated cryogen is then allowed to drop to ambient pressure so that the cryogen may be vented, or recycled, at block **334**. In other embodiments, the remaining pressurized cryogen at block **326** may also return along a path to block **310** to recycle rather than vent the cryogen at ambient pressure.

Cryoablation Catheters

Embodiments of the cryoablation apparatus of the present invention may have a wide variety of configurations. For example, one embodiment of the present invention is a flexible catheter **400** as shown in FIG. **5A**. The catheter **400** includes a proximally disposed housing or connector **410** adapted to fluidly connect to a fluid source (not shown).

A plurality of fluid transfer tubes **420** are shown extending from the connector **410**. These tubes include a set of inlet fluid transfer tubes **422** for receiving the inlet flow from the connector and a set of outlet fluid transfer tubes **424** for discharging flow from the connector **410**.

In embodiments each of the fluid transfer tubes is formed of material that maintains flexibility in a full range of temperatures from -200° C. to ambient temperature. In embodiments, the fluid transfer tubes **420** are formed of

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annealed stainless steel or a polymer such as polyimide. In such configurations, the material may maintain flexibility at near critical temperature. In embodiments, each fluid transfer tube has an inside diameter in a range of between about 0.1 mm and 1 mm (preferably between about 0.2 mm and 0.5 mm). Each fluid transfer tube may have a wall thickness in a range of between about 0.01 mm and 0.3 mm (preferably between about 0.02 mm and 0.1 mm).

An end cap **440** is positioned at the ends of the fluid transfer tubes to provide fluid transfer from the inlet fluid transfer tubes to the outlet fluid transfer tubes. The endcap **440** is shown having an atraumatic tip. The endcap **440** may be any suitable element for providing fluid transfer from the inlet fluid transfer tubes to the outlet fluid transfer tubes. For example, endcap **440** may define an internal chamber, cavity, or passage serving to fluidly connect tubes **422, 424**.

With reference to FIG. 5B, an outer sheath **430** is shown surrounding the tube bundle **420**. The outer sheath serves to hold the tubes in a tubular arrangement, and protect the construct from being penetrated or disrupted by foreign objects and obstacles.

A temperature sensor **432** is shown on the surface of the distal section. Temperature sensor may be a thermocouple to sense a temperature corresponding to the adjacent tissue, and sends the signal back through a wire in the tube bundle to the console for processing. Temperature sensor may be placed elsewhere along the shaft or within one or more of the fluid transport tubes to determine a temperature difference between inflow and outflow.

There are many configurations for tube arrangements. In embodiments the fluid transfer tubes are formed of a circular array, wherein the set of inlet fluid transfer tubes comprises at least one inlet fluid transfer tube **422** defining a central region of a circle and wherein the set of outlet fluid transfer tubes **424** comprises a plurality of outlet fluid transfer tubes spaced about the central region in a circular pattern. In the configuration shown in FIG. 5B, the fluid transfer tubes **422, 424** fall within this class of embodiments.

During operation, the cryogen/cryogenic fluid arrives at the catheter through a supply line from a suitable cryogen source at a temperature close to -200°C . The cryogen is circulated through the multi-tubular freezing zone provided by the exposed fluid transfer tubes, and returns to the connector. Cryogen flows into the freeze zone through the inlet fluid transfer tube **422** and flows out of the freeze zone through the outlet fluid transfer tubes **424**.

In embodiments, the nitrogen flow does not form gaseous bubbles inside the small diameter tubes under any heat load, so as not to create a vapor lock that limits the flow and the cooling power. By operating at the near critical condition for at least an initial period of energy application, the vapor lock is eliminated as the distinction between the liquid and gaseous phases disappears. After initially operating under near critical conditions, e.g., for nitrogen, at a temperature near the critical temperature of -147.15°C . and a pressure near the critical pressure of 3.396 MPa, the operating pressure may be decreased as is disclosed and described in commonly assigned U.S. patent application Ser. No. 14/919,681 entitled "PRESSURE MODULATED CRYOABLATION SYSTEM AND RELATED METHODS," filed Oct. 21, 2015 by Alexei Babkin, the contents of which are incorporated herein by reference in their entirety for all purposes.

A multi-tube design may be preferably to a single-tube design because the additional tubes can provide a substantial increase in the heat exchange area between the cryogen and tissue. Depending on the number of tubes used, cryo-

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instruments can increase the contact area several times over previous designs having similarly sized diameters with single shafts/tubes. However, embodiments of the invention are not intended to be limited to a single or multi-tubular design except where specifically recited in the appended claims.

Cryoablation Console

FIG. 6 illustrates a cryoablation system **950** having a cart or console **960** and a cryoablation catheter **900** detachably connected to the console via a flexible elongate tube **910**. The cryoablation catheter **900**, which shall be described in more detail below in connection with FIG. 7, contains one or more fluid transport tubes to remove heat from the tissue.

The console **960** may include or house a variety of components (not shown) such as, for example, a generator, controller, tank, valve, pump, etc. A computer **970** and display **980** are shown in FIG. 6 positioned on top of cart for convenient user operation. Computer may include a controller, timer, or communicate with an external controller to drive components of the cryoablation systems such as a pump, valve or generator. Input devices such as a mouse **972** and a keyboard **974** may be provided to allow the user to input data and control the cryoablation devices.

In embodiments computer **970** is configured or programmed to control cryogen flowrate, pressure, and temperatures as described herein. Target values and real time measurement may be sent to, and shown, on the display **980**.

FIG. 7 shows an enlarged view of distal section of cryoablation apparatus **900**. The distal section **900** is similar to designs described above except that treatment region **914** includes a flexible protective cover **924**. The cover serves to contain leaks of the cryogen in the event one of the fluid transport tubes is breached. Although a leak is not expected or anticipated in any of the fluid delivery transport tubes, the protective cover provides an extra or redundant barrier that the cryogen would have to penetrate in order to escape the catheter during a procedure. In embodiments the protective cover may be formed of metal.

Additionally, a thermally conducting liquid may be disposed within spaces or gaps between the transport tubes and the inner surface of the cover to enhance the device's thermal cooling efficiency during treatment. In embodiments the thermally conductive liquid is water.

Cover **924** is shown being tubular or cylindrically shaped and terminates at distal tip **912**. As described herein, the cooling region **914** contains a plurality of fluid delivery and fluid return tubes to transport a cooling fluid through the treatment region **914** causing heat to be transferred/removed from the target tissue. In embodiments, the cryogen is transported through the tube bundle under physical conditions near the fluid's critical point in the phase diagram. The cover serves to, amongst other things, contain the cooling fluid and prevent it from escaping from the catheter in the event a leak forms in one of the delivery tubes.

Although a cover is shown in FIGS. 6-7, the invention is not intended to be so limited except as where recited in the appended claims. The apparatus may be provided with or without a protective cover and used to cool a target tissue. Tube within Tube

FIG. 8 shows a partial view of a cryoablation catheter **1010** according to another embodiment of the invention having a protective means to mitigate leaks in the event a cooling fluid/cryogen escapes from the cryogen delivery tubes described above. In particular, catheter **1010** comprises a plurality or bundle **1012** of flexible multi-layer cryoenergy transfer tubes, each of which comprises two tubes in a coaxial arrangement, namely a tube within a tube.

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FIG. 9A shows a cross-sectional view taken along line 9A-9A of FIG. 8. The bundle 1012 of multilayer tubes is shown with the fluid delivery tubes 1014 and the fluid return tubes 1015 assembled in a parallel arrangement. The tube bundle 1012 is shown having 12 tubes/lines including four (4) fluid return tubes 1015a-1015d and eight (8) fluid delivery tubes 1014a-1014h. The fluid delivery tubes 1014a-1014h form a perimeter around the fluid return tubes 1015a-1015d. This arrangement ensures that colder delivery fluid/cryogen is adjacent to the tissue to be ablated/frozen and warmer return fluid/cryogen is shielded from the tissue to be ablated/frozen.

FIG. 9B shows an enlarged cross-sectional view of fluid delivery tube 1014d of FIG. 9A. The first or inner tube 1013 is shown coaxially surrounded by a second or outer tube 1018. A space or gap 1020 between the exterior surface of the inner tube 1013 and the interior surface of the outer tube 1018 is capable of being filled with a thermally conductive media 1021 as described herein. In embodiments, the gap 1020 has an annular shape. All of the fluid delivery tubes 1014 as well as the fluid return tubes 1015 can have a similar tube within a tube construction.

In the event of a leak of the cooling fluid 1016 or breach of the inner tube 1013 during use, the cooling fluid 1016 is contained within the gap 1020 between the inner tube 1013 and the outer tube 1018. This tube within a tube feature adds an additional safety element to the device as any leaking fluid/cryogen 1016 is contained within the catheter and is prevented from entering the patient. In some embodiments, a pressure sensor/device or gauge may be incorporated to monitor the pressure of the thermally conductive media 1021 in the gap 1020. Therefore, if fluid/cryogen 1016 breaches the inner tube 1013 and leaks into the gap 1020, the pressure in the gap 1020 and hence, the conductive media 1021 will increase. Should a change in pressure occur above a threshold limit, the system can be programmed to halt ablation thereby preventing potential harm to a patient and/or notify the user/physician of this change in pressure.

The inner tube 1013 may be fabricated and made from materials as described herein in connection with other flexible tubes for transporting the cooling fluid.

The outer tube 1018 material should also be flexible to enable elastic deflection of the distal treatment section to allow the distal treatment section to transform its shape as disclosed herein. In some embodiments, the outer tube is not inflatable, distensible nor expandable such that its size and shape remains substantially unaffected by the presence of the thermally conductive media 1021 contained therein. Non-limiting exemplary materials for the outer tube 1018 include polymers and metals or alloys. An example of an outer tube 1018 material is Nitinol or polyimide.

The number of tubes forming the tubular bundle 1012 may vary widely. In some embodiments, the tubular bundle 1012 includes 5-15 tubes, and more preferably, includes between 8-12 tubes comprising fluid delivery tubes 1014 and fluid return tubes 1015.

The cross-sectional profile of the tube bundle 1012 may also vary. Although FIG. 9A shows a substantially circular profile, in embodiments, the profile may be rectangular, square, cross or t-shaped, annular or circumferential, or another shape profile, including some of the arrangements described above. The tubes may also be braided, woven, twisted, or otherwise intertwined together, as depicted in FIGS. 9, 14 and 16 of commonly assigned U.S. patent application Ser. No. 14/915,632 entitled "ENDOVASCULAR NEAR CRITICAL FLUID BASED CRYOABLATION CATHETER AND RELATED METHODS," filed

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Sep. 22, 2014 by *Alexei Babkin, et al.*, the entire contents of which are incorporated herein by reference for all purposes.

The diameter of the freezing section or tubular bundle may vary. In embodiments, the diameter of the bundle ranges from about 1-3 mm, and is preferably about 2 mm.

FIG. 9C shows a cross-section of a cryoablation catheter having another tubular arrangement 1017. The eight (8) tubular elements (1019a-1019d and 1023a-1023d) are spaced or distributed circumferentially about a core element 1025. Preferably, as shown, fluid delivery elements/tubes (1019a-1019d) and fluid return elements/tubes (1023a-1023d) alternate along the circumference of the catheter.

Each inner tubular element (e.g., 1019a) includes an outer tubular element (e.g., 1027a) coaxially surrounding the inner tubular element thereby creating a space or gap which can be filled with a thermally conductive media/fluid as described with respect to FIG. 9B.

Steering elements, sensors and other functional elements may be incorporated into the catheter. In embodiments, steering elements are incorporated into a mechanical core such as the mechanical core 1025 shown in FIG. 9C.

FIG. 10A shows an enlarged cut-away view of the catheter at detail 10A in FIG. 8, illustrating tube bundle 1012 fluidly connected to the end portion 1040 of an intermediate section of the catheter 1010.

FIG. 10B shows an exploded view of a proximal section of the tube bundle 1012 and the intermediate section of catheter 1040. Tube bundle 1012, having inner tubular elements 1013a-1013d extending beyond outer tubular elements/covers 1018a-1018d of fluid delivery lines 1014, can be inserted into intermediate section of catheter 1040.

With reference to FIGS. 10A-10B, fluid delivery lines 1014 are shown bundled together and inserted/joined to main line 1032. An adhesive plug 1042 or seal, gasket, or stopper, etc. may be applied to facilitate and ensure a fluid seal between the tube members. The cooling power fluid (CPF) is transported to the fluid delivery lines 1014 from the fluid delivery main line 1032.

The proximal ends of outer tubular elements/covers 1018a-d, which are offset from proximal ends of inner tubular elements 1013a-d, are shown inserted into intermediate section 1040 of catheter such that the thermally conductive fluid (TCF) within lumen 1050 can fill gaps 1020 (FIG. 9B) of each of the multi-layer cryoenergy tubular elements. An adhesive plug 1044 (weld or bond) may be applied to facilitate a fluid tight and robust connection. Press fits, heat, and other fabrication techniques can be applied to join components as is known to those of skill in the art.

FIG. 11 shows another cryoablation catheter 500 including a distal treatment section 510, a handle 520, and an umbilical cord 530. The proximal end of the umbilical cord 530 terminates in connector 540, which is inserted into receptacle port 560 on console 550.

One or more ancillary connector lines 570 are shown extending proximally from the handle 520. The tubular lines 570 may serve to provide various functionality including without limitation (a) flushing; (b) vacuum; (c) thermally conductive liquid described above; and/or (d) temperature and pressure sensor conductors.

The catheter 500 is also shown having electrical connector 580 extending proximally from the handle 520. Electrical connector 580 may be coupled to an EP recording system for analyzing electrical information detected in the distal treatment section 510. Examples of systems for analyzing the electrical activity include, without limitation, the GE Healthcare CardioLab II EP Recording System, manufactured by GE Healthcare, USA and the LabSystem PRO EP Recording

System manufactured by Boston Scientific Inc. (Marlborough, MA). The recorded electrical activity may also be used to evaluate or verify the continuous contact with the target tissue as described in commonly assigned International Patent Application No. PCT/US16/51954, entitled “TISSUE CONTACT VERIFICATION SYSTEM”, filed Sep. 15, 2016 by Alexei Babkin, et al., the entire contents of which are incorporated herein by reference for all purposes.

FIG. 12 shows an enlarged view of a portion of the distal section 510 of the catheter 500. Ring-shaped electrodes 602, 604 are circumferentially disposed about shaft 606. Although two electrodes are shown, more or less electrodes may be present on the shaft for sensing electrical activity. In embodiments, up to 12 electrodes are provided on the shaft. In one embodiment, 8 electrodes are axially spaced along the shaft 606.

FIG. 13 is a cross section of the catheter shown in FIG. 12 taken along line 13-13. The catheter shaft is shown having a mechanical core 620 extending along the central axis, and a plurality of energy delivering tube constructs 630 extending parallel and circumferentially disposed about the mechanical core.

Each tube construct 630 is shown having dual layers as described above in connection with FIGS. 8-9 and a thermally conductive liquid layer disposed there between.

A tubular line 624 is shown for housing conducting wires 626 for the various sensors described herein.

The mechanical core 620 may be constructed to provide a preset shape to the catheter distal treatment section. With reference to FIG. 13, the mechanical core includes a metal tubular member 622 having a preset shape. The preset shape matches the target anatomy to make continuous contact with the target anatomy. An exemplary material for the preset tubular element 622 is Nitinol. FIG. 13 also shows an exterior layer or cover concentrically surrounding the Nitinol tube. The exterior cover may be a flexible polymer such as, for example, PET. Collectively, the inner PET layer 620 and outer shaft layer 606 form a fluidly-sealed annular chamber to house the plurality of tubular constructs 630.

With reference to FIGS. 14-15, a catheter 608 is shown being deployed from an outer sheath 642. Initially, catheter distal section 606 is disposed within a lumen of external sheath 642, and prohibited from assuming its preset shape. The distal section 606 and external sheath 642 are moved axially relative to one another. For example, the catheter may be ejected from the sheath. Once the catheter is free from constraint, it assumes the preset shape as shown in FIG. 15.

Mechanical core assembly biases the shape of the catheter distal section 608, forcing the energy delivering elements into a curvilinear shape. In embodiments, the catheter shape is adapted to create lesions in the right atrium useful in treating atrial flutter. The shape shown in FIG. 15, for example, is a single loop or elliptical shape which has curvature to match target zones of tissue in the right atrium useful in treating atrial flutter. Additional apparatus and methods for treating atrial flutter are described in commonly assigned U.S. Patent Application No. 61/981,110, filed Apr. 17, 2014, now International Patent Application No. PCT/US2015/024778, filed Oct. 21, 2015 entitled “ENDOVASCULAR NEAR CRITICAL FLUID BASED CRYOABLATION CATHETER HAVING PLURALITY OF PREFORMED TREATMENT SHAPES,” the contents of both of which are incorporated herein by reference in their entireties for all purposes.

FIG. 16 shows another cryoablation catheter 700 including a distal treatment section 710, a handle 720, and an

umbilical cord 730 which terminates in connector 740. Similar to the system described above in connection with FIG. 11, connector 740 may be inserted into a receptacle port on a console.

Additional lines 742, 744 are shown extending proximally from handle. Lines 742, 744 provide various functionalities to the distal treatment section 710 during a procedure. Example functionalities include, without limitation, temperature, EP recording, pressure, fluid flush, source liquids, etc.

FIG. 17 is an enlarged view of the catheter distal section following deployment. The treatment section is shown having a generally looped or elliptical shape 714. An intermediate section 716 is shown providing a bend or articulation from central axis 718. Such functionality aids in positioning the treatment section in continuous direct contact with the tissue. In embodiments, the shape is configured to create complete PVI in the left atrium.

FIG. 18 is an enlarged cross-sectional view of a portion of the distal treatment section. The catheter shaft is shown having a mechanical core 750 extending along the central axis, and a plurality of energy delivering tube constructs 752 extending parallel and circumferentially about the mechanical core. One or more spare tubular elements 754, 758 can be incorporated into the perimeter space in combination with energy delivery elements. Tubular element 754 holds a plurality of electrical conductors to transmit electrical activity from sensors or ring electrodes 756 present on the distal treatment section. Tubular element 758 may provide vacuum or liquid to the catheter for various functions described herein.

Mechanical core 750 is shown extending axially through the treatment section and comprising a plurality of members 760, 762 which extend through the distal treatment section to bias the distal section into a preset shape such as the loop shape shown in FIG. 17. In particular, in embodiments, the mechanical core can include a biased shape element 760 such as a Nitinol wire, and an axially movable control member 762 connected to a distal tip of the treatment section to adjust the curvature of the preset shape. Core may include additional lumens 766, 768 if desired. The mechanical core acts to shape the distal treatment section to a first preset loop shape, and can be further adjusted by the control member to make continuous contact with a target tissue surface.

FIGS. 19A-19D illustrate sequentially deployment of an ablation catheter 810 from a first arcuate shape having a slight bend to a second configuration having a complete ring or circular shape 820. The shape is assumed once the catheter treatment section is not constrained by the outer sheath 812.

FIGS. 20A-20B show an enlarged view of the catheter 800 of FIG. 19D except that the loop has been adjusted by reducing its diameter ϕ_1 . As described herein, a control member extending through the shaft of the distal treatment section is pulled to reduce the diameter of the preset loop ϕ_1 to diameter ϕ_2 as shown in FIG. 20A. FIG. 20B shows the loop adjusted to an even smaller diameter ϕ_3 than that shown in FIG. 20A.

The diameter ϕ of the loop may vary. In embodiments, the diameter of the loop is controlled to range from 2 cm to 5 cm, and in embodiments, preferably about 2-3 cm.

FIGS. 21A-21C show sequentially articulation of the intermediate section 814 of the catheter. The intermediate section 814 is shown having an outer support or reinforcing structure 816. In embodiments, the support layer 816 is a spring or coil.

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FIG. 21A shows catheter intermediate section **814** substantially straight or aligned with the shaft axis.

FIG. 21B shows catheter intermediate section having a slight articulation forming angle θ_1 with shaft axis.

FIG. 21C shows catheter intermediate section having further articulation θ_2 with shaft axis. The degree of articulation may vary and be adjusted by the physician as described below. In embodiments, the degree of articulation is up to 120 degrees from the central shaft axis, and more preferably up to about 90 degrees.

FIGS. 22A-22B show examples of components/structures for articulating the intermediate section. The components include a coil **832**, second pull wire **834**, and spine **836**. The pull wire **834** is fixed to a distal location of the intermediate section. Pulling on the pull wire results in deflecting or articulating the coil **832**. Spine **836** is shown diametrically opposite the pull wire. The spine serves to bias the direction that the catheter bends when the pull wire is retracted and serves to return the catheter to its straightened position when the pull wire is released. In particular, when the pull wire is retracted, the catheter bends towards the pull wire along a plane including the pull wire, central coil axis, and the spine.

The various articulating components/structures may be made of a wide variety of materials. Exemplary materials include without limitation Nitinol, stainless steel, or other materials having the functionality described herein. Additionally, the components may be fabricated from wire, tubular elements, or sheets of stock material. In one embodiment, the coil and spring are integrally formed from a sheet of metal alloy. The desired shape may be machined or laser cut to create the spine and rib elements, allowing for biased articulation. See also US Patent Publication No. 2003/0195605, filed May 30, 2003, entitled "Cryogenic Catheter with Deflectable Tip" to Kovalchek et al. for further details describing catheters comprising a spring, pull wire and spine for controlling deflection.

FIG. 23A shows a perspective view of a handle **852** of an ablation catheter. A flexible catheter shaft **854** extends from a distal section **856** of the handle. Umbilical cord **858** and various other functional lines and connectors **859** are shown extending proximally from a proximal section **860** of handle.

Handle **852** is shown having an ergonomic design including a smooth gently curved intermediate section **862** that allows a user to conveniently hold the handle.

Handle is shown comprising a knob **864** which may be rotated relative to the handle body to control the diameter of the deployed loop as described above. An axially movable hub **866** is shown proximal to the knob. Movement of the hub **866** forward or backwards serves to adjust or articulate the deployed shaft as described above. Additionally, handle may be rotated as a whole to steer the catheter in one direction or another. Collectively, the handle provides a convenient and semi automatic apparatus to turn, articulate, and control the diameter or size of the deployed structure.

FIG. 23B shows a partial perspective view of the handle shown in FIG. 23A with the exterior removed for clarity. A segment of an external thread or teeth **872** are shown. The teeth **872** mate with grooves or thread in the knob **864**. The teeth are linked to a first control member described above for changing the shape or diameter of the loop. As the knob is rotated, the pull wire is moved simultaneously.

Slider **874** is also shown in handle. Slider **874** is joined to hub **866** such that movement of the hub causes the slider to move. Slider is also linked to a second control member as

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described above for articulating the catheter shaft. When the exterior hub is moved by the physician, the second control member articulates the shaft.

Although the handle is shown having a knob, hub, and slider, the invention is not intended to be so limited. The invention can include other levers, gears, buttons, and means for causing the above described functionality.

Depicted in FIG. 24 is an ablation catheter **880** according to another embodiment of the invention. In this embodiment, the ablation catheter **880** comprises two main components—(a) an ablation shaft/sleeve **881** for delivering ablation energy to a site of interest within the human body and (b) a stylet **882** that is capable of being inserted into an internal hollow cavity within the ablation shaft/sleeve **881**. As will be discussed in more detail below, at least a portion of the ablation shaft/sleeve **881** is made of a flexible material such that this portion of the ablation shaft/sleeve **881** can assume a shape of the stylet **882** that is inserted therein and that is constructed from a shape memory alloy. While the ablation catheter **880** will be described herein for use as a cryoablation catheter that creates lesions by freezing tissue with any suitable cryogen (for example, and not limited to, nitrogen, argon, neon, helium, hydrogen, and oxygen), in other embodiments, the ablation catheter can be used with other ablation energies such as, for example, radiofrequency, microwave, laser, and high frequency ultrasound (HIFU).

As depicted in FIG. 24, the ablation shaft/sleeve **881** includes a handle portion (not shown and which may be constructed in accordance with any of the handle embodiments disclosed herein), a first shaft portion **883**, a flexible shaft portion **884**, a flexible distal ablation portion **885** and a distal ablation tip **886**. In some embodiments, the ablation catheter **880** may also include a plurality of electrodes **887** on the flexible distal ablation portion **885** that may be used to detect electrical activity in the target tissue in order to evaluate or verify continuous contact of the flexible distal ablation portion **885** with the target tissue as described in commonly assigned International Patent Application No. PCT/US16/51954, entitled "TISSUE CONTACT VERIFICATION SYSTEM", filed Sep. 15, 2016 by Alexei Babkin, et al., the entire contents of which are incorporated herein by reference for all purposes. In some embodiments, electrodes **887** may be included on the distal ablation tip **886**. In some embodiments, the first shaft portion **883** may be flexible, semi-flexible, semi-rigid or rigid. In some embodiments, the first shaft portion **883** is less flexible than the flexible shaft portion **884**, however, the first shaft portion **883** will still be flexible such that it can be delivered through the venous system of the body to the target tissue.

In some embodiments, the ablation shaft/sleeve **881** may comprise a handle portion, a flexible shaft portion **884**, a flexible distal ablation portion **885** and a distal ablation tip **886**. That is, the ablation shaft/sleeve **881** may be flexible along its entire length,

FIG. 25A depicts a cross-sectional view of the ablation catheter **881** taken along line 24A-24A in FIG. 24 with the stylet **882** not being inserted into the ablation shaft/sleeve **881**. As can be seen in the cross-sectional view, the ablation shaft/sleeve **881** includes a plurality of multilayer cryogen delivery tubes/lumens **888** for transporting the cryogen to the flexible distal ablation portion **885** and a plurality of multilayer cryogen return tubes/lumens **889** for transporting the cryogen away from the flexible distal ablation portion **885**. Also shown are a plurality of service tubes/lumens **891** that may include catheter control wires, electrode wires **892**, or any other elements that may be desired. The plurality of multilayer cryogen delivery tubes/lumens **888**, the plurality

of multilayer cryogen return tubes/lumens **889** and the plurality of service tubes/lumens **891** are arranged in a circular array around a hollow tube/lumen **890** that is adapted to receive the stylet **882** therein. The hollow tube/lumen **890** extends along the length of the ablation shaft/sleeve **881** from the handle to at least the flexible distal ablation portion **885**.

While FIG. 25A depicts four (4) multilayer cryogen delivery tubes **888**, four (4) multilayer cryogen return tubes **889** and four (4) service tubes/lumens **891**, the embodiments of the invention are not intended to be so limited and may include any number of multilayer cryogen delivery tubes **888**, multilayer cryogen return tubes **889** and service tubes/lumens **891** depending on the desired ablating power of the catheter or the condition that the catheter will be used to treat. Additionally, while FIG. 25A depicts a certain configuration of the multilayer cryogen delivery tubes **888**, the multilayer cryogen return tubes **889** and the service tubes/lumens **891**, specifically that pairs of multilayer cryogen delivery tubes **888** and multilayer cryogen return tubes **889** are located adjacent to one another and separated with a service tubes/lumens **891**, the embodiments of the invention are not intended to be so limited and may include any number of different configurations for the multilayer cryogen delivery tubes **888**, the multilayer cryogen return tubes **889** and the service channels/tubes **891**.

Air Gap Elimination

In some embodiments, the annular space **813** defined between the inner surface **823** of the outer sheath **821** and outer surface **822** of the hollow tube/lumen **890** (see FIG. 25A) is filled with a thermally conductive liquid (not shown), eliminating air gaps/bubbles between the thermal elements **888**, **889** and the outer sheath **821**. Air gaps between the outer sheath and the cryogen delivery and return tubes/thermal elements **888**, **889** are undesirable because such gaps decrease thermal conduction from the thermal elements to the target tissue.

An example of a suitable thermally conductive liquid is water. The thermally conductive liquid may be delivered to the space **813** through a working line such as lines **570** and **742** described above with reference to FIG. 11 and FIG. 16, respectively. Flushing the space with the thermally conducting water is performed until all air is removed from the space **813**.

Additionally, although water is described as a suitable thermally conductive media for flushing the space **813**, as described herein, embodiments of the invention include use of other thermally conductive materials to eliminate the air gaps and increase the thermal conducting properties within the ablation portion of the catheter.

With reference to FIG. 25B, a cross section of another embodiment of catheter **881** is shown. Particularly, and unlike the cross section shown in FIG. 25A, the space **813** is shown in FIG. 25B filled with a solid thermally conductive material or liner **833** that comprises a base material and a filler having a high thermal conductivity. Exemplary base materials for the thermally conductive liner **833** include, and are not limited to, thermoplastic elastomers (TPE) or thermoplastic urethanes (TPU) adapted to conduct thermal energy. Exemplary filler materials include, and are not limited to, aluminum, copper, silver and gold.

In some embodiments, the filler can be a ceramic material that has good thermal conductivity and preferably is electrically insulating in order to isolate any electrodes that are included on the catheters ablation portion as discussed herein. Exemplary ceramic filler materials include, and are not limited to, BN (boron nitride), AlN (aluminum nitride),

Si₃N₄ (silicon nitride), SiC (silicon carbide), Al₂O₃ (aluminum oxide) and ZnO (zinc oxide)

As will be readily understood by those of skill in the art, any material having a high thermal conductivity can be used as the filler material.

The TPEs or TPUs (base materials) are loaded with a thermally conductive agent or filler to increase the base material's thermal conductivity. An example of a TPE is polyether block amide (PEBA), which is also known under the tradename PEBAX® manufactured by Arkema (France). An example of a TPU is PELLETHANE® manufactured by Lubrizol (Wickliffe, Ohio). An example of a conductive agent or filler is aluminum oxide. Preferably, the conductive filler or agent increases or substantially increases the thermal conductivity of the compounded material, described further below.

In particular embodiments, the material is PEBA loaded with about 10-70% aluminum oxide (Al₂O₃) by weight, and in a particularly preferred embodiments, the material is PEBAX® 35D loaded with about 65-75% Al₂O₃ by weight.

The base material (e.g., TPE or TPU) can be loaded with the thermally conductive filler by compounding the base material with filler into a pellet form material, and then extruding the loaded material into the desired tubing suitable for lining the thermal elements **888**, **889** described herein.

In embodiments, the base material is loaded such that the thermal conductivity (K) of the liner **833** is at least three to five times greater, and more preferably at least five times greater, than the base material without being modified with the thermal conductive filler. In embodiments, the thermal conductivity (K) of the liner **833** is at least 1 W/m-K at 23° C., and ranges from 0.5 W/m-K to 3 W/m-K.

The thermally conductive liner **833** can be incorporated or otherwise assembled into the catheter in various ways. In some embodiments, the liner **833** is heat fused over the thermal energy elements **888**, **889** and service tubes **891**. During the heat fusing step, the liner melts and flows around the thermal elements **888**, **889** and service tubes **891** until the air gaps between the thermal elements **888**, **889**, and service tubes **891** are completely filled and eliminated. In some embodiments, an outer sheath is subsequently applied over the thermal elements and liner leaving no gaps. In some embodiments, an outer sheath is not necessary when a conductive liner **833** is heat fused over thermal energy elements **888**, **889**. In these embodiments, the liner acts as the sheath. Adding a sheath may only increase stiffness of the device or catheter. Also, if the sheath is not made of a conductive material, the sheath may act as a thermal barrier.

The presence of the thermal conductive liner **833** around the thermal elements **888**, **889** has a number of additional benefits including simplifying the medical procedure and saving time because the physician is no longer required to flush the catheter space **813** with a liquid to eliminate the air gaps.

Additionally, the thermally conductive liner **833** makes the ablation portion of the catheter kink resistant. Absent the thermally conductive liner **833** between the outer sheath **821** and the thermal elements **888**, **889**, the thermal elements are not supported except where the thermal elements contact the outer sheath **821**, which generally occurs at the apex of the thermal elements. On the other hand, when the space **813** is filled with the thermally conductive liner **833** described herein, the presence and properties of the liner mechanically support the thermal elements **888**, **889** and service tubes **891** and provide greater kink resistance.

With reference to FIG. 25C, a cross section of another embodiment of the catheter **881** is shown. Particularly, and

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unlike the cross section shown in FIG. 25B, catheter 881 in FIG. 25C includes a thermally conductive liner 843 disposed only on the outer side/perimeter of the thermal elements 888, 889 and service tubes 892. Consequently, an air-filled space 853 is present between the working lumen 890 and the inner thermal elements 888, 889 and service tubes 892. This space acts as a thermal barrier or insulator because air has a low thermal conductivity. However, the thermally insulating space 853 is not disadvantageous because there is no desire to conduct thermal energy to the hollow tube 890. Additionally, leaving the space 853 open/free of the conductive liner 833 allows for more freedom of movement of the thermal elements 888, 889. Thus, the thermal elements 888, 889 are not bound/connected to the hollow tube/lumen 890. This greatly increases the flexibility of the ablation catheter and allows one to achieve a tighter bend radius and more complex shapes.

Additionally, the thermoplastic liners described herein can be adapted to bond various components of the catheter together and withstand the temperature variations during operation. The liners can be adapted to have a relatively low linear coefficient of thermal expansion (LCTE) by loading the liners with certain low LCTE agents such as, for example, quartz. In embodiments, PEBAX is loaded with quartz to about 60% by weight. The enhanced liner, which may be in the form of a tube, can be used to heat fuse/bond various components of the catheter together. Preferably the linear coefficient of thermal expansion is reduced to 0.5/K or less. Examples of components to be joined include mating thermal elements, liquid tubes such as the liquid main to the liquid sub-main or branches, and other lumens for delivering a liquid and that are susceptible to thermal expansion arising from temperature changes.

The low LCTE enhanced bonds described herein do not require use of an epoxy adhesive. Thus, the low LCTE enhanced bonds do not require an adhesive cure cycle, making the low LCTE enhanced bonds faster than the epoxy adhesive counterpart. Indeed, the type and configuration of the thermally conductive media, liner or tubing may vary widely from various liquids to thermally conducting solids, and from partially filling the space 813 to entirely filling the space.

Although, the conductive liner 833 has been described with respect to the ablation catheter embodiments depicted in FIGS. 24 and 25, the conductive liner 833 can be used in any of the ablation catheter embodiments disclosed and described herein. Additionally, it will be readily understood that the conductive liner 833 embodiments described and disclosed herein are not limited to uses with ablation catheters but can be used in any application to increase the conductivity of an item/element.

As will be readily understood by those of skill in the art, the disclosed and described embodiments of the conductive liner 833 can be used in any application to increase thermal conductivity whether it be to increase cooling/freezing or warming/heating. Thus, the disclosed and described embodiments of the conductive liner 833 can be used with many different ablation technologies including, and not limited to, cryoablation, radiofrequency, microwave, laser, and high frequency ultrasound (HIFU).

FIG. 26 shows an enlarged cross-sectional view of the multilayer cryogen delivery tubes 888 and multilayer cryogen return tubes 889 of FIG. 25A. The first or inner tube 893 is shown coaxially surrounded by a second or outer tube 894. The lumen 895 of the inner tube 893 is designed to receive the flow of cryogen. The inner tube 893 and outer tube 894 are arranged such that a space or gap 896 is created

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between the exterior surface of the inner tube 893 and the interior surface of the outer tube 894. This gap 896 is capable of being filled with a thermally conductive media 897 as described herein. In some embodiments, the gap 896 has an annular shape. All of the multilayer cryogen delivery tubes 888 as well as the multilayer cryogen return tubes 889 can have a similar tube within a tube construction.

In the event of a leak of the cryogen flowing through lumen 895 or breach of the inner tube 893 during use, the leaking cryogen is contained within the gap 896 between the inner tube 893 and the outer tube 894. This tube within a tube construction adds an additional safety element to the device as any leaking fluid/cryogen is contained within the catheter and is prevented from entering the patient. In some embodiments, a pressure sensor/device or gauge may be incorporated to monitor the pressure of the thermally conductive media 897 in the gap 896. Therefore, if fluid/cryogen breaches the inner tube 893 and leaks into the gap 896, the pressure in the gap 896 and hence, the pressure of the conductive media 897 will increase. Should a change in pressure occur above a threshold limit, the system can be programmed to (a) halt ablation thereby preventing potential harm to a patient and/or (b) notify the surgeon of this change in pressure.

The inner tubes 893 may be fabricated and made from materials as described herein in connection with other flexible tubes for transporting the cryogen/cooling fluid. The outer tubes 895 may also be manufactured from a flexible material to enable elastic deflection of the flexible shaft portion 884 and the flexible distal ablation portion 885 of the ablation shaft/sleeve 881 to allow these portions to transform their shapes to assume the shape of the stylet 882 as disclosed herein. In some embodiments, the outer tube 895 is not inflatable, distensible nor expandable such that its size and shape remains substantially unaffected by the presence of the thermally conductive media 897 contained therein. Non-limiting exemplary materials for the outer tube 895 include polymers and metals or alloys. An example of an outer tube 894 material is polyimide.

The diameter of the flexible distal ablation portion 885 may vary. In some embodiments, the diameter of the flexible distal ablation portion 885 ranges from about 1-3 mm, and is preferably about 2 mm.

FIG. 27A and FIG. 27B depict an embodiment of the ablation catheter 880 with the stylet 882 fully inserted into the ablation shaft/sleeve 881.

FIG. 28A shows a cross-sectional view of the ablation catheter 880 of FIG. 27 taken along line 27A-27A in FIG. 27A. As can be seen in FIG. 28A, the stylet 882 is inserted into the hollow tube/lumen 890 of the ablation shaft/sleeve 881. As previously disclosed, the stylet 882 is made from a shape memory alloy such as, for example, nickel titanium (Nitinol).

FIGS. 28B-28C are identical cross sectional views to the cross sectional views shown in FIGS. 25B-25C respectively except that FIGS. 28B-28C show stylet 882 inserted into the hollow tube/lumen 890 of the ablation shaft 881.

Depicted in FIG. 29 are sample shapes that can be pre-set into the distal portion 898 of the stylet 882. In some embodiments, the length of the distal portion 898 corresponds to at least a portion of the length of the flexible distal ablation portion 885 of the ablation shaft/sleeve 881. Thus, when the stylet 882 is in place in the hollow tube/lumen 890 of the ablation shaft/sleeve 881 and the flexible distal ablation portion 885 is positioned at the ablation site within the patient, the distal portion 898 of the stylet 882 transforms

into its pre-set shape causing the flexible distal ablation portion **885** to transform to a corresponding shape as depicted in FIG. 27B.

The shape of the distal portion **898** of the stylet **882** can be based on the type of procedure/treatment that the ablation catheter **880** will be used to perform as well as the patient's anatomy where the treatment is being performed. Thus, if a procedure is performed with one stylet **882** having a specific shape/orientation and the ablation was not successful because of incomplete lesion formation, for example, the surgeon can simply remove the stylet **882** from the ablation shaft/sleeve **881** while leaving the ablation shaft/sleeve **881** in place in the patient. The surgeon can then (a) choose a different stylet **882** having a distal portion **898** with a different size and/or shape than that of the previously-used stylet **898**, (b) insert this new stylet **882** into the hollow tube/lumen **890** of the ablation shaft/sleeve **881** and (c) continue with the ablation procedure. The surgeon can do this as many times as is necessary to achieve a successful ablation, e.g., complete lesion formation.

In some embodiments, a portion **899** of the stylet **882** can be set with a pre-determined articulation angle, which can be helpful in directing the flexible distal ablation portion **885** into contact with the target tissue for the ablation. In some embodiments, the articulation portion **899** of the stylet **882** corresponds to the flexible shaft portion **884** of the ablation shaft/sleeve **881**.

In some embodiments, the stylet **882** can be designed to have different flexibilities along its length. As depicted in FIG. 30, in one embodiment, the stylet **882** can be designed to have three (3) portions identified as portions "A," "B" and "C" with different flexibilities. For example, portion "A" can have a first flexibility, portion "B" can have a second flexibility and portion "C" can have a third flexibility. In some embodiments, portion "B" is more flexible than portions "A" and "C" as it may be necessary for portion "B" and its associated portion of the ablation shaft/sleeve **881** to articulate such that portion "A" and its associated portion of the ablation shaft/sleeve **881** can be manipulated into contact with the target tissue within the heart to be ablated. It may be necessary for portions "A" and "C" and their associated portions of the ablation shaft/sleeve **881** to be less flexible/more rigid or stiffer than portion "B" such that pressure/force can be applied during delivery of the ablation shaft/sleeve **881** and transferred to the flexible distal ablation portion **885** of the ablation shaft/sleeve **881** such that the flexible distal ablation portion **885** can be manipulated into the proper position against the target tissue and held in place.

In some embodiments, portions of the stylet **882** can be designed to have a flexibility similar to the flexibility of corresponding portions of the of the ablation shaft/sleeve **881**. In some embodiments, the ablation shaft/sleeve **881** can be designed to have a uniform flexibility, however, the flexibility of specific portions the ablation shaft/sleeve **881** can be adjusted or controlled based on the flexibility of corresponding portions of the stylet **882**. Thus, the stylet **882** may be responsible for controlling the flexibility of the catheter **880**.

The flexibility along the length of the stylet **882** can be changed or altered in various ways. For example, in some embodiments, the properties of the shape memory material from which the stylet **882** is constructed, can be altered. One property that can be altered is the transition temperature of the shape memory alloy. Thus, a shape memory alloy that may have a certain flexibility at one temperature can have a different flexibility at the same temperature due to an altered transition temperature.

As depicted in FIG. 31A and FIG. 31B, in one embodiment, the flexibility along the length of the stylet **882** can be altered by changing the diameter of the stylet **882**. FIG. 31B, which is a detail of View A in FIG. 31A, shows that material can be removed from stylet **882** such that portions of the stylet **882** have a diameter "d1" while other portions of the stylet **882** have a diameter "d2," which is less than diameter "d1." Thus, portions of the stylet **882** that have either diameters that alternate between "d1" and "d2" or that have extended lengths "L2" with a diameter "d2," are more flexible than portions of the stylet **882** that have a consistent diameter "d1." In some embodiments, the flexibility can be altered based on lengths "L1" and "L2" of the larger diameter portions "d1" and smaller diameter portions "d2," respectively. Thus, portions of the stylet **882** having lengths "L2" of smaller diameter portions "d2" that are greater in length than the length "L1" of larger diameter portions "d1" will be more flexible than portions of the stylet **882** having lengths "L2" of smaller diameter portions "d2" that are shorter in length than the length "L1" of larger diameter portions "d1." In other embodiments, any number of different diameter stylet portions, i.e., "d1," "d2," "d3," "d4," etc., of any lengths may be designed to impart the desired flexibility on the stylet **882** and these different diameter stylet portions may be arranged in any order and/or configuration to impart the desired flexibility on the stylet **882**.

In some embodiments, as depicted in FIGS. 32A-32C, the flexibility of portions of the stylet **882** can be altered with the inclusion of a plurality of circumferential grooves **5000**, a plurality of longitudinal grooves **5010**, or a plurality of holes **5020**. In the embodiment depicted in FIG. 32A, the flexibility of the stylet **882** can be altered based on the width "W1" of the circumferential grooves **5000**, the spacing "S1" between adjacent grooves **5000** and the spacing "L2" between adjacent sets **5030** of circumferential grooves **5000**. Thus, (a) embodiments having circumferential grooves **5000** that have a width "W1" that is greater than a width "W1" of circumferential grooves **5000** in other embodiments, (b) embodiments having circumferential grooves **5000** that have a closer spacing "S1" between adjacent grooves **5000** than spacing "S1" between circumferential grooves **5000** in other embodiments and (c) embodiments having sets **5030** of circumferential grooves **5000** that have a shorter distance "L2" between adjacent sets **5030** of circumferential grooves **5000** than in other embodiments, will be more flexible than in the other embodiments. Various combinations of widths "W1", spacings "S1" and distances "L2" can be designed to achieve the desired flexibilities of different portions of the stylet **882**.

In the embodiment depicted in FIG. 32B, the flexibility of the stylet **882** can be altered based on the width "W2" of the longitudinal grooves **5010**, the spacing "S1" between adjacent grooves **5010**, the spacing "L2" between adjacent sets **5040** of longitudinal grooves **5010** and the length "L3" of the longitudinal grooves **5010**. Thus, (a) embodiments having longitudinal grooves **5010** that have a width "W2" that is greater than a width "W2" of longitudinal grooves **5010** in other embodiments (b) embodiments having longitudinal grooves **5010** that have a length "L3" that is greater than a length "L3" of longitudinal grooves **5010** in other embodiments, (c) embodiments having longitudinal grooves **5010** that have a closer spacing "S1" between adjacent longitudinal grooves **5010** than spacing "S1" between adjacent longitudinal grooves **5010** in other embodiments and (d) embodiments having sets **5040** of longitudinal grooves **5010** that have a shorter distance "L2" between adjacent sets **5040** of longitudinal grooves **5010** than in other embodiments,

will be more flexible than in the other embodiments. Various combinations of widths "W2", lengths "L3," spacings "S1" and distances "L2" can be designed to achieve the desired flexibilities of different portions of the stylet **882**.

In the embodiment depicted in FIG. 32C, the flexibility of the stylet **882** can be altered based on the diameter "D3" of the holes **5020**, the spacing "S1" between adjacent holes **5020** in the X-direction, the spacing "S2" between adjacent holes **5020** in the Y-direction and the spacing "L2" between adjacent sets **5050** of holes **5020**. Thus, (a) embodiments having holes **5020** that have a diameter "D3" that is greater than a diameter "D3" of holes **5020** in other embodiments, (b) embodiments having holes **5020** that have a closer spacing "S1" between adjacent holes **5020** in the X-direction than spacing "S1" between adjacent holes **5020** in the X-direction in other embodiments, (c) embodiments having holes **5020** that have a closer spacing "S2" between adjacent holes **5020** in the Y-direction than spacing "S2" between adjacent holes **5020** in the Y-direction in other embodiments and (d) embodiments having sets **5050** of holes **5020** that have a shorter distance "L2" between adjacent sets **5050** of holes **5020** than in other embodiments, will be more flexible than in the other embodiments. Various combinations of diameters "D3", spacings "S1," spacings "S2" and distances "L2" can be designed to achieve the desired flexibilities of different portions of the stylet **882**.

In most embodiments, the degree of flexibility correlates to the amount of stylet material that is removed or that remains in the portions of the stylet **882** where altered flexibilities are desired. Portions of the stylet **882** having more material removed will be more flexible than portions of the stylet **882** having less material removed.

In the stylet embodiments disclosed herein, combinations of alterations may be used. For example, desired flexibilities can be achieved by combining smaller diameter portions with circumferential grooves **5000** and/or longitudinal grooves **5010** and/or holes **5020**.

The multiple flexibilities in the embodiments disclosed herein are due to a removal of material in portions of the stylet along its length. The removed material can be in the form of smaller diameter portions, circumferential grooves, longitudinal grooves and/or holes and any other shapes as will be readily apparent to those skilled in the art.

In some embodiments, the ablation catheter **880** may be packaged as a kit with multiple stylets **882** having various shapes and sizes thereby giving the physician different options regarding the size and shape of the lesions to be created during the ablation procedure. These kits can be treatment specific. Therefore, only stylets having shapes and sizes for the specific procedure can be included in the kits. Thus, the ablation catheter **880** of this embodiment allows a single, universal ablation shaft/sleeve **881** to be designed and constructed that can be used for a multitude of various ablation procedures based only on providing stylets **882** specific for the procedure being performed. Constructing a single, universal ablation shaft/sleeve **881** is more cost efficient and provides for higher production rates than having to construct multiple ablation catheters that are designed to have different shapes and different handle functionality.

In some embodiments, the ablation shaft/sleeve **881** can be used to perform ablations without a stylet **882** inserted therein.

As will be discussed in more detail below, in use, the ablation shaft/sleeve **881** is delivered to an area of interest with the body, in some embodiments, for example, the left atrium of the heart to treat atrial fibrillation or the right atrium to treat atrial flutter or the right and left ventricles to

treat ventricular tachycardia, through a delivery catheter. After the ablation shaft/sleeve **881** is in position and depending on the ablation treatment being performed and the patient's anatomy, the surgeon chooses a stylet **881** to use. The surgeon then inserts this stylet **881** through the catheter handle and into the hollow tube/lumen **890** of the ablation shaft/sleeve **881** until the distal portion **898** of the stylet **882** is in place within the flexible distal ablation portion **885**. Once in place, the shape memory characteristics of the distal portion **898** of the stylet **882** cause the distal portion **898** to transform into its pre-set shape thereby causing the flexible distal ablation portion **885** to transform into a corresponding shape. The surgeon can then proceed with the ablation treatment.

Applications

Embodiments of the cryoablation apparatus (catheters, probes, etc.) described herein have a wide range of diagnostic and therapeutic applications including, for example, endovascular-based cardiac ablation and more particularly, the endovascular-based cardiac ablation treatment of atrial fibrillation.

FIG. 33 shows examples of target ablation lesions in a pulmonary vein isolation (PVI) procedure for the treatment of atrial fibrillation.

The basic structures of the heart **1** are shown in FIG. 33 including the right atrium **2**, the left atrium **3**, the right ventricle **4** and the left ventricle **5**. The vessels include the aorta **6** (accessed through the femoral artery), the superior vena cava **6a** (accessed through the subclavian veins) and the inferior vena cava **6b** (accessed through the femoral vein).

Exemplary target lesions for a PVI procedure include lesion **8** which surrounds and isolates all left pulmonary veins (PVs), and lesion **9** which surrounds and isolates all right pulmonary veins (PVs). As described further herein, the invention may include application or creation of additional lesions to increase the effectiveness of the treatment. Also, it is to be understood that although the following discussion primarily focuses on embodiments for performing PVI, the technology and procedure described herein for producing these lesions can be used to create other lesions in an around the heart and other organs such as that described in international patent application nos. PCT/US2012/047484 to Cox et al. and PCT/US2012/047487 to Cox et al. corresponding to International Publication Nos. WO2013/013098 and WO2013/013099 respectively, the contents of each of which is hereby incorporated by reference in their entirety.

FIG. 34 illustrates one technique to reach the left atrium with the distal treatment section of a catheter. The procedure may be performed under conscious sedation, or general anesthetic if desired.

A peripheral vein (such as the femoral vein FV) is punctured with a needle. The puncture wound is dilated with a dilator to a size sufficient to accommodate an introducer sheath, and an introducer sheath with at least one hemostatic valve is seated within the dilated puncture wound while maintaining relative hemostasis.

With the introducer sheath in place, the guiding catheter **10** or sheath is introduced through the hemostatic valve of the introducer sheath and is advanced along the peripheral vein, into the target heart region (e.g., the vena cavae, and into the right atrium **2**). Fluoroscopic imaging can be used to guide the catheter to the selected site.

Once in the right atrium **2**, the distal tip of the guiding catheter is positioned against the fossa ovalis in the intra-atrial septal wall. A needle or trocar is then advanced distally

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through the guide catheter until it punctures the fossa ovalis. A separate dilator may also be advanced with the needle through the fossa ovalis to prepare an access port through the septum for seating the guiding catheter. The guiding catheter thereafter replaces the needle across the septum and is seated in the left atrium through the fossa ovalis, thereby providing access for devices through its own inner lumen and into the left atrium.

Placement of the above tools may be carried out with guidance from one or more of the following: fluoroscopy, intracardiac pressures, transesophageal echocardiography (TEE), and intracardiac echocardiography (ICE).

FIGS. 35-38 illustrate a method for deploying a ring-shaped catheter in the left atrium and around pulmonary vein entries for treating various heart conditions such as atrial fibrillation.

With reference first to FIG. 35, a cross sectional view of the heart includes the right atrium RA 2, left atrium LA 3, left superior pulmonary vein LSPV entry, and left inferior pulmonary vein LIPV entry. Guide catheter 2100 is shown extending through the septum and into the left atrium.

Though not shown, mapping catheters may be positioned in the entry to the LSPV of the left atrium for monitoring electrical signals of the heart. The mapping catheters may be placed in other locations, such as, for example the coronary sinus (CS). Examples of mapping catheters include the WEBSTER® CS Bi-Directional Catheter and the LASSO® Catheter, both of which are manufactured by Biosense Webster Inc. (Diamond Bar, CA 91765, USA). Another example of mapping and cryo-treatment system is described in US Patent Publication No. 2015/0018809 to Mihalik.

Optionally, an esophageal warming balloon may be placed in the esophagus to mitigate collateral damage arising from creating the lesions. An esophageal warming balloon prevents the cold temperatures from reaching the inner layer of cells of the esophagus, and can prevent formation of, e.g., an atrio-esophageal fistula. An example of a suitable esophageal warming balloon apparatus that may be used is described in commonly assigned U.S. patent application Ser. No. 15/028,927, entitled "ENDOESESOPHAGEAL BALLOON CATHETER, SYSTEM, AND RELATED METHOD," filed Oct. 12, 2014 by Alexei Babkin, et al., the contents of which is incorporated herein by reference in its entirety for all purposes.

FIG. 36 illustrates a distal section of the cryoablation catheter 2116 advanced through the guide sheath 2100. The energy element 2118 is shown having a circular shape formed as disclosed and described herein and urged against the endocardium. As described herein the shape may be adjusted to make continuous contact with the tissue, and to form an elliptical or circular-shaped continuous lesion (such as lesion 8 shown in FIG. 33) which encloses all the left PV entries.

In embodiments the shape is modified by reducing the diameter of loop, articulating the intermediate section of the shaft, and rotating or steering the catheter distal section. Collectively, the steps of deployment, diameter control, steering and articulation can place the entire circumference of the loop in continuous contact with the endocardium tissue. When energy is applied to the distal treatment section such as, for example, by flowing a cryogen through the distal treatment section, a continuous elongate ring-shaped lesion (frozen tissue) is formed such as the lesion 8 shown in FIG. 33, enclosing all left pulmonary vein entries.

FIG. 37 illustrates formation of a ring-shaped lesion around the right superior pulmonary vein (RSPV) entries and the right inferior pulmonary vein (RIPV) entries such as,

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for example, lesion 9 shown in FIG. 33. In contrast to the somewhat linear (straight shot) positioning shown in FIGS. 35-36, the catheter neck region 2116 shown in FIG. 37 is deflected nearly 180 degrees to aim towards the right pulmonary veins. Energy element portion 2118 is positioned around the RSPV and RIPV entries.

FIG. 37 shows the energy element 2118 deployed in a circular shape and contacting the endocardium. As described herein the shape may be adjusted to make better contact with the tissue in order to form an elongate ring-shaped, continuous lesion that engulfs or surrounds the RSPV and RIPV entries.

A similar elongate ring-shaped, continuous lesion can be formed to surround the left superior pulmonary vein (LSPV) entries and the left inferior pulmonary vein (LIPV) entries.

FIG. 38 shows the catheter 2116 deflected to aim towards the posterior wall of the left atrium. Energy element portion 2118 is manipulated to form a loop and urged against the posterior wall, overlapping with previously-formed right and left lesions.

Optionally, and not shown, guidewires can be advanced from the guide sheath and used to navigate the catheter treatment section into position.

The shape of the lesion and pattern may vary. In embodiments, and with reference to FIG. 39, a "box-shaped" lesion 900 is shown surrounding multiple pulmonary vein entries in a PVI procedure. The box-shaped lesion surrounds the pulmonary vein entries on both the left and right sides of the left atrium.

The box-shaped lesion 900 may be formed in various ways. In some embodiments, the box-shaped lesion is formed by overlapping a combination of lesions, which can have similar or different shapes (e.g., oval, ellipse, ring, etc.) to form an overall larger continuous lesion, which may have a box-like shape 900 as shown in FIG. 39.

With reference to the illustration shown in FIG. 40, and the corresponding flow diagram shown in FIG. 41, a method 1000 for forming a box-shaped lesion in the left atrium that encircles/encloses all pulmonary vein (RSPV, RIPV, LSPV and LIPV) entries, is described.

Step 1010 states to advance the cryoablation catheter into the left atrium, which can be performed using a guide sheath, for example.

Step 1020 states to navigate the treatment section (energy element portion 2118) of the catheter to one side of the left atrium and into the antrum of the superior and inferior pulmonary veins on that side of the atrium.

Step 1030 states to manipulate the treatment section (energy element portion 2118) of the catheter to form a loop-like shape and to adjust the size of the loop to make full circumference tissue contact with tissue to enclose the superior and inferior vein entries on that side of the atrium.

Step 1040 states to verify tissue contact. This step may be performed using, for example, electrodes mounted on the distal treatment section as disclosed and described in commonly assigned International Patent Application No. PCT/US16/51954, entitled "TISSUE CONTACT VERIFICATION SYSTEM", filed Sep. 15, 2016 by Alexei Babkin, et al., the entire contents of which are incorporated herein by reference for all purposes. The tissue electrocardiograms (ECGs) may be displayed using an EP recording system.

Optionally, an esophageal balloon (EBB) (as discussed above) is advanced into the esophagus in the vicinity of the heart. The EBB is inflated and a thermally conducting liquid is circulated through the balloon for the duration of the ablation treatment. As described herein, the EBB minimizes

collateral damage to tissue adjacent the ablation zone by warming the tissue during the ablation cycle.

Step 1050 states to perform the ablation by freezing the tissue to create a first continuous lesion enclosing/surrounding the pulmonary vein entries on the first side of the left atrium, for example, the left side lesion 901 in FIG. 40. The duration of the tissue freeze may be up to 3 minutes or more, and generally ranges from about 1 to 3 minutes, and preferable is about 2 minutes. In embodiments, the freeze step comprises a single application of uninterrupted ablation energy.

In some embodiments, the duration of the energy application ranges from approximately 10 to 60 seconds, and sometimes is less than or equal to approximately 30 seconds.

The duration of the freeze cycle may vary. A physician or electro physiologist can elect to terminate the freeze cycle as desired (e.g., before or after the anticipated time period has passed). Examples of reasons for early termination include: a desire to reposition the catheter, a desire to improve catheter-tissue contact, or a safety concern.

Step 1060 states to confirm ablation is complete. Electrical activity from the electrodes on the distal treatment section may be monitored. During freezing, the electrocardiograms (ECG) will present abnormal signals due to freezing of the tissue and blood in contact with the freezing tip. After freezing is completed, however, the ECGs should not show any signal or evidence of a voltage potential in the tissue due to tissue necrosis.

If, however, the ECG signals/signatures reappear after the freezing step indicating that there is still electrical activity in the tissue, this is evidence that the ablation was not complete and that PVI may not have been achieved. In the event PVI was not achieved, the above described applicable steps can be repeated.

In some embodiments, another freeze in the same location can be commenced. Or, the catheter may be repositioned or otherwise adjusted to make better contact with the target tissue. Then, an additional freeze may be performed.

Performing an additional freeze can be beneficial especially if the distance between the pulmonary veins is unusually large. When the distance between the pulmonary veins is unusually large, isolating the pulmonary vein entries with only one continuous lesion is a challenge. In a sub population of patients with unusually enlarged hearts, forming an additional lesion around the pulmonary vein entries increases the likelihood of a complete and durable PVI.

Additionally, in some situations, it may be desirable to narrow the ablation loop to accommodate a single vein. In embodiments, the method comprises performing a single vein isolation around the ostium of the single vein. The diameter of the catheter loop is reduced from the relatively large size for isolating multiple veins to the applicable size of the single vein. In embodiments, the single vein isolation is performed subsequent to the larger multiple vein isolations.

Step 1070 states to repeat the applicable steps for the pulmonary veins on the other side of the left atrium. That is, for example, after the left vein antrum is isolated, the catheter loop will be navigated to the right vein antrum and all relevant steps should be repeated to create a second, right side lesion (e.g., lesion 902 of FIG. 40).

Step 1080 states to repeat the applicable above described steps for the posterior wall lesion (lesion 903 in FIG. 40). Once both the LSPV and LIPV antrum and the RSPV and RIPV vein antrum are isolated, the looped treatment section of the catheter is navigated to the posterior wall of the left atrium.

Optionally, the EBB is inflated in the esophagus and activated prior to ablation of the posterior wall. The other applicable steps for placing the left and right lesions are repeated for the posterior lesion. The posterior lesion 903 is more centrally located, and shown in FIG. 40 overlapping the left and right antrum lesions (901 and 902, respectively). Lesion 903 is also shown extending from the floor to the ceiling of the left atrium.

Although the method describes a particular order to create the left pulmonary vein, right pulmonary vein and posterior wall lesions, embodiments of the invention are not intended to be so limited except where specifically recited in the appended claims. The order that the lesions are created may vary. For example, in embodiments, the right side or posterior lesion may be performed prior to the left side lesion.

As can be seen in FIGS. 39 and 40, collectively, the plurality of independent lesions (901, 902, 903) form a composite box-like shaped continuous lesion 900 (FIG. 39) that encloses all the pulmonary vein entries on all sides (left, right, top and bottom) of the left atrium. In embodiments, the sum of the sub-lesions form an enclosure in the shape of a box, square, or rectangle. Performing the ablations to form this composite, continuous lesion 900 effectively electrically isolates all the pulmonary vein entries in the left atrium.

In patients that have atrial flutter in addition to paroxysmal atrial fibrillation and in patients that have non-paroxysmal atrial fibrillation, in addition to forming the lesions (901, 902, 903) discussed above with reference to FIGS. 39-41, it will be necessary to form an additional lesion to isolate the mitral valve. In these patients, as depicted in FIG. 42, there is electrical activity/current 950 that flows around the mitral valve 960. Therefore, the flow of this electrical activity/current 950, must be interrupted and stopped/prevented in order to treat these patients. Depicted in FIGS. 43A and 43B are embodiments of lesions that can be formed to interrupt the flow of current 950. As can be seen in the figures, this mitral lesion 975 connects to the box-like lesion 900 formed by the left pulmonary vein lesion 901, the right pulmonary vein lesion 902 and the posterior wall lesion 903.

As depicted in FIG. 43A, in one embodiment, the mitral lesion 975 extends from the vicinity of the mitral valve 960 (the mitral valve annulus) and intersects with the flow path of the current 950 and lesion 900. In this and other embodiments, it important that the mitral lesion 975 at least intersects with the flow path of the current 950 and lesion 900. Therefore, the mitral lesion 975 can be formed at various locations within the left atrium as long as it intersects the flow path of the current 950 and connects to lesion 900. This type of lesion can be formed by modifying the shape of the treatment section of the catheter.

In the embodiment depicted in FIG. 43B, the same loop-like treatment section of the catheter used to create the left pulmonary vein lesion 901, the right pulmonary vein lesion 902 and the posterior wall lesion 903 can be used to create the mitral lesion 975. As can be seen in FIG. 43B, creating a loop-like or circular mitral lesion 975 cause the lesion 975 to intersect the flow path of the current 950 and lesion 900 at multiple points (A, B, C, D) thereby increasing the likelihood of a successful procedure.

If necessary, the mitral lesion 975 can be created after the box-like lesion 900 described above with respect to FIG. 41 is formed. A method 1100 for performing a procedure that includes forming the mitral lesion 975 as step 1090 after the box-like lesion 900 is formed is set forth in the flow diagram shown in FIG. 44. It will be readily apparent to those skilled in the art that the steps used in the procedure for forming the left pulmonary vein lesion 901, the right pulmonary vein

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lesion 902, the posterior wall lesion 903 and the mitral lesion 975 can be performed in any order as long as following the procedure, all the pulmonary vein entries are isolated and the flow path of current 950 is interrupted.

In another embodiment, in some patients that suffer from persistent atrial fibrillation, a linear lesion in the right atrium 2 may be necessary. As depicted in FIG. 45, this linear lesion 2500 is created to connect the entrance of the Inferior Vena Cava (IVC) 6b and the annulus of the Tricuspid Valve (TV) 2510 and extends through the Cava Tricuspid Isthmus (CTI) 2520. This CTI lesion is used to prevent/interrupt the majority of potential re-entry circuits in the right atrium such as, for example, right atrial flutter and/or other arrhythmias that originate in the right atrium. This type of lesion is described in commonly assigned U.S. patent application Ser. No. 15/304,524, entitled "ENDOVASCULAR NEAR CRITICAL FLUID BASED CRYOABLATION CATHETER HAVING PLURALITY OF PREFORMED TREATMENT SHAPES," filed Oct. 15, 2016 by Alexei Babkin, et al., the contents of which is incorporated herein by reference in its entirety for all purposes.

In some embodiments, for certain patients, in addition to forming the lesions (901, 902, 903) discussed above with reference to FIGS. 39-41, it will be necessary to form the CTI lesion 2500 discussed above with reference to FIG. 45. It will be readily apparent to those skilled in the art that the steps used in the procedure for forming the left pulmonary vein lesion 901, the right pulmonary vein lesion 902, the posterior wall lesion 903 and the CTI lesion 2500 can be performed in any order as long as following the procedure, all the pulmonary vein entries are isolated and the majority of the potential re-entry circuits in the right atrium are interrupted/prevented.

In some embodiments, for certain patients, in addition to forming the lesions (901, 902, 903) discussed above with reference to FIGS. 39-41 and the mitral lesion 975 discussed above with reference to FIGS. 43A, 43B and 44, it will be necessary to form the CTI lesion 2500 discussed above with reference to FIG. 45. It will be readily apparent to those skilled in the art that the steps used in the procedure for forming the left pulmonary vein lesion 901, the right pulmonary vein lesion 902, the posterior wall lesion 903, the mitral lesion 975 and the CTI lesion 2500 can be performed in any order as long as following the procedure, all the pulmonary vein entries are isolated, the flow path of current 950 is interrupted and the majority of the potential re-entry circuits in the right atrium are interrupted/prevented.

Many modifications and variations of the present invention are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims the invention may be practiced otherwise than as specifically described.

The invention claimed is:

1. An ablation apparatus for creating a lesion in target tissue, the ablation apparatus comprising:
 - a handle;
 - an elongate shaft extending from the handle to a distal tip, the shaft comprising:
 - a first portion;
 - an ablation portion distal to the first portion; and
 - a plurality of ablation energy elements disposed along the ablation portion, wherein a space is formed between each of the ablation energy elements; and
 - a solid thermally conductive liner disposed within the space and affixed to each of the ablation energy elements by a heat-fusion bond, and wherein the

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solid thermally conductive liner individually surrounds at least a portion of each of the ablation energy elements.

2. The ablation apparatus of claim 1, wherein the plurality of ablation energy elements comprise at least one ablation energy delivery lumen and at least one ablation energy return lumen.

3. The ablation apparatus of claim 2, wherein each of the at least one ablation energy delivery lumen and the at least one ablation energy return lumen comprise an inner tube having an outer tube surrounding the inner tube thereby defining a gap between the inner tube and the outer tube.

4. The ablation apparatus of claim 3, wherein the gap is adapted to being filled with a thermally conducting media.

5. The ablation apparatus of claim 2, further comprising a plurality of ablation energy delivery lumens and a plurality of ablation energy return lumens.

6. The ablation apparatus of claim 1, further comprising at least one electrode on an exterior surface of the distal portion.

7. The ablation apparatus of claim 1, further comprising at least one service lumen.

8. The ablation apparatus of claim 1, wherein the ablation energy arises from flowing a cryogen through the plurality of ablation energy elements.

9. The ablation apparatus of claim 8, wherein the cryogen is near critical nitrogen.

10. The ablation apparatus of claim 1, further comprising a stylet lumen that extends substantially along a length of the shaft from the handle to at least the ablation portion; and a stylet capable of being inserted into the stylet lumen.

11. The ablation apparatus of claim 10, wherein the stylet comprises a shape-memory material, and wherein the stylet has a plurality of flexibilities along its length.

12. The ablation apparatus of claim 10, wherein at least a distal portion of the stylet is pre-set with a shape that corresponds to a desired shape of the lesion to be formed.

13. The ablation apparatus of claim 1, wherein the ablation apparatus is used to treat a condition selected from the group consisting of atrial fibrillation, atrial flutter and ventricular tachycardia.

14. The ablation apparatus of claim 1, wherein the thermally conductive liner is a thermoplastic elastomer (TPE).

15. The ablation apparatus of claim 14, wherein the TPE is loaded with a thermally conductive material.

16. The ablation apparatus of claim 15, wherein the TPE is a polyether block amide (PEBA) loaded with aluminum oxide.

17. The ablation apparatus of claim 15, wherein the TPE is PEBA loaded with aluminum oxide in the range from about 10% to about 70% by weight.

18. The ablation apparatus of claim 15, wherein the TPE is a PEBA loaded with boron nitride.

19. The ablation apparatus of claim 15, wherein the TPE is PEBA loaded with boron nitride in the range from about 10% to about 70% by weight.

20. The ablation apparatus of claim 1, wherein the thermally conductive liner is disposed in the space by flow melting to substantially fill the space and surround the at least one ablation energy element.

21. The ablation apparatus of claim 1, wherein ablation energy in the at least one ablation energy element decreases the temperature of the target tissue sufficient to cause ablation.

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22. The ablation apparatus of claim 1, wherein ablation energy in the at least one ablation energy element increases the temperature of the target tissue sufficient to cause ablation.

23. The ablation apparatus of claim 1, further comprising an outer sheath.

24. An ablation system for creating a lesion in target tissue, the ablation system comprising:

a catheter comprising a handle, a distal tip, and an elongate shaft extending from the handle to the distal tip, the shaft comprising:

a first portion;

an ablation portion distal to the first portion; and

at least one ablation energy element disposed along the ablation portion, wherein a space is formed between each of the at least one ablation energy elements; and

a first solid thermally conductive liner disposed within said space; and

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an energy generator coupled to the catheter to deliver and control the ablation energy from the at least one ablation energy element to the target tissue.

25. An ablation catheter for creating a lesion in target tissue comprising:

an elongate shaft extending from the handle to a distal tip, the shaft comprising an ablation portion distal to the first portion;

at least one ablation energy element disposed along the ablation portion and defining a space between each of the at least one ablation energy elements;

a solid thermally conductive liner disposed within the space; and

a heat-fusion bond joining each of the at least one ablation energy elements to the solid thermally conductive liner.

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